Smooth isometric bending of a closed sphere

Boyang Hu

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Abstract

A nonradial disk with a rotational metric can rotate inside a fixed shell without changing that shell's intrinsic metric. The difficulty is to make the disk exactly planar along its attaching rim. We construct the required annulus and obtain a smooth embedded sphere with a smooth periodic isometric bending and noncongruent images. An inner disk and a complementary cap stay fixed; area, enclosed volume and total oriented mean curvature stay constant.

The construction starts with a radial infinitesimal deformation. Infinitely many changes of curvature allow it to reach two flat ends. Explicit one-dimensional models select one angular mode, while a mass–resistance description controls the accumulating ends. Spectral matching and characteristic return maps impose different nonresonance conditions; both are needed for an inverse on the whole annulus. Residual corrections put the nonlinear linearizations on one fixed geometric domain. A Nash–Moser argument then produces an actual height with flat endpoint extensions. Angular averaging also gives twice the initial decay exponent for the mean and metric-control corrections, quantifying how the final metric differs from its radial reference. The fixed cap supplies the closed bending.

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1 A closed shell with a moving annulus

Imagine a round disk rotating inside a stationary shell. If the disk is a surface of revolution, its image does not change. If its height depends on angle, the image does change, but a rotation will generally move its attaching boundary. We arrange a third possibility: the metric is rotational, the embedded disk is not, and a whole collar of its boundary is planar. An intrinsic rotation followed by the opposite ambient rotation then fixes the collar pointwise. A stationary cap closes the disk and turns its relative rotation into a change of shape of a closed surface.

The geometrical motion is elementary once such a disk exists. Most of this paper is devoted to its existence, and especially to the exact flatness required at the attaching collars.

Theorem A. *There are a smooth surface Σ diffeomorphic to S^2, a smooth Riemannian metric g_Σ, and a jointly smooth map $\mathcal{F} : \Sigma \times \mathbb{R} \longrightarrow \mathbb{R}^3$ such that, writing $F_\tau = \mathcal{F}(\,\cdot\,, \tau)$,*

$$F_\tau \text{ is an embedding}, \qquad F_\tau^* g_{\mathrm{E}} = g_\Sigma, \qquad F_{\tau+1} = F_\tau.$$

At least two images $F_\tau(\Sigma)$ are not congruent under the Euclidean isometry group. An inner disk and a complementary cap are fixed pointwise throughout the family. The area, enclosed volume and total oriented mean curvature are independent of τ.

Here a closed surface is compact without boundary. A bending refers to a family of isometric embeddings of one fixed metric; nontriviality in this paper always means noncongruence of images, not just motion of material points. Mean curvature is $(k_1 + k_2)/2$, with the outward orientation. These conventions matter because a relabelling of a fixed surface is an isometric motion of its material points, but does not change its shape.

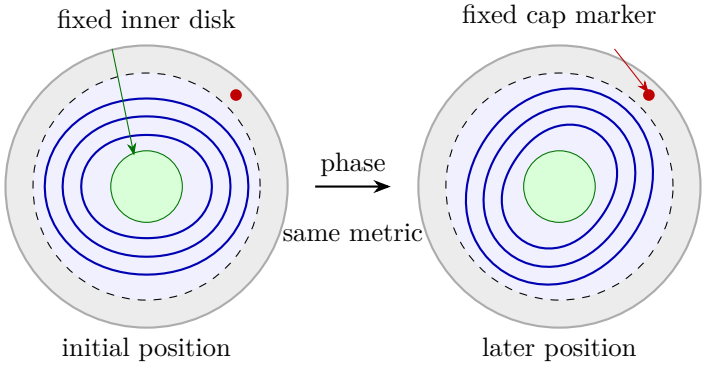


Figure 1: Relative rotation, viewed from above. The blue curves are schematic height contours, not numerical data for the constructed surface. The circular collars and the green inner disk stay fixed pointwise. The cap's red marker stays fixed while the nonradial part turns, so an ambient rotation cannot account for the entire motion.

1.1 Where the question sits in surface theory

Ghomi asks whether a closed C^2 surface in $\mathbb{R}^3$ can be flexible in Problem 1.1 of [6]. The problem requires a continuous family. Replacing a dimple by its reflection across a flat rim gives two different isometric realizations, but gives no path between them. Theorem A provides such a path in the smooth category.

The distinction from infinitesimal bending is just as substantial. A normal vector field supported inside a planar patch has zero first variation of the metric. This observation already

appears in Cohn-Vossen's discussion of closed infinitesimally nonrigid surfaces [2, footnote 2]. It does not solve the nonlinear isometry equations. Efimov's classical survey [4] and his work on deformation at zero-curvature points [5] belong to the historical setting of this distinction; the later survey [13, Sections 1.1–1.4] treats finite bendings, their infinitesimal equations and higher-order extension. Our first deformation is also infinitesimal. The analytic argument below is what turns it into an actual annular height.

On the rigid side, the hypotheses on curvature and boundary data are decisive. For example, Ghomi and Spruck [7, Theorem 1.1] prove that two properly oriented $C^{2,1}$ positively curved isometric immersions which agree on a curve segment agree everywhere on their connected domain. The present annulus has both signs of curvature, with infinitely many transitions at either flat end. Thus neither strict positivity nor a finite collection of nondegenerate interfaces describes its geometry.

Connelly's flexible embedded polyhedral sphere [3] establishes flexibility in the polyhedral category. Rounding its edges would change the metric, so it does not furnish a smooth bending. Nash's and Kuiper's C^1 theorems [16, 14] concern another regularity regime. In the smooth problem the second fundamental form and the type of the Darboux equation must be controlled at every scale.

Our analytic method has closer connections with smooth isometric embedding. Smoothing compensates for derivative loss already in Nash's argument [17]. Han, Hong and Lin [10, Theorem 1.1] use a Nash–Moser–Hörmander scheme for nonpositive curvature under a specified degeneracy condition. Han and Khuri treat local mixed-type problems with finite-order changes of type [11, 12]. In particular, the residual modification of the linearization in [11, pp. 650–653] is a precedent for keeping useful geometric features fixed before the nonlinear equation is solved. Here the extra issue is simultaneous control of infinitely many interfaces and both flat ends on a single closed domain.

The three invariants in Theorem A follow directly from our rotation construction. They should be read with their respective scopes: Sabitov's volume theorem [19] is polyhedral; Almgren and Rivin [1] prove the smooth bending invariance of the mean-curvature integral. Curvature-precession motions also arise in the shell model of Hamouche, Maurini, Vidoli and Vincenti [9]. That mechanical comparison does not supply the flat collars or the closed embedded surface here.

1.2 How the construction is organized

Let $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ and choose a radial metric

$$g = (1 + w(r)^2)\, dr^2 + r^2\, d\theta^2, \qquad h'(r) = w(r).$$

The radial graph of h has this metric. We seek a different height $z = h + qu$, where q is a small construction parameter. The condition that $g - dz \otimes dz$ be positive and flat is the Darboux equation. It is enough to solve that scalar equation with a nonradial u and constant endpoint extensions. Section 2 proves this geometric reduction completely, including injectivity and noncongruence.

Three decisions then determine the analysis. First, a monotone flat end kills every nonzero flat angular Jacobi field. We therefore use an infinite sequence of rising and falling pieces, and design their scales so that one angular mode survives while all jets still vanish at the endpoint. Second, propagation of characteristics and matching of endpoint solution lines have different resonances. We separate them in a finite model before making the infinite construction. Third, the inverse is needed for nearby nonlinear states, on the same function

space. We adjust the linearization by terms proportional to the unsolved equations, keeping the turning circles and characteristic returns fixed. These terms vanish at a solution.

The estimates are arranged to spend precision where these decisions are implemented. The model spectra and elementary operator facts are standard inputs; the matching of infinitely many intervals, the fixed domain of the perturbed inverse, and the improvement of endpoint decay receive the detailed proofs. In particular the numbers $2n^3/3$, $n^{5/2}$, 12 and $3/4$ are introduced with the inequalities they must satisfy, rather than as unexplained choices.

Once a nonzero q has been fixed, the height and the metric are fixed. The bending parameter is a different variable:

$$F_\tau(r,\theta) = R_{-2\pi\tau} F(r, \theta + 2\pi\tau). \tag{1}$$

The rest of the proof returns to this formula. No nonlinear integration of an infinitesimal bending in the variable τ remains to be done.

2 The geometric mechanism

This section isolates what the analysis has to produce. Throughout, $0 < r_- < r_+$, $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$, and

$$g = E(r)\, dr^2 + r^2\, d\theta^2, \qquad E = 1 + w^2. \tag{2}$$

A function is flat at an endpoint when it extends smoothly there with every derivative zero. A height has a constant endpoint extension when its value has a limit and every positive-order derivative of its difference from that limit vanishes. Extending these functions and slightly enlarging the annulus then gives actual open collars on which $w = 0$ and the height is constant. Thus endpoint flatness, rather than a merely small error at the boundary, is the appropriate analytic target.

2.1 Removing one coordinate from the metric

An embedding of the form $F = (P, z)$ is isometric exactly when

$$P^* g_{\mathbb{R}^2} = \bar{g} := g - dz \otimes dz.$$

We need $\bar{g}$ to be a positive flat metric and its planar development to be injective. The local condition is a scalar equation. For a symmetric tensor B, write $\det_g B = \det(B_{ij})/\det(g_{ij})$.

Lemma 2.1. *For a smooth height z with $v = |dz|_g^2 < 1$,*

$$K_{g - dz^2} = \frac{K_g(1 - v) - \det_g \nabla_g^2 z}{(1 - v)^2}. \tag{3}$$

In particular, the positive metric $g - dz^2$ is flat precisely when

$$\det_g \nabla_g^2 z = K_g(1 - |dz|_g^2). \tag{4}$$

Proof. In a g-orthonormal frame with its first covector parallel to dz, $\bar{g}$ has matrix $\mathrm{diag}(1-v, 1)$. Regard $(x, z(x))$ as a graph in $(\Sigma, \bar{g}) \times \mathbb{R}$, with induced metric g. Its upward unit normal has vertical component $\sqrt{1-v}$, and its second fundamental form satisfies $\nabla_g^2 z = \sqrt{1-v}\,\mathrm{II}$. The ambient sectional curvature of its tangent plane is $(1-v)K_{\bar{g}}$, because horizontal projection multiplies squared area by $1 - v$. The Gauss equation therefore reads

$$K_g = (1 - v)K_{\bar{g}} + (1 - v)^{-1} \det_g \nabla_g^2 z,$$

which gives the formula. □

This proof also explains the equation: it is exactly the compatibility condition for the two horizontal coordinates after a vertical coordinate has been chosen. No local embedding theorem is being used to bypass the global reconstruction.

2.2 Why the development is a graph over a round disk

Flatness on an annulus alone leaves a holonomy problem. Filling the inner collar by a planar disk removes it. Half-turn symmetry then ensures that this disk is centered correctly.

Proposition 2.2 (Planar reconstruction)**.** *Suppose w vanishes on open collars of both ends and z is constant there, with $z(r, \theta + \pi) = z(r, \theta)$. If $g - dz^2$ is positive and flat, extend $w = 0$ and $z = c_L$ across the inner disk. There is an orientation-preserving diffeomorphism $P : D_{r_+} \longrightarrow D_{r_+}$ with*

$$P^* g_{\mathbb{R}^2} = g - dz^2, \qquad P = \mathrm{id} \text{ on the outer collar}, \qquad P = R_\gamma \text{ on the inner disk}$$

for some angle γ. Consequently (P, z) is an isometric embedded disk.

Proof. The extended metric is smooth on the disk, including its center, where it is Euclidean. Take a global oriented orthonormal coframe η_1, η_2, equal to dx, dy on the outer collar, and let α be its connection form, with $d\eta_1 = \alpha \wedge \eta_2$ and $d\eta_2 = -\alpha \wedge \eta_1$. Flatness and simple connectivity give $\alpha = d\psi$, normalized by $\psi = 0$ on that collar. Thus $e^{i\psi}(\eta_1 + i\eta_2)$ is closed. Its primitive P, normalized to be the identity on the collar, is an oriented local isometry to the plane.

The half-turn $I : x \mapsto -x$ preserves the metric. Both P and $-P \circ I$ are normalized developments, so they agree: two oriented planar developments of a connected flat surface differ by a fixed Euclidean motion, as follows by differentiating their relative rotation. Hence P is odd. On the inner Euclidean disk it is a Euclidean motion, and oddness removes its translation. This gives the asserted rotation there.

Extend P by the identity outside the disk. It is a proper local diffeomorphism of $\mathbb{R}^2$, hence a covering. It has one sheet outside a compact set, so it is a diffeomorphism. Since the exterior is fixed, the disk maps onto itself. The height graph (P, z) is therefore embedded, with metric $\bar{g} + dz^2 = g$. □

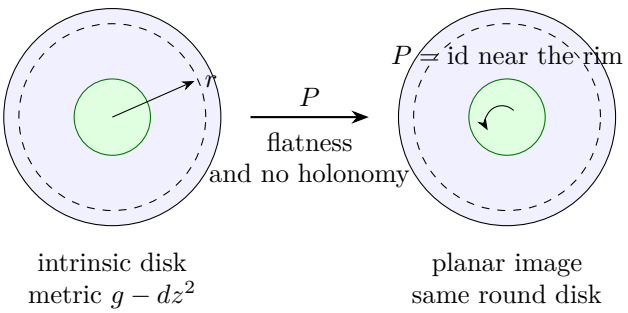


Figure 2: The inner fill removes the annular holonomy. Normalization fixes the outer collar; half-turn symmetry fixes the center of the inner disk. Properness then upgrades the local development to a global diffeomorphism. The sketch records these constraints, not the pointwise distortion of P.

2.3 A stationary cap which distinguishes shapes

We need more than a cap that closes the graph. A completely symmetric cap could hide the relative rotation. The following explicit marker reduces the possible congruences to a finite set.

Lemma 2.3. *A smooth graph over D_{r_+} which is horizontal near its boundary extends to a smooth embedded sphere. The complementary cap may be chosen to contain the unique global maximum of Gaussian curvature, with distinct principal curvatures at that point.*

Proof. Translate the boundary height to zero and extend the horizontal collar to a radius $R > r_+$. Choose a smooth nondecreasing function $\chi : [0,1] \to [0,1]$, constant at its endpoint values near the endpoints and satisfying $\chi(1-t) = 1-\chi(t)$. The unit-speed meridian starting at $(R, 0)$ with

$$r'(s) = \cos(\pi\chi(s/L)), \qquad z'(s) = -\sin(\pi\chi(s/L)), \qquad 0 \le s \le L,$$

stays at radius at least R and returns to radius R with inward horizontal tangent. Its height decreases by a positive constant times L. Take L large enough to pass below the minimum of the graph, and finish with a horizontal disk to the axis. Rotation of this meridian gives a smooth embedded cap disjoint from the graph except at its attaching collar.

Place a bump in the unused horizontal annulus $r_+ < r < R$. In local Cartesian coordinates choose $A, B > 0$ with $A \neq B$, put $\xi = Ax^2 + By^2$, and set

$$u(x,y) = \frac{k}{2}\int_{\xi}^{\rho^2} e^{-s/(\rho^2-s)}\, ds \quad (\xi < \rho^2), \qquad u = 0 \quad (\xi \ge \rho^2).$$

This is smooth across its support boundary. If $\Xi = \xi/(\rho^2 - \xi)$, the curvature of its graph is

$$K = \frac{ABk^2 e^{-2\Xi}(1 - 2\xi\Xi')}{\left[1 + k^2 e^{-2\Xi}(A^2x^2 + B^2y^2)\right]^2}. \tag{5}$$

At the center it equals ABk^2. Elsewhere its numerator is either nonpositive or strictly smaller, and its denominator is at least one. Choose k to make this value larger than the curvature of the rest of the sphere. Then choose ρ small enough to fit the support and keep its height, at most $k\rho^2/2$, away from the bottom cap. The principal curvatures at the center are $-kA, -kB$ for the upward normal, so they are distinct. □

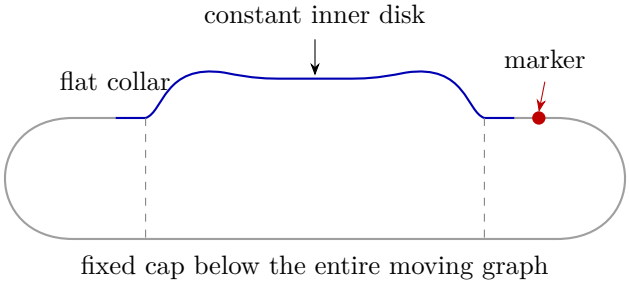


Figure 3: A meridional sketch of the closure. The actual blue graph depends on angle; its whole rotation stays inside the central cylinder and above the bottom disk. The marker lies in the stationary horizontal extension.

2.4 The annulus implies the theorem

Theorem 2.4 (Geometric closure)**.** *If a nonradial, half-turn invariant height z for (2) is constant on two collars where $w = 0$, and $g - dz^2$ is positive and flat, then all the conclusions of Theorem A hold.*

Proof. Develop the filled disk by Proposition 2.2 and attach the cap of Lemma 2.3. On the disk set $F_\tau = R_{-2\pi\tau} \circ F \circ R_{2\pi\tau}$, and leave the cap fixed. The intrinsic rotation preserves g. On the inner disk and outer collar its action cancels the ambient rotation, since P is a centered rotation there and z is constant. The formula therefore glues smoothly in both variables and is one-periodic. Each disk image is a rotation of a graph over the same round disk; the separated cap closes it without introducing intersections.

The disk images cannot all coincide. Otherwise $F^{-1}R_\vartheta F$ would be an intrinsic isometry equal to R_ϑ on the outer collar. An isometry of a connected Riemannian manifold is determined by its value and derivative at an interior point, by geodesic uniqueness. Thus it would equal R_ϑ everywhere. Its height component would force $z(r, \theta + \vartheta) = z(r, \theta)$ for every ϑ, contrary to nonradiality.

Any congruence between two closed images fixes the unique curvature maximum in the cap. It preserves the outward normal and the two distinct principal lines there. There are at most four orthogonal maps with these properties. If all the closed images were congruent, their continuous family would therefore lie in a finite set and would be constant. Its disk portion would be constant too, which we have just excluded.

Finally, the moving disk is an ambient rotation of itself, while the rest of the surface is stationary. This proves invariance of area and of the oriented mean-curvature integral. If Z_τ is the height over the planar disk and c_B is the bottom height, the part of the enclosed volume inside its cylinder is

$$\int_{D_{r_+}} (Z_\tau(y) - c_B)\, dy = \int_{D_{r_+}} (Z_0(R_{2\pi\tau}y) - c_B)\, dy,$$

which is constant by rotation invariance of planar area. The remaining volume is stationary. □

The task left for the analysis is now precise: construct one nonradial annular height with a positive flat remainder metric and constant endpoint extensions. The family itself then follows without an additional existence argument.

3 What the height equation requires

The geometric reduction leaves considerable freedom: we may choose the rotational metric as well as its nonradial height. This section identifies the conditions that guide those choices. A connection form gives the scalar equation and its conservation law; its linearization reveals the changes of type; a finite model then separates the two resonance problems that the construction must solve.

3.1 A curvature residual with a symmetric derivative

Write $h' = w$, $E = 1 + w^2$, and introduce the local height jets

$$k = z_r, \qquad p = z_\theta/r, \qquad D = E(1 - p^2) - k^2, \qquad Q = k + Ep_\theta. \tag{6}$$

The metric $g - dz^2$ is positive if and only if $D > 0$. A useful orthonormal coframe for it is

$$\eta_1 = \frac{\sqrt{D}}{\sqrt{1-p^2}}\,dr, \qquad \eta_2 = r\sqrt{1-p^2}\,d\theta - \frac{kp}{\sqrt{1-p^2}}\,dr.$$

In this polar frame the Euclidean connection is $d\theta$. We use $dr\,d\theta$ for all scalar adjoints; the source measure must remain explicit when coordinates are changed later.

Proposition 3.1. *For $D > 0$, the connection form and the curvature residual are*

$$\alpha = D^{-1/2}\left(d\theta + \frac{k}{1-p^2}\,dp\right), \tag{7}$$

$$r\mathcal{N}_h(z) = D^{-3/2}\left[Qk_r - Erp_r^2 - \frac{E'}{2}(1-p^2+kp_\theta)\right], \tag{8}$$

where $\mathcal{N}_h(z) = -\sqrt{D}\,K(g-dz^2)$ and $d\alpha = r\mathcal{N}_h(z)\,dr\wedge d\theta$. Holding w fixed, the derivative is

$$rD_z\mathcal{N}_h(z)\eta = \partial_r(a\eta_r + b\eta_\theta) + \partial_\theta(b\eta_r + c\eta_\theta), \tag{9}$$

with

$$a = QD^{-3/2}, \quad b = -Ep_rD^{-3/2}, \quad c = r^{-1}(Ek_r - E'k/2)D^{-3/2}. \tag{10}$$

Its determinant satisfies

$$ac - b^2 = \frac{K_0}{D^2} + ED^{-3/2}\mathcal{N}_h(z), \qquad K_0 = \frac{ww'}{r}. \tag{11}$$

Proof. Equality of mixed height derivatives gives

$$rp_r = k_\theta - p. \tag{12}$$

Substitution in the two structure equations for η_1, η_2 gives (7). Here is a direct check of its curvature numerator, which also fixes all signs. Use $D_r = E'(1-p^2) - 2Epp_r - 2kk_r$ and $D_\theta = -2Epp_\theta - 2kk_\theta$. The coefficient of $dr\wedge d\theta$ in $D^{3/2}d\alpha$ is

$$-\tfrac{1}{2}D_r + \frac{D}{1-p^2}(k_rp_\theta - k_\theta p_r) - \frac{k}{2(1-p^2)}(D_rp_\theta - D_\theta p_r).$$

Replacing $k_\theta - p$ by rp_r reduces it to the bracket in (8). This is also the Darboux numerator:

$$Qk_r - Erp_r^2 - \frac{E'}{2}(1-p^2+kp_\theta) = 0. \tag{13}$$

For example the Christoffel symbols are $\Gamma^r_{rr} = E'/2E$, $\Gamma^r_{\theta\theta} = -r/E$, $\Gamma^\theta_{r\theta} = 1/r$, and

$$K_g = E'/(2rE^2). \tag{14}$$

These formulas give the same numerator from (4).

For the variation, set $B_1 = k/((1-p^2)\sqrt{D})$. The identities $(B_1)_k = ED^{-3/2}$ and $(B_1)_E = -k/(2D^{3/2})$ give

$$\delta\alpha = d(B_1\delta p) + (a\eta_r + b\eta_\theta)\,d\theta - (b\eta_r + c\eta_\theta)\,dr, \qquad \delta p = \eta_\theta/r. \tag{15}$$

Taking the exterior derivative removes the exact term and proves (9). In particular the normal flux is $a\eta_r + b\eta_\theta$.

Finally, let N denote the bracket in (8). Since $kQ + D = E(1-p^2+kp_\theta)$,

$$ac - b^2 = \frac{E}{rD^3}\left(N + \frac{E'D}{2E}\right).$$

Using $N = rD^{3/2}\mathcal{N}_h(z)$ proves the determinant identity. □

At the radial height $z = h$, these expressions simplify to $D = 1$, $a = w$, $b = 0$, $c = w'/r$. Thus

$$\mathscr{L}_w u = (wu_r)_r + \frac{w'}{r} u_{\theta\theta}. \tag{16}$$

For $w > 0$ it is elliptic where $w' > 0$ and hyperbolic where $w' < 0$. A simple zero of w' is a turning circle. Notice that it is the slope of the profile that changes sign, not the profile itself.

3.2 Why the endpoint must keep changing type

It is tempting to take an uncomplicated bump for w, solve in its middle, and flatten its ends. The next observation explains why that does not preserve a nonzero infinitesimal height.

Proposition 3.2 (The monotone-end obstruction)**.** *Let $w > 0$ on (r_-, r_+) be C^1, with $w(r_-) = 0$ and $w' \geq 0$ near r_-. If a continuous endpoint extension of a solution satisfies*

$$(wf')' - j^2 w' f/r = 0, \qquad f(r_-) = 0, \qquad \lim_{r \downarrow r_-} wf' = 0,$$

then $f = 0$. The analogous assertion holds at a right endpoint with $w(r_+) = 0$ and $w' \leq 0$ nearby.

Proof. Integrating the flux equation and putting $m(r) = \max_{r_- \leq t \leq r} |f(t)|$ gives

$$|f'(r)| \leq \frac{j^2}{w(r)} \int_{r_-}^{r} \frac{w'(t)}{t} |f(t)|\, dt \leq \frac{j^2}{r_-} m(r).$$

Integration once more and Gronwall's inequality give $m = 0$ on a collar; ordinary interior uniqueness continues the zero. At the right endpoint integrate backwards and use $\int_r^{r_+} |w'| = w(r)$. The proof also applies to complex f. □

Consequently a positive profile with only finitely many changes of monotonicity cannot carry our endpoint-flat nonzero angular mode. The infinitely many turning circles are forced by this obstruction within the present construction. Their amplitudes and widths will decrease fast enough that the profile nevertheless extends smoothly.

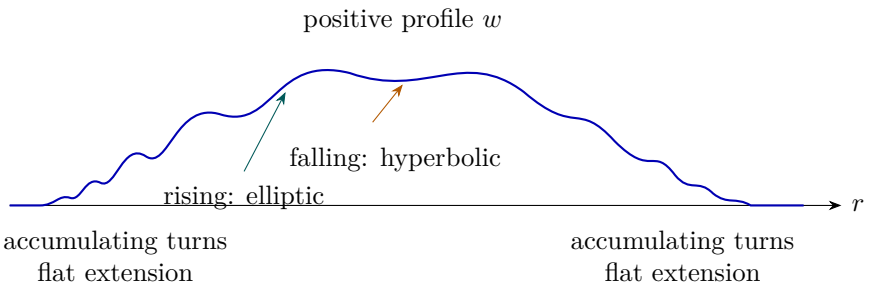


Figure 4: Why there are infinitely many interfaces. The drawing shows only finitely many representative turns, without their actual scales. On each rising interval the linearized equation is elliptic; on each falling interval it is hyperbolic. Every derivative still tends to zero at the accumulation endpoints.

3.3 Two denominators in one finite model

The next calculation is a guide to the spectral construction, not an approximation to its full inverse. Set $\epsilon = \sqrt{2}/100$, $\rho(X) = \epsilon^2 X(1-X)$, and on $[-1,2]$ solve

$$r_{XX} = -\rho r, \qquad r(0) = 1, \qquad r_X(0) = 1/10. \tag{17}$$

Its integral equation bounds $R := \max|r|$ by $R \le 1.2 + 8\epsilon^2 R$, so $R < 1.203$, $r > 0.798$, and $|r_X - .1| \le 4\epsilon^2 R < .001$. Hence $w(r(X)) = r_X(X)$ is a genuine positive radial profile. It has two simple turning points, at $X = 0, 1$. For an angular mode j and $J = j\epsilon$, its equation becomes

$$L_j U = \frac{1}{r_X}(U_{XX} + J^2 X(1-X)U). \tag{18}$$

The characteristics have slopes $d\theta/dX = \pm\epsilon\sqrt{X(1-X)}$ on $0 < X < 1$. Their travel angle is

$$T = \epsilon \int_0^1 \sqrt{X(1-X)}\, dX = \epsilon\pi/8. \tag{19}$$

The difference of the two endpoint translations acts by

$$\mathcal{D}_T f = f(\theta + T) - f(\theta - T), \qquad \mathcal{D}_T \sin(j\theta) = 2\sin(jT)\cos(j\theta). \tag{20}$$

Thus straightening the two characteristic returns costs the divisor $\sin(jT)$. For this quadratic irrational travel angle the integer $2j^2 - 640000m^2 \neq 0$, where m is nearest to $j\sqrt{2}/800$, yields

$$|\sin(jT)| \ge (320000j)^{-1}. \tag{21}$$

Indeed factor that integer, use $j\sqrt{2}/800 + |m| \le j$, and then $|\sin(\pi x)| \ge 2\,\mathrm{dist}(x, \mathbb{Z})$. The inverse of $\mathcal{D}_T$ therefore loses one angular derivative.

Now impose a radial boundary condition. Under $x = \sqrt{J}(X - 1/2)$ the homogeneous equation becomes

$$(-\partial_x^2 + x^2)U = (J/4)U. \tag{22}$$

At each end of the finite interval take the solution line obtained by restricting the corresponding decaying solution of this whole-line oscillator. These are legitimate boundary lines for (18); the radius itself need not extend outside $[-1,2]$. The two lines agree exactly at an oscillator eigenvalue. Since those eigenvalues are $2m+1$ [20, Section 8.3], the resonance is

$$J/4 = 2m + 1, \qquad \text{or equivalently} \qquad jT = (m + \tfrac{1}{2})\pi. \tag{23}$$

For an inhomogeneous source, multiply by $-r_X/J$, extend by zero outside $[-3\sqrt{J}/2, 3\sqrt{J}/2]$, apply the whole-line oscillator resolvent, and restrict back. This proves that the prescribed boundary problem has precisely these spectral denominators; it is not just a comparison of formal phases.

The half-step shift also has a local interpretation. Near $X = 0$, put $y = J^{2/3}X$. Then

$$U_{yy} + yU = J^{-2/3}y^2 U.$$

On any fixed y-interval the leading solutions are $\mathrm{Ai}(-y), \mathrm{Bi}(-y)$, with an error $O(J^{-2/3})$ in value and first derivative for fixed initial data. Variation of constants against their bounded fundamental matrix on that interval, followed by Gronwall, gives this estimate. Their standard asymptotics carry the phase shift $\pi/4$ at each turn [18, 9.7.5–9.7.12]. The exact oscillator calculation, rather than this local heuristic, proves (23).

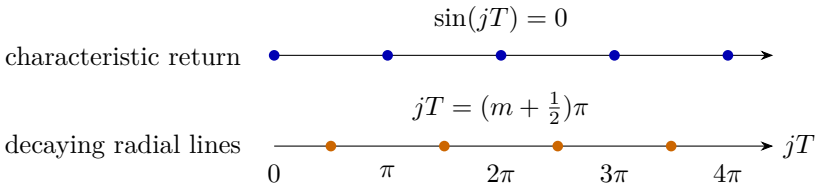


Figure 5: Two distinct resonance sets in the exact finite model. The horizontal variable is the phase, with marks spaced by π. Controlling one row does not control the other. Both must stay separated when the cells of the infinite construction are joined.

3.4 Normalization, conservation and the first residual correction

The nonlinear unknown is $z = h + qu$. Divide its residual by the small parameter, using the removable definition

$$\mathcal{A} = \mathcal{F}_q(w, u) = \begin{cases} r\mathcal{N}_h(h + qu)/q, & q \neq 0, \\ \mathscr{L}_w u, & q = 0. \end{cases} \tag{24}$$

Smoothness at zero follows from the exact formula

$$\mathcal{F}_q(w, u) = \int_0^1 r D_z \mathcal{N}_h(h + tqu) u \, dt. \tag{25}$$

We work with heights even in θ and invariant under $\theta \mapsto \theta + \pi$. Nonlinear products preserve this class. For functions constant at the endpoint collars, Stokes' theorem applied to the connection gives

$$\int_{r_-}^{r_+} \langle \mathcal{F}_q(w, u) \rangle \, dr = 0, \qquad \langle f \rangle = \frac{1}{2\pi} \int_0^{2\pi} f \, d\theta. \tag{26}$$

Both boundary periods equal 2π. Every differential of this identity, including metric differentials, has the same compatibility. Constants in u do not affect the equation.

The mean is more transparent at the level of the connection period. For $U = z - h$ set

$$\mathfrak{F}_w(U) = \langle \alpha_\theta \rangle - 1 = \left\langle \frac{1 + kp_\theta/(1 - p^2)}{\sqrt{D}} \right\rangle - 1. \tag{27}$$

Its derivative is $r\langle \mathcal{N}_h(h + U) \rangle$, and its endpoint value is zero. If $U = Z + m(r)$ with $\langle Z \rangle = 0$, the mean equation is therefore $\mathfrak{F}_w(Z + m) = 0$. Formula (15) gives

$$\partial_{m'} \mathfrak{F}_w(U) = \langle a \rangle, \qquad D_U \mathfrak{F}_w(0)[U] = w \langle U_r \rangle. \tag{28}$$

This linear cancellation is the reason the mean can later gain twice the decay of the nonconstant height.

The true linearization has one inconvenient feature: away from a solution, the residual term in (11) moves the turning circles. We remove that term while preserving the normal flux.

Lemma 3.3 (Fixing the type-change set)**.** *On the neighborhood $D \geq 1/2$, $Q \geq 7w/8 > 0$, put*

$$B = E/(rQ), \qquad \bar{c} = c - Br\mathcal{N}_h(z), \qquad \bar{L} = \partial_r(a\partial_r + b\partial_\theta) + \partial_\theta(b\partial_r + \bar{c}\partial_\theta).$$

Then

$$a\bar{c} - b^2 = K_0/D^2, \qquad D_u \mathcal{F}_q \eta = \bar{L}\eta + q\partial_\theta(B\mathcal{A}\eta_\theta). \tag{29}$$

The corrected operator is symmetric for $dr\, d\theta$, kills constants, and has the same normal flux as the true derivative.

Proof. The change in determinant is $aBr\mathcal{N}_h = ED^{-3/2}\mathcal{N}_h$, exactly the unwanted term. The change of operator is $\partial_\theta(Br\mathcal{N}_h\partial_\theta)$; substitute $r\mathcal{N}_h = q\mathcal{A}$. Only the tangential coefficient changed, proving the remaining assertions. □

The correction vanishes with the equation being solved. Its error is bilinear in the residual and the increment, the form needed in Hamilton's implicit theorem. A second correction will fix the characteristic returns. For reference the present characteristic slopes on a negative component are

$$\frac{d\theta}{dr} = d \pm v_{\mathrm{char}}, \qquad d = \frac{E(p-k_\theta)}{rQ}, \qquad v_{\mathrm{char}} = \frac{\sqrt{-ww'/r}\sqrt{D}}{Q}. \tag{30}$$

At a simple turning circle this speed is a smooth positive factor times the square root of inward distance. The subsequent coordinate construction must respect that degeneracy; merely treating all intervals as uniformly elliptic would lose the geometry of the problem.

4 Designing a profile with one surviving angular mode

The geometric construction asks for a nonradial height which becomes exactly constant at two ends. Its first-order version already tells us what the radial metric must do. For

$$u(r,\theta) = u_0(r) + \sum_{\substack{j\geq 2\\ j\ meven}} u_j(r)e_j(\theta), \qquad e_j(\theta) = \sqrt{2}\cos(j\theta),$$

the linearized equation is a family of ordinary differential equations:

$$L_j u = (wu')' - j^2\frac{w'}{r}u, \qquad \mathscr{L}_w u = (wu_r)_r + \frac{w'}{r}u_{\theta\theta}. \tag{31}$$

We want exactly one endpoint-flat solution among the nonzero angular modes. We choose its frequency to be two, the first frequency compatible with the half-turn symmetry used in the geometric closure. The nonlinear equation will create the higher even frequencies, so they must all have inverses on this same profile.

There are two freedoms here. The shape of a local profile determines its oscillation and its spectrum. Multiplying that profile by a positive constant changes its flux scale without changing its homogeneous solutions. We use the first freedom to obtain explicit local models, and the second to prevent a resonance when neighboring models are joined. This separation is the organizing idea of the construction.

4.1 What the radial construction must supply

We record the analytic interface before choosing the models. The estimates are stated in the original radius, so no change of source measure is implicit. Angular brackets denote normalized angular average.

Definition 4.1 (The three endpoint scales)**.** The cells approaching either endpoint are indexed by $n \geq N$. Set

$$A_n \asymp e^{-n^2}, \qquad S_n = \frac{2}{3}n^3, \quad H_n = e^{-S_n}, \qquad q_n = n^{5/2}\log(e+n), \quad \mathcal{E}_n = e^{q_n}. \tag{32}$$

A coefficient estimate of order s on cell n means an estimate by $C_s\mathcal{E}_n^{C_s}$. The same notation on the finitely many central pieces denotes fixed positive constants. A cell includes fixed portions of its neighboring positive intervals; a bounded enlargement changes its index by at most a fixed integer.

The length scale A_n measures how much room is available. The much smaller quantity H_n will measure the decay of the distinguished solution. The intermediate weight $\mathcal{E}_n$ pays for taking reciprocals and derivatives of the coefficients. Section 5 derives all three scales from a continuous construction; they are not independent hypotheses about an unknown profile.

Choose a smooth partition ϑ_m with bounded overlap on the cells and the finite middle; m records both the endpoint and the cell index. For fixed $0 < \gamma < 1$, write

$$\|(f,c)\|^2_{Y_{\gamma,s}} = |c|^2 + \sum_m \mathcal{E}_m^{2M_0} H_m^{-2\gamma} \|\vartheta_m f\|^2_{L^2(dr)H^s(\mathbb{T})}. \tag{33}$$

The output norm at angular order zero replaces $\mathcal{E}_m^{2M_0}$ by $\mathcal{E}_m^{-2M_1}$. In that norm the zero mode is written $a_L\chi_L + v_0$, where χ_L is a fixed smooth function equal to one near the left endpoint and zero near the right. We include $|a_L|$ and the scalar unknown below in the output norm.

Proposition 4.2 (The fixed radial reference)**.** *There is a positive smooth profile w_* on a finite interval (r_-, r_+), flat at both endpoints, with only simple interior zeros of w_*'. It has the following properties.*

There is a nonzero endpoint-flat function ϕ_* with $L_2\phi_* = 0$. This is the whole endpoint-flat kernel at frequency two; the kernel is zero at every even frequency $j \geq 4$. Every component of $\{w_*' < 0\}$ has travel angle T_m satisfying

$$|\sin(kT_m)| \geq \gamma_m/|k| \quad (k \geq 1), \qquad \gamma_m^{-1} \leq \mathcal{E}_m^C, \qquad T_m/\pi \text{ is quadratic irrational.} \tag{34}$$

There is a smooth χ_* supported in a positive component such that

$$g_* = L'_{2,w_*}[w_*\chi_*]\phi_*, \qquad \int g_*\phi_*\, dr = 1, \qquad \ell_* u = \int g_* u_2\, dr. \tag{35}$$

The bordered operator

$$\mathbb{L}_{w_*}(u,\nu) = (\mathscr{L}_{w_*}u + \nu g_* e_2, \ell_* u) \tag{36}$$

has a two-sided inverse on compatible sources, $\int \langle f\rangle\, dr = 0$, with right endpoint height zero and one free left endpoint constant. For every compact interval $[\gamma_-, \gamma_+] \Subset (0,1)$ there are fixed M_0, M_1 for which this inverse is bounded from $Y_{\gamma,0}$ to the output norm above, uniformly for γ in that interval. One radial derivative costs at most one additional angular derivative of the source. The profile, its kernel and the border are the same for all these exponents.

Here $L'_{2,w}[v]\phi = (v\phi')' - 4v'\phi/r$ is the derivative with respect to the radial profile; the source is still a density in dr. The proof occupies this section and the next two. Its conclusion is a base inverse estimate, including the endpoint realization. Higher total orders will be recovered from the differential equation after the weighted spaces have been introduced. Keeping these two tasks separate avoids making the choice of geometry depend on the eventual regularity order.

4.2 Turning a spectral potential into a radial metric

It is easier to design the spectrum before designing the radius. Let X be a model coordinate, let $r_X = v > 0$, and choose a positive density $\mathcal{P}$. Put

$$\begin{gathered} w(r(X)) = \lambda\mathcal{P}(X)v(X), \qquad \rho = -\frac{r_{XX} + (\mathcal{P}_X/\mathcal{P})r_X}{r}, \qquad V_0 = \frac{(\sqrt{\mathcal{P}})_{XX}}{\sqrt{\mathcal{P}}}, \\ L_j(\mathcal{P}^{-1/2}\psi) = -\frac{\lambda\sqrt{\mathcal{P}}}{v}(-\psi_{XX} + (V_0 - j^2\rho)\psi). \end{gathered} \tag{37}$$

Indeed $wu_r = \lambda \mathcal{P} u_X$, and differentiating this expression and using the definition of ρ gives the identity. Thus a Schrödinger model determines the radial equation once its radius satisfies the displayed linear equation and remains increasing. The factor λ is absent from the homogeneous model.

Our two choices have complementary jobs. The central oscillator retains the desired zero mode. The Morse models near the ends have explicit nonzero gaps, but are short enough in radius to accumulate at a finite point. Figure 6 illustrates the potentials; it is their solvable spectra, rather than the qualitative shapes alone, that make them useful.

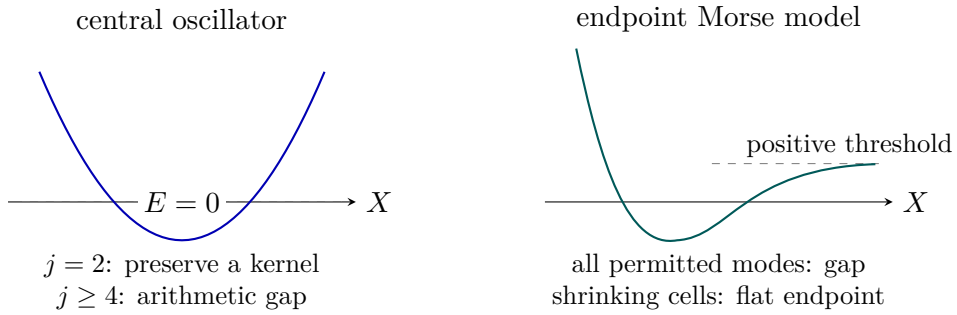


Figure 6: The two full-line spectral models, drawn schematically with an arbitrary vertical scale. Only finite pieces are used in the surface. Positive end regions separate those pieces when they are joined.

4.3 The central oscillator

Set

$$A_c = 1/\sqrt{2}, \qquad a_c = \sqrt{4A_c - 2}, \qquad c_c = 3A_c - 2 > 0, \qquad \alpha_c = (c_c + 1)/2. \tag{38}$$

Take $\mathcal{P} = e^{-2a_c X}$ and $\rho = A_c - X^2$. Then the model operator is

$$H_{j,c} = -\partial_X^2 + j^2 X^2 + a_c^2 - j^2 A_c.$$

Its standard oscillator spectrum is

$$E_{m,j} = j(2m+1) + a_c^2 - j^2 A_c, \qquad m = 0, 1, \ldots; \tag{39}$$

see [20, §8.3, equations (8.33)–(8.42)]. The choice of a_c sets $E_{0,2} = 0$. The irrational choice of A_c keeps the other even frequencies away from zero.

Here is the quantitative part of that last assertion. Put $\Phi_j = jA_c - a_c^2/j$. For even $j \geq 4$,

$$N_j := j^2/2 - \Phi_j^2 = 2A_c a_c^2 - a_c^4/j^2$$

lies between $13/4 - 3\sqrt{2}/2$ and $4 - 2\sqrt{2}$, so its distance from the integers is at least $\delta = \frac{3}{4}(3 - 2\sqrt{2}) > 0$. If $|E_{m,j}| < 1$, then $2m + 1 = \Phi_j + E_{m,j}/j$ and $2m + 1 + \Phi_j < 2j$. Since $j^2/2 - (2m+1)^2$ is an integer,

$$\delta \leq |(2m+1)^2 - \Phi_j^2| < 2|E_{m,j}|.$$

This also yields the bound when $|E_{m,j}| \geq 1$. At $j = 2$ the levels are $4m$. Consequently

$$\|H_{j,c}^{-1}\| \leq 2/\delta \quad (j \geq 4 \text{ even}), \qquad \ker H_{2,c} = \operatorname{span}\{e^{-X^2}\}, \tag{40}$$

and the reduced inverse at $j = 2$ has norm $1/4$. We use the closed quadratic-form realization in these statements.

We still need an increasing radius; an arbitrary solution of the radius equation would not suffice. An explicit positive one is

$$\psi_1(X) = e^{-X^2/2} \int_0^\infty t^{\alpha_c - 1} e^{-t^2/2 + \sqrt{2}Xt}\, dt, \qquad r = e^{a_c X}\psi_1, \qquad \upsilon = r_X/r. \tag{41}$$

An integration by parts in t shows that $\psi_1'' = (X^2 + c_c)\psi_1$. The function is positive, while ψ_1, ψ_1' tend to zero as $X \to -\infty$. Hence $\psi_1'' > 0$ implies $\psi_1' > 0$ and $\upsilon = a_c + \psi_1'/\psi_1 > a_c$. Using $c_c - a_c^2 + A_c = 0$ gives $r_{XX} - 2a_c r_X + (A_c - X^2)r = 0$, exactly the equation in (37). In particular this spectral model really is a radial metric.

For later cuts, the needed end estimates are

$$\upsilon = \begin{cases} -X + a_c - \alpha_c/X + O(X^{-3}), & X \to -\infty, \\ X + a_c + (\alpha_c - 1)/X + O(X^{-3}), & X \to +\infty. \end{cases} \tag{42}$$

These estimates hold after any fixed number of derivatives. For $X = -L$, substitute $t = z/(\sqrt{2}L)$ in the integral; the remaining integral is $\Gamma(\alpha_c)(1 + O(L^{-2}))$ and can be differentiated under its exponentially integrable majorant. For $X > 0$, complete the square and expand $t^{\alpha_c - 1}$ about $\sqrt{2}X$ against the Gaussian; the part $|t - \sqrt{2}X| > X/2$ is exponentially small. Logarithmic differentiation gives (42). Thus r ranges from zero to infinity, and $\upsilon \asymp 1 + |X|$.

The surviving radial height and its logarithmic slope are

$$\phi_0 = e^{a_c X - X^2}, \qquad \frac{\partial_y \phi_0}{\phi_0} = \frac{a_c - 2X}{\upsilon} \longrightarrow 2 \text{ on the left}, \quad -2 \text{ on the right}, \qquad y = \log r. \tag{43}$$

Writing $w = re^f$, the radius equation also gives

$$1 + f_y = \frac{X^2 - A_c}{\upsilon^2} \longrightarrow 1. \tag{44}$$

We may therefore cut far enough out that both end regions are positive and have uniformly bounded logarithmic-coordinate jets. There are just two simple turning points in the retained middle. Its travel angle is $\pi A_c/2 = \pi/(2\sqrt{2})$, whose sine divisors satisfy $|\sin(kT)| \geq c/k$ by the integer norm $|k^2 - 8m^2| \geq 1$.

There is one measure conversion worth settling now. Define

$$\varpi = 12w/r^2, \quad B_j = -\varpi^{-1} L_j, \qquad \|u\|^2_{L^2(\varpi\, dr)} = 12\lambda \|\upsilon\psi\|_2^2, \quad u = \mathcal{P}^{-1/2}\psi. \tag{45}$$

The inverse in this measure contains $\upsilon H_{j,c}^{-1}\upsilon$, not merely $H_{j,c}^{-1}$. It is uniformly bounded. Indeed $V_j \geq cj^2\upsilon^2 - Cj^2$ and the spectral gap first give $\|\upsilon H_{j,c}^{-1}\| \leq C$; taking adjoints gives the bound on the other side. For $\psi = H_{j,c}^{-1}(\upsilon F)$,

$$cj^2 \|\upsilon\psi\|_2^2 \leq \|\upsilon\psi\|_2 \|F\|_2 + Cj^2 \|F\|_2^2,$$

and absorption yields the two-sided weight bound. At $j = 2$, add the projection onto e^{-X^2}, apply the same argument, and subtract that projection. Its weighted norm is finite.

We will temporarily border mode two by a compact function g_0 with $\int g_0 \phi_0\, dr = 1$ and row $\ell_0 u = \int g_0 u\, dr$. The source obstruction is removed by pairing with ϕ_0; adding a multiple of ϕ_0 fixes the row. Thus this bordered inverse has the same bounds. Normalize its Hilbert scalar so that under a height rescaling $w \mapsto \lambda w$ one has $g_0 \mapsto \sqrt{\lambda} g_0$, $\phi_0 \mapsto \phi_0/\sqrt{\lambda}$. Both the row and column then have bounded Hilbert norm. This is an auxiliary border, changed to (35) after all geometric choices have been made.

4.4 Morse models: two quantitative gaps from one irrationality

For the tails take $\mathcal{P} = 1$ and

$$\rho(X) = A(e^{-X} - e^{-2X}) - B, \qquad H_{j,M} = -\partial_X^2 - j^2\rho, \qquad A > 4B > 0.$$

The essential spectrum starts at j^2B; below it the simple eigenvalues are

$$E_{m,j} = j^2B - (\alpha_j - m)^2, \qquad \alpha_j = (j\sqrt{A} - 1)/2, \qquad 0 \le m < \alpha_j. \tag{46}$$

This is the Morse spectrum [15]. For completeness, the factorization which fixes its normalization is short. With $\omega = j\sqrt{A}$ and $D_\alpha = \partial_X + \alpha - \omega e^{-X}$,

$$H_\alpha = D_\alpha^* D_\alpha + j^2B - \alpha^2, \qquad H_{\alpha-1} = D_\alpha D_\alpha^* + j^2B - \alpha^2.$$

The null function of D_α is $e^{-\alpha X - \omega e^{-X}}$, square integrable exactly for $\alpha > 0$. Descending by integers removes one ground state at a time. At $-1 < \alpha \le 0$, $H_\alpha - j^2B = D_0^* D_0 - 2\omega\alpha e^{-X} \ge 0$; there are no further levels. Reversing the intertwining recovers every level in (46). The quadratic-form realization justifies these identities by cutoff integration. At the left end the potential is confining, and at the right it approaches j^2B exponentially; the usual compact-perturbation argument gives the stated essential threshold [20, Theorem 6.19]. These facts suffice for the resolvent estimates below, all at energy zero.

Choose a nonsquare integer d and set

$$A = (d^{-1/2} + 2d^{-9/2})^2, \qquad B = d^{-9}. \tag{47}$$

For positive integers j and odd M,

$$|j/\sqrt{d} - M| \ge (4j\sqrt{d})^{-1}.$$

If the difference is smaller than $1/2$, this follows by factoring the nonzero integer $j^2 - dM^2$; otherwise it is immediate. At a discrete level write $\kappa = (j\sqrt{A} - M)/2 > 0$. Since $\sqrt{A} - 2\sqrt{B} = d^{-1/2}$,

$$|j^2B - \kappa^2| \ge \frac{j\sqrt{B}}{2}|j/\sqrt{d} - M| \ge \frac{1}{8d^5}.$$

Together with the threshold j^2d^{-9} this proves

$$\|H_{j,M}^{-1}\| \le 8d^9 \qquad (j \ge 2). \tag{48}$$

The same irrationality controls the characteristic return. The change $t = e^{-X}$ gives

$$T = \int_{\rho>0} \sqrt{\rho}\, dX = \pi(\sqrt{A}/2 - \sqrt{B}) = \frac{\pi}{2\sqrt{d}}. \tag{49}$$

To evaluate the integral, write the positive quadratic as $A(t - t_-)(t_+ - t)$ and use $t = c + h\cos\xi$; the elementary integral is $\pi(c - \sqrt{c^2 - h^2})$. Here $t_- + t_+ = 1$ and $t_-t_+ = B/A$. Factoring the integer $k^2 - 4dm^2$, with m the nearest integer to $k/(2\sqrt{d})$, yields

$$|\sin(kT)| \ge (3k\sqrt{d})^{-1} \qquad (k \ge 1). \tag{50}$$

For $k < \sqrt{d}$ use $m = 0$ directly; otherwise the factorization gives distance at least $(6k\sqrt{d})^{-1}$ from the integers. Thus one parameter choice supplies both local spectral separation and return-angle separation. They remain distinct conditions when the models are joined: a junction can create a new spectral resonance without changing any individual travel angle. The next section deals with that remaining freedom.

5 Reaching a flat endpoint with an infinite chain

A finite collection of turning circles cannot produce the endpoint behavior we need. The construction instead repeats a small local mechanism infinitely often. On one part of each cell the solution has almost constant value and its flux changes; on the next part its flux is almost constant and its value changes. The exact continuous equation then has a discrete recurrence with a large diagonal. Its decaying solution loses approximately e^{-2n^2} on crossing cell n. Accumulating those losses gives $e^{-2n^3/3}$, fast enough to survive every fixed number of derivatives in the shrinking radial intervals.

This section makes that picture precise. We first build the actual smooth profile, then derive the recurrence without discarding its error terms. Only after those estimates are established do we choose the height ratios which prevent matching resonances.

5.1 The budget of lengths, amplitudes and errors

Let d_n be the least nonsquare integer not smaller than e^{n^2}. Then $d_n = e^{n^2} + O(1)$. Use (47) with this integer, writing the resulting parameters A_n and B_n; thus $A_n \asymp e^{-n^2}$ and $B_n \asymp A_n^9$. Initially let $\lambda_n = e^{-n^{5/2}}$. Later we allow

$$\lambda_n = e^{-n^{5/2}} \prod_{k=N}^{n} \sigma_k, \qquad 1 \le \sigma_k \le 2. \tag{51}$$

Their logarithms change by at most $O(n)$, which does not disturb the scale comparisons below.

The choices can be understood before the details of the profile are computed. A finite-order derivative on a cell of size e^{-n^2} costs $e^{C_s n^2}$. Its amplitude must decay faster; the exponent $n^{5/2}$ does this. The solution itself must decay faster still in order to pay for inverse coefficients. Its cubic exponent will come from the recurrence. These requirements leave the interval $2 < 5/2 < 3$, rather than singling out an optimal numerical exponent.

Quantity	Size	Purpose
Radial cell length	$A_n \asymp e^{-n^2}$	Summable lengths at a finite endpoint.
Profile amplitude	$e^{-n^{5/2}+O(n)}$	Flatness after every fixed derivative cost $e^{C_s n^2}$.
Positive joining length	$\asymp A_n^{12}$	A large height change has negligible transfer error.
Allowed matching angle	$\delta_n = c_0 e^{-n^2}$	Summable bad sets over all frequency bands.
Solution attenuation	e^{-S_n}, $S_n = 2n^3/3$	Integrability and flat extension of the selected solution.
Coefficient allowance	$\mathcal{E}_n = e^{n^{5/2}\log(e+n)}$	Absorbs fixed coefficient and coordinate losses while remaining smaller than $e^{\varepsilon S_n}$.

Table 1: The scale choices are a budget for distinct jobs. Their compatibility is verified in the construction and recurrence below.

In particular, for fixed $a, b, \varepsilon > 0$,

$$\mathcal{E}_n^a e^{bn^2} \le C_{a,b,\varepsilon} e^{\varepsilon S_n}, \qquad n^2 = o(q_n), \qquad q_n = o(S_n). \tag{52}$$

These comparisons concern fixed derivative orders. No assertion of uniform bounds as the derivative order tends to infinity is being made.

5.2 A cell of the actual radial profile

On a Morse interval solve

$$r_X = v > 0, \qquad v_X = -\rho r, \qquad \rho = A(e^{-X} - e^{-2X}) - B, \qquad w = \lambda v. \tag{53}$$

The last identity in (37) becomes $L_j = -(\lambda/v)H_{j,M}$. Let X_R be the right zero of ρ. Prescribe $r(X_R) = 1$, $v(X_R) = \delta A^5$ with fixed small $\delta > 0$. Retain the model from $X_c = -\log(3/2)$ to $X_R + c_R A^{-4}$, for a fixed sufficiently small $c_R > 0$.

Here are the estimates which ensure this is a genuine increasing radius. On $[X_c, X_R]$ the weighted integral of $|\rho|$ is $O(A)$, so the reverse Volterra equation

$$r(X) = 1 - v(X_R)(X_R - X) - \int_X^{X_R} (Y - X)\rho(Y)r(Y)\, dY$$

gives $r = 1 + O(A)$. At the left cut it yields

$$v(X_c) = 3A/8 + O(A^2), \qquad r(X_c) = 1 - 15A/16 + O(A^2). \tag{54}$$

The two leading integrals, after $t = e^{-X}$, are $\int_0^{3/2}(1-t)\, dt = 3/8$ and $\int_0^{3/2}\log((3/2)/t)(1-t)\, dt = 15/16$. The constant B contributes only $O(B\log^2(A/B)) = o(A^2)$. On the negative component, backward integration from X_R keeps v positive; on the short left positive component its minimum is the positive value at X_c. On the long right positive component, comparison with $r_{XX} = Br$ gives

$$\begin{gathered} r = 1 + O(A), \qquad cA^5 \le v \le CA, \qquad v \asymp A^5 \text{ on the long right part,} \\ \Delta r \asymp A, \qquad \int_{X_c}^{X_R} v^2\, dX \asymp A^2, \qquad \int_{\text{each positive cap}} \sqrt{-\rho}\, dX \asymp \sqrt{A}. \end{gathered} \tag{55}$$

For the last long-cap estimate use $\sqrt{B}\, A^{-4} \asymp \sqrt{A}$; on the short cap $-\rho \ge cA$ on a fixed subinterval. Taking c_R small ensures that r stays in a fixed positive range throughout this argument.

Multiply r, v by a radius factor $\mathfrak{r}_n$. To pass from a cell to its next neighbor in the increasing-r direction, insert a positive interval of length

$$\ell_{n,m} = c_C r_o \min(A_n, A_m)^{12}, \qquad m = n \pm 1, \tag{56}$$

where r_o is its initial radius. Choose the neighboring radius factor so that its left endpoint is $r_o + \ell_{n,m}$. The ratio of successive radius factors is $1 + O(A_n + A_m)$. The recursion converges on either end, with a prescribed limiting radius on the left. Thus the radii stay in a compact subset of $(0, \infty)$ and all cells accumulate at finite endpoints.

At this joining interval the profile increases from W_L to W_R, with

$$\log(W_R/W_L) = 4n^2 + O(n^{3/2} + n). \tag{57}$$

To match the complete smooth germs, put $f = \log w$ and prescribe

$$f' = \chi_L f_L' + \chi_R f_R' + C_1(1 - \chi_L - \chi_R), \qquad C_1 = \frac{\log(W_R/W_L) - \int \chi_L f_L' - \int \chi_R f_R'}{\int (1 - \chi_L - \chi_R)}. \tag{58}$$

The end cutoffs have disjoint supports and equal one near the respective ends. Their derivative integrals are $O(\ell_{n,m}/\min(A_n, A_m))$, whereas the required logarithmic height change is comparable to n^2. Hence $C_1 > 0$ and, after the fixed initial index is increased, $w'/w \ge 2/r$

on the joining interval. Integrating f' with the left value matches the right value and all endpoint derivatives.

This explains the apparently extravagant exponent 12. The height contrast costs $e^{4n^2+o(n^2)}$. At the largest frequency assigned to this cell, j^2 costs e^{n^2} times a polynomial. The interval length contributes $e^{-12n^2+o(n^2)}$. Their product is $e^{-7n^2+o(n^2)}$, much smaller than the allowed matching angle e^{-n^2}. The same choice makes both the mass and resistance of the join negligible in the estimates below. No minimal choice of this exponent is needed.

Flatness and coefficient bounds. The length of cell n is comparable to A_n, and $A_{n+1}/A_n = e^{-2n-1}(1+o(1))$. If d_∂ is distance to the endpoint, then on that cell and its assigned joins

$$cA_{n+1} \le d_\partial \le CA_n, \qquad \log(d_0/d_\partial) = n^2 + O(n). \tag{59}$$

All interval lengths and their reciprocals are bounded by e^{Cn^2}. On a model, $\partial_r = v^{-1}\partial_X$ and $v \ge c\mathfrak{r}_n A_n^5$. The differential equations for r, v bound every fixed derivative of $\log v$ by $e^{C_s n^2}$. Formula (58) gives the same bound on the joins. Since each derivative of $e^{\pm \log w}$ contains only one undifferentiated exponential, we obtain

$$\begin{aligned} e^{-n^{5/2}-Cn^2} \le w \le e^{-n^{5/2}+Cn^2}, \\ |\partial_r^s w| \le C_s e^{-n^{5/2}+C_s n^2}, \qquad |\partial_r^s (w^{-1})| \le C_s e^{n^{5/2}+C_s n^2}. \end{aligned} \tag{60}$$

Comparison with (59) proves that extension by zero is smooth and flat. It also proves the fixed-order coefficient bounds in Definition 4.1, uniformly in all height choices (51).

5.3 Why a mass–resistance chain appears

Use $y = \log r$. For the mode-two comparison density $\mathsf{p} = wr^7$, divide a cell into a mass interval I_n (the short positive cap and the negative component) and a resistance interval J_n (the long positive cap and its following short join). Define

$$\mu_n = \int_{I_n} \mathsf{p}\, dy, \quad R_n^I = \int_{I_n} \mathsf{p}^{-1}\, dy, \quad \nu_n = \int_{J_n} \mathsf{p}\, dy, \quad R_n = \int_{J_n} \mathsf{p}^{-1}\, dy. \tag{61}$$

For an equation $-(\mathsf{p}V_y)_y = \Lambda \mathsf{p} V$, the change of flux across an almost constant-value region is its value times $-\Lambda$ times the mass. The change of value across an almost constant-flux region is the flux times the resistance. This gives the picture in Figure 7; the estimates following it turn the picture into an exact reduction.

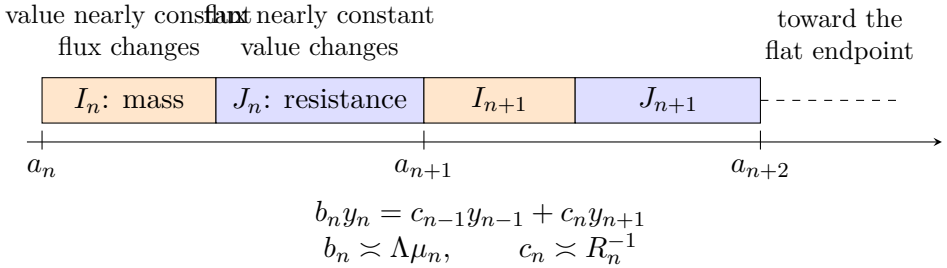


Figure 7: The continuous intervals and their exact node equation. The drawing is schematic; $a_n = (-1)^n y_n$ for the decaying solution. Large products $\Lambda \mu_n R_n$ make successive node values small.

Since $dr = v\,dX$, the two measures in (61) are $wr^6 dr$ and $dr/(wr^8)$. Using (55) gives

$$\mu_n \asymp \lambda_n A_n^2, \qquad R_n^I \le C\lambda_n^{-1}\log(A_n/B_n), \qquad \nu_n \le C\lambda_n A_n^6, \qquad R_n \asymp \lambda_n^{-1}A_n^{-4}. \tag{62}$$

A radius dilation multiplies the mass by its eighth power and the resistance by its inverse eighth power; the products below are unchanged. The joining interval contributes at most $W_R\ell$ to mass and ℓ/W_L to resistance. Their relative errors are respectively $e^{-7n^2+O(n^{3/2}+n)}$ and $e^{-11n^2+O(n)}$, so

$$\mu_n R_n \asymp A_n^{-2}, \qquad \mu_n R_n^I \le CA_n^2\log(A_n/B_n), \qquad \nu_n R_n \le CA_n^2, \tag{63}$$

while

$$\log(\mu_{n+1}R_n) = 2n^2 + O(n^{3/2}+n). \tag{64}$$

The same structure handles a general frequency. The exact gauge $u = r^{j^2}V$ gives

$$(\mathsf{p}_j V_y)_y + \Lambda \mathsf{p}_j V = 0, \qquad \mathsf{p}_j = wr^{2j^2-1}, \qquad \Lambda = j^2(j^2-1). \tag{65}$$

After removing its constant endpoint factor, $\mathsf{p}_j/\mathsf{p} = (r/r_\partial)^{2(j^2-4)}$. On the tail beyond N its logarithm is bounded by Cj^2A_N. Thus the mass and resistance comparisons remain uniform whenever j^2A_N is bounded. The stronger condition used below is

$$j^2 A_N\sqrt{\log(A_N/B_N)} \le c. \tag{66}$$

5.4 The exact continuous-to-discrete reduction

Orient $x = \log r$ on the right and $x = -\log r$ on the left, so x always increases toward the flat endpoint. On the left the outward order is $J_N, I_N, J_{N+1}, I_{N+1}, \ldots$; therefore, within this reduction only, the resistance following I_n is the original J_{n+1}. This one-index change affects only the lower-order terms in (64). The initial J_N is treated as a separate resistance edge.

The following conditions isolate precisely what is needed from the profile. For a positive density p, assume

$$\begin{gathered}\Lambda\mu_n R_n^I + \Lambda\nu_n R_n \le \eta \ll 1, \qquad \Lambda\mu_n R_n,\ \Lambda\mu_{n+1}R_n \ge K \gg 1,\\ \nu_n \le \eta\min(\mu_n, \mu_{n+1}).\end{gathered} \tag{67}$$

An initial resistance edge has the same large-product conditions and $\Lambda\nu_0 R_0 \le \eta$, $\nu_0 \le \eta\mu_N$. Call this a *deep tail*. Equations (62)–(66) imply these conditions for p_j after the initial index is large enough.

Lemma 5.1 (Deep inverse and unique decaying line)**.** *On a deep tail let $A_\mathsf{p} = -\mathsf{p}^{-1}(\mathsf{p}\partial_x)'$ act in $\mathcal{H} = L^2(\mathsf{p}\,dx)$, with Dirichlet condition at the finite entrance and the Friedrichs condition at the outer endpoint. Then*

$$\|(A_\mathsf{p} - \Lambda)^{-1}\| \le C/\Lambda.$$

The homogeneous equation has exactly one square-integrable line at the outer endpoint. At successive mass entrances its nonzero values are $a_n = (-1)^n y_n$, with $y_n > 0$ and

$$\frac{y_{n+1}}{y_n} = \frac{1 + O(\eta) + O(K^{-1})}{\Lambda\mu_{n+1}R_n}, \qquad \frac{\sqrt{\mu_{n+1}}y_{n+1}}{\sqrt{\mu_n}y_n} \le C/K < 1/2. \tag{68}$$

For the constructed Morse tails this line, transferred back to u, is smooth and flat in the original radius.

Proof. We give the reduction on the continuous equation, so that the endpoint condition and the operator norm follow from the same argument. Introduce resistance coordinate $ds = dx/\mathsf{p}$ on an edge $E_n = I_n \cup J_n$. Its length is $L_n = R_n^I + R_n$ and its density is $\omega = \mathsf{p}^2$, so $\mathsf{p}\,dx = \omega\,ds$ and

$$(A_{\mathsf{p}} - \Lambda)u = f \quad \Longleftrightarrow \quad -u_{ss} - \Lambda\omega u = \omega f.$$

The zero-potential Dirichlet kernel is $G_0(s,t) = \min(s,t)[L_n - \max(s,t)]/L_n$. The small quantity in this equation is

$$\delta_n^{\mathrm{edge}} = \Lambda \int \omega(s) G_0(s,s)\,ds \le 3\eta, \qquad R_n^I/L_n \le \eta/K.$$

The kernel inequality $G_0(s,z)G_0(z,t) \le G_0(s,t)G_0(z,z)$ bounds its resolvent series term by term. Hence the actual kernel and the two boundary-height functions satisfy

$$G_0 \le G_\Lambda \le (1-\delta_n^{\mathrm{edge}})^{-1} G_0, \qquad l_i \le v_i \le (1-\delta_n^{\mathrm{edge}})^{-1} l_i, \quad l_0 = 1 - s/L_n, l_1 = s/L_n. \tag{69}$$

The analogous inequality for the Poisson functions follows by replacing one kernel factor by l_i. For a zero-boundary u, $\Lambda \int \omega |u|^2 \le \delta_n^{\mathrm{edge}} \int |u_s|^2$. Thus the direct sum G_D of the Dirichlet edge inverses has norm at most $C\eta/\Lambda$.

The outward derivative matrix of this exact edge problem is

$$\begin{pmatrix} d_{n,0} & -c_n \\ -c_n & d_{n,1} \end{pmatrix}, \qquad c_n = L_n^{-1} + \Lambda \int \omega l_0 v_1, \qquad d_{n,i} = L_n^{-1} - \Lambda \int \omega l_i v_i.$$

Since $l_0 l_1 = G_0(s,s)/L_n$, we have

$$c_n = R_n^{-1}(1 + O(\eta)), \qquad b_n := -d_{n-1,1} - d_{n,0} = \Lambda\mu_n(1 + O(\eta) + O(K^{-1})). \tag{70}$$

For the second estimate, the mass on I_n sees $l_0 = 1 + O(R_n^I/L_n)$. The preceding edge contributes at most $\nu_{n-1} + \mu_{n-1}(R_{n-1}^I/L_{n-1})^2$; the second term is at most $\eta^2 K^{-2}\mu_n$ by (67). The two L^{-1} terms are at most $2\Lambda\mu_n/K$. These calculations also cover the initial resistance edge.

For node values $a = (a_n)$ put $(Ta)|_{E_n} = a_n v_0 + a_{n+1} v_1$ and let $\mathfrak{h} = \ell^2(\mu_n)$. Equation (69) gives

$$c\|a\|_{\mathfrak{h}}^2 \le \|Ta\|_{\mathcal{H}}^2 \le C\|a\|_{\mathfrak{h}}^2. \tag{71}$$

For example, on I_n the squared neighbor term costs at most $C\mu_n(R_n^I/L_n)^2|a_{n+1}|^2$; after an index shift its sum is $O(\eta^2/K^2)\|a\|_{\mathfrak{h}}^2$. The mass of J_n is handled by its last inequality in (67).

The shifted quadratic form on these interpolants is

$$(\mathcal{D}a, a)_{\mathfrak{h}} = -\sum b_n |a_n|^2 - 2\,\mathrm{Re}\sum c_n a_n \overline{a_{n+1}}.$$

In coordinates $\sqrt{\mu_n}a_n$, the diagonal has size Λ, while each of the two neighboring entries is bounded by $C/(R_n\sqrt{\mu_n\mu_{n+1}}) \le C\Lambda/K$. Diagonal dominance therefore gives $\|\mathcal{D}^{-1}\| \le C/\Lambda$, independently of the number of edges.

These algebraic formulas give an operator inverse because they respect the form domain. The interpolation energy is bounded by $C\Lambda\|a\|_{\mathfrak{h}}^2$ using its shifted form and (71). Conversely, averaging $u(x_n) - u(x)$ on I_n gives

$$\mu_n |u(x_n)|^2 \le 2\|u\|_{I_n}^2 + 2\mu_n R_n^I \int_{I_n} |u_s|^2.$$

Thus every form-domain function has square-summable node data and splits as $Ta + v$, where v vanishes at every node. The two parts are orthogonal for the shifted form, by the homogeneous edge equation. Consequently the exact inverse is

$$(A_{\mathsf{p}} - \Lambda)^{-1} = G_D + T\mathcal{D}^{-1}T^*, \qquad (T^* f)_n = \mu_n^{-1} \int \mathsf{p}\,(T\mathbf{e}_n) f \, dx. \tag{72}$$

Finite Dirichlet truncations have this same formula and converge strongly: the uniformly convergent diagonal resolvent series for $\mathcal{D}^{-1}$ converges on finite compressions, and the direct edge terms do as well. Their limit is the Friedrichs realization just identified by the form decomposition. This proves the inverse bound.

For the homogeneous line, substituting $a_n = (-1)^n y_n$ gives $b_n y_n = c_{n-1} y_{n-1} + c_n y_{n+1}$. Set $\epsilon_n = c_n^2/(b_n b_{n+1}) \le C/K^2$ and $t_n = b_{n+1} y_{n+1}/(c_n y_n)$. Backward elimination is

$$t_n = (1 - \epsilon_{n+1} t_{n+1})^{-1}.$$

For large K these maps preserve $[1, 2]$ and contract by at most $1/2$. Finite outer walls therefore give a unique bounded limiting sequence. Equation (70) yields (68), and (71) makes its interpolation square integrable. An independent node solution has the Wronskian form

$$z_n = y_n \left(C + \mathcal{W} \sum_{k=N}^{n-1} \frac{1}{c_k y_k y_{k+1}} \right), \qquad \mathcal{W} \neq 0.$$

Successive summands grow by at least cK^2, so eventually $|z_{n+1}/z_n| \asymp \Lambda \mu_n R_n$ and $\sqrt{\mu_{n+1}}|z_{n+1}| \ge cK\sqrt{\mu_n}|z_n|$. On the mass interval its neighbor term is at most $C\eta$ times its own-node term. Its continuous L^2 mass therefore diverges. This proves uniqueness of the square-integrable line and the limit-point property.

Finally, summing (64) in (68) gives, for $k \ge m$,

$$\left| \frac{a_k}{a_m} \right| \le \mathcal{E}_k^C \mathcal{E}_m^C e^{-(S_k - S_m)}, \qquad 2 \sum_{n=m}^{k-1} n^2 = \frac{2}{3}(k^3 - m^3) - (k^2 - m^2) + \frac{1}{3}(k - m). \tag{73}$$

The accumulated $O(n^{3/2} + n)$ error is absorbed by $q_k + q_m$. Every fixed radial derivative costs at most $e^{C_s n^2}$, by the profile equation and the interval lengths. The normalized gauge $(r/r_\partial)^{j^2}$ is bounded with bounded inverse in the deep range. Thus the constructed homogeneous solution and all its fixed radial derivatives are bounded by $\mathcal{E}_n^{C_s} e^{-S_n}$ after a fixed entrance normalization. They extend flatly at the endpoint. The same contraction also shows continuity of the endpoint line in the product topology of all the height ratios. □

The localized Green estimate. We will use a little more than the operator norm. Let B be the positive node matrix with diagonal b_n and neighboring entries $-c_n$, and let $W = \operatorname{diag}(\mu_n)$. Define $\Gamma_{nm} = (\mu_n \mu_m)^{-1/2} (W^{-1/2} B W^{-1/2})^{-1}_{nm}$. Its diagonal lies between b_m^{-1} and $C b_m^{-1}$ by the positive diagonal series. Uniqueness of the decaying recurrence gives

$$\Gamma_{nm} = \Gamma_{mm} \frac{y_n}{y_m} \le \frac{C}{\Lambda \mu_m} \prod_{k=m}^{n-1} \frac{1 + C(\eta + K^{-1})}{\Lambda \mu_{k+1} R_k} \quad (n \ge m). \tag{74}$$

Symmetry handles the reverse direction. The interpolation kernel in (72) has the additional sign $-(-1)^{n+m}$, which is irrelevant to absolute estimates. The continuous kernel adds the same-edge Dirichlet term and at most two interpolation functions at each endpoint. Hence its localized absolute bound is

$$C \mathcal{E}_n^C \mathcal{E}_m^C e^{-|S_n - S_m|}. \tag{75}$$

For the original source convention, put $g = (r/r_\partial)^{j^2}$ and $\mathsf{p}^\circ = g^2 w/r$. A direct calculation gives

$$L_j(gV) = -\frac{gw}{r^2}(A_{\mathsf{p}^\circ} - \Lambda)V, \qquad \mathsf{p}^\circ dx = \frac{g^2 w}{r^2}\, dr, \qquad G_{L_j}(r,r') = -g(r)g(r')G_A(x,x').$$

The bounded gauge factors preserve (75) in the dr pairing. Also, for $B_j = -\varpi^{-1}L_j$,

$$B_j = \frac{1}{12}(A_{\mathsf{p}_j} - j^2(j^2-1)), \qquad \|B_{j,D}^{-1}\| \le Cj^{-4} \tag{76}$$

on a deep tail. At its entrance the derivative matrix gives $\mathsf{p}u_x/(\Lambda\mu_n u) = 1 + O(\eta) + O(K^{-1})$. If an initial resistance edge comes first, the impedance is instead $-(1+O(\eta)+O(K^{-1}))/R_0$. In either case the endpoint Green pairing vanishes: at the nodes it is bounded by $C\Lambda\mu_n|a_n|^2 \to 0$, and finite mass and energy identify this with the endpoint limit.

5.5 One transition cell at every frequency

The useful decomposition depends on frequency, although the geometry does not. Put

$$\mathcal{H}_n = C_* q_n A_n^{-1/2}, \qquad \mathcal{I}_n = \{j : \mathcal{H}_{n-1} \le j < \mathcal{H}_n\}. \tag{77}$$

At the initial index N, include every remaining even frequency $2 \le j < \mathcal{H}_N$ in the first band. For $j \in \mathcal{I}_n$, all earlier cells have large positive-cap action $j\sqrt{A_m} \ge C_* q_m$. Cell n is the transition cell; the next one is already deep, since

$$j^2 A_{n+1}\sqrt{\log(A_{n+1}/B_{n+1})} \le C q_n^2 (A_{n+1}/A_n)(n+1) \longrightarrow 0. \tag{78}$$

Thus there is no unexamined frequency range between the local model inverse and the deep inverse. This one-transition-cell decomposition will be crucial when the inverse is assembled.

At each finite tail entrance attach an auxiliary positive cone $w = cr$. For the right tail it continues to $r = 0$, and for the left tail to $r = \infty$. A short positive interpolation with $rw'/w \ge 1$ is constructed by applying (58) to $\log(w/r)$. The cone merely specifies an incoming line. The final surface retains only a finite piece of it. In $y = \log r$,

$$B_j = \tfrac{1}{12}(-\partial_y^2 + j^2), \qquad \varpi\, dr = 12c\, dy;$$

the incoming lines are r^j and r^{-j} respectively, and the Dirichlet inverse has norm at most $12/j^2$.

5.6 Choosing the heights so that the lines do not meet

For $j \in \mathcal{I}_n$, select the positive join immediately before cell n in the outward order. Its inner height W_{in} does not depend on σ_n, while its outer height is $\sigma_n W^0_{\mathrm{out}}$. Write

$$\Xi_n = W^0_{\mathrm{out}}/W_{\mathrm{in}}, \qquad \Upsilon_n = \max(\Xi_n, \Xi_n^{-1}), \qquad \log\Upsilon_n = 4n^2 + O(n^{3/2} + n). \tag{79}$$

The radius recursion and all cut locations are independent of the height choices. At the first join choose the auxiliary cone amplitude independently of σ_N, with the same contrast as in (79). All later pieces are multiplied by the same σ_n, so their homogeneous equations, and hence their natural endpoint line, are unchanged. The incoming line is determined before this join and is also independent of σ_n. What changes is the flux scale by which the two lines are compared.

We first control the incoming slope. Set

$$\zeta_j = ru_r/j^2 - u. \tag{80}$$

The solution coming from a Dirichlet cut in the preceding positive cap can be normalized to $u = 1$ at the join, and then

$$a := \zeta_j/u \in [-3/2, -1/2]. \tag{81}$$

Indeed on that cap $u_{XX} = j^2|\rho|u$. Comparison of u_X/u with the constant-potential Riccati solution gives $|u_X/u| \le Cj\sqrt{b}\coth(cj\sqrt{A})$, where the cap scales are $(b, h, v) \asymp (A, 1, A)$ or (A^9, A^{-4}, A^5). In both cases $h\sqrt{b} \asymp \sqrt{A}$ and $v \asymp \sqrt{Ab}$, so $|ru_r/(j^2u)| \le C/(j\sqrt{A_{n-1}}) < 1/2$ after choosing C_* large. Reversing the left orientation changes only the sign. In the first band the cone gives $ru_r/(j^2u) = \pm 1/j$, proving the same assertion even at $j = 2$.

The derivative w'/w can be very large on the short join. It disappears from the transfer equation if we use the right state. The identity $(w\zeta_j)' = -(j^2-1)wu_r/j^2$ gives, for $Y = (u, w\zeta_j/W_{\mathrm{in}})$,

$$Y_r = \frac{1}{r}\begin{pmatrix} j^2 & j^2W_{\mathrm{in}}/w \\ -(j^2-1)w/W_{\mathrm{in}} & -(j^2-1)\end{pmatrix} Y. \tag{82}$$

The transfer therefore differs from the identity by at most $e^{\eta_{n,j}} - 1$, where

$$\eta_{n,j} = Cj^2(\ell_n/r_{\min})\Upsilon_n \le CC_*^2 q_n^2 e^{-7n^2+O(n^{3/2}+n)} = o(e^{-n^2}). \tag{83}$$

This follows directly from the Volterra series for the displayed first-order system. It uses only the size of w and w^{-1}, which is why the sharp interpolation causes no difficulty here.

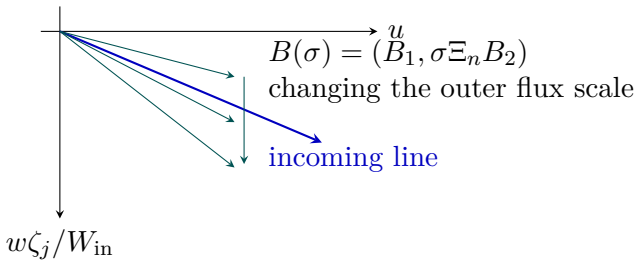


Figure 8: Height selection in the two-dimensional boundary-state plane. The incoming slope stays in a fixed negative interval. A small angle can therefore occur only for a short interval of the height parameter, even when the height contrast Ξ_n is very large or very small.

Normalize the natural endpoint line in its local (u, ζ_j) frame to (B_1, B_2). In the common frame it is $B(\sigma) = (B_1, \sigma\Xi_n B_2)$; this is the geometric content of Figure 8. If $B_1 = 0$, its angle with $(1, a)$ is bounded below. Otherwise write $b_\sigma = \sigma\Xi_n B_2/B_1$. The ideal angle sine is

$$\frac{|a - b_\sigma|}{\sqrt{1+a^2}\sqrt{1+b_\sigma^2}}.$$

A small value forces b_σ into a fixed negative compact interval, where $|b'_\sigma| = |b_\sigma|/\sigma$ is bounded below. Thus the set where this angle is less than $2\delta_n$, with $\delta_n = c_0 e^{-n^2}$, lies in an interval of length at most $C\delta_n$. The true incoming state is $(1, a) + O(\eta_{n,j})$; normalization is Lipschitz near these vectors. Equation (83) therefore gives the same excluded-interval bound for a true matching angle below δ_n. All constants are independent of the other height ratios.

Lemma 5.2 (Simultaneous height selection)**.** *At each endpoint one can choose all $\sigma_n \in [1, 2]$ so that for every even $j \in \mathcal{I}_n$ the incoming and natural endpoint lines have angle sine at least $\delta_n = c_0 e^{-n^2}$ in the common state of (82).*

Proof. The union of the excluded sets has measure less than one, because

$$\sum_{n\geq N} C\#\mathcal{I}_n\delta_n \leq CC_*c_0 \sum_{n\geq N} q_n e^{-n^2/2} < \tfrac{1}{2} \tag{84}$$

for sufficiently large N. More explicitly, for a finite set of bands fix all later ratios to one and use Lebesgue measure on the finite cube of remaining ratios. Fubini's theorem applies the uniform one-coordinate exclusion estimate to each condition, and the union bound gives a good choice. Extend those finite choices by ones and take a coordinatewise convergent subsequence. At any fixed frequency the finite-prefix transfer is continuous, and the deep continued fraction proved above contracts uniformly over all height choices. Hence the natural line is continuous in the product topology. Each good-angle condition is closed and survives the limit. The two endpoints can be chosen independently. □

We have now constructed smooth flat tails with a quantitative endpoint line at every frequency. We have not yet claimed that gluing them to the central oscillator preserves all the spectral gaps. That global question is addressed next. Its answer uses both parts of the present construction: the selected angle at the transition cell, and the strong attenuation through the preceding positive cap.

6 An inverse on one fixed annulus

Local spectral gaps do not automatically survive gluing. A homogeneous solution can leave one piece, return through its neighbor, and satisfy a new matching condition. We must therefore estimate the global matching problem, not just list the inverses of its pieces.

The frequency decomposition from the preceding section makes this possible. At a fixed frequency there are three regions: an inner collection of well-separated models, one transition cell at each end, and two deep tails. The selected height ratios control the matching angle at the transition cells. A positive interval just before each transition prevents its possibly large response from reaching the fixed interior. Figure 9 shows this division of labor. The position of the division changes with frequency; the surface itself will be chosen once.

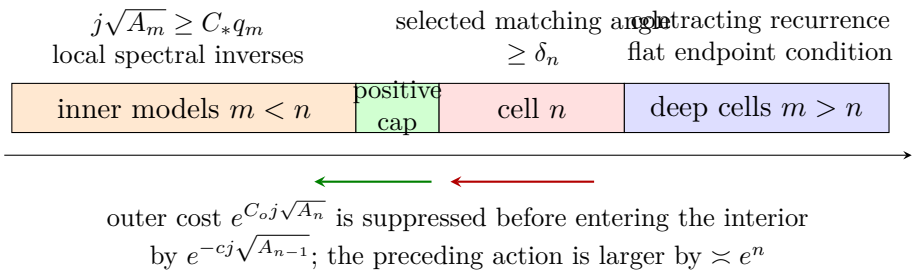


Figure 9: The frequency-dependent decomposition of one end. The other end is treated simultaneously. Costs whose output remains in cell n or farther out can be absorbed by the coefficient weight there; costs returning to the interior require the preceding positive cap.

6.1 Joining the distinguished solution first

We first finish the actual geometry, keeping its second mode exactly. On a positive interval put $y = \log r$, $w = re^f$, and prescribe the logarithmic slope $k = \phi_y/\phi$ of a nonvanishing

second-mode solution. The equation $L_2\phi = 0$ is equivalent to

$$f_y = \frac{k^2 + k_y - 4}{4 - k}, \qquad 1 + f_y = \frac{k^2 - k + k_y}{4 - k}. \tag{85}$$

Thus choosing $k < 4$ and $k^2 - k + k_y > 0$ constructs both the solution and a positive-curvature connector: integrate f_y and then $\phi_y = k\phi$. This is more reliable than interpolating w and hoping to retain its kernel.

The central slopes tend with all fixed derivatives to 2 on the left and -2 on the right, by (43). Cut sufficiently far out and interpolate k to the respective constant on a unit y interval. At either constant the second expression in (85) equals one, so C^1 closeness ensures $1 + f_y \geq 3/4$. Smooth cutoffs match all endpoint germs. The connector ends in an exact cone, where $k = \pm 2$.

On the auxiliary cone of a selected tail, the natural second-mode solution is $C_+ r^2 + C_- r^{-2}$. At the right tail the incoming line is r^2, so the selected nonzero angle implies $C_- \neq 0$. Far toward $r = 0$ the natural slope therefore tends to -2. At the left tail the incoming line is r^{-2}, and the slope tends to 2 toward $r = \infty$. In both cases the solution is nonzero sufficiently far along the cone. The same interpolation of k connects it to an exact cone while retaining $1 + f_y \geq 3/4$. Arbitrarily long exact cone segments may be placed between these short connectors.

A radius change

$$\widetilde{r} = Rr, \qquad \widetilde{w}(Rr) = Rw(r) \tag{86}$$

places each tail at the required radius and preserves w/r, B_j and $\varpi\, dr$. A subsequent positive height rescaling matches the additive constant in f without changing homogeneous solutions or travel angles. Scaling the tail solution matches its nonzero value. We obtain one smooth profile w_* and one globally defined, endpoint-flat second-mode solution ϕ_*. We write $w = w_*$ for the rest of this section. Only positive pieces have been added, so all negative travel angles remain those already computed. What remains to prove is that there is no additional kernel and that the inverse estimates are uniform.

6.2 What a positive buffer does to a model inverse

A useful local fact is that a distant boundary condition produces an exponentially small change at a cut inside a positive region. We need this fact for the operator and for its boundary derivatives, including the bordered central model.

Lemma 6.1 (Cutting inside a positive buffer)**.** *Let $H_0 = -\partial_X^2 + V$ have a self-adjoint full-model realization. Allow also a compact symmetric scalar border, supported away from all cuts. Suppose the resulting operator $\mathcal{H}$ is invertible with $\|\mathcal{H}^{-1}\| \leq B_0$. At each of at most two cuts i, suppose $V \geq m_i^2$ on the exterior and on an inward interval of length $(S_i + 6)/m_i$. Suppose $V \leq C m_i^2$ on the cut layer and the inner cutoff layer, each of width $4/m_i$; any finite exterior wall is at least $4/m_i$ away.*

Let α_i^{in} be the unit-boundary energy of the inward positive interval with an artificial Dirichlet wall, and α_i^{out} that of the exterior with its natural condition. Then $\alpha_i^{\mathrm{in}}, \alpha_i^{\mathrm{out}} \asymp m_i$. With $D = \operatorname{diag}(\alpha_i^{\mathrm{in}} + \alpha_i^{\mathrm{out}})$, the retained Dirichlet operator $\mathcal{H}_D$, its row-zero Poisson map T, and its outward derivative matrix Λ satisfy

$$\begin{aligned} \|\mathcal{H}_D^{-1}\| &\leq B_0 + C \sum_i m_i^{-2}, \qquad \|T D^{-1/2}\| \leq C\Big(\sum_i m_i^{-2}\Big)^{1/2}, \\ D^{-1/2}(\Lambda - \operatorname{diag} \alpha_i^{\mathrm{in}}) D^{-1/2} &= O(E), \qquad |E_{ik}| \leq C B_0 m_i m_k e^{-S_i - S_k}, \end{aligned} \tag{87}$$

provided $B_0 m_i^2 e^{-S_i}$ is sufficiently small at every cut. In particular Λ is positive.

Proof. The unit-boundary energy is at least m_i by expanding $\int |v' \pm m_i v|^2$. A test function supported within distance $2/m_i$ gives the matching upper bound. Join the inward and exterior unit solutions at the cut; after division by $\alpha_i^{\mathrm{in}} + \alpha_i^{\mathrm{out}}$ this is its local point-source Green function. Cut it off just before the inner artificial wall, obtaining K_i. Positive comparison with $-v'' + m_i^2 v = 0$ gives

$$\mathcal{H}(K_i, 0) = (\delta_{c_i} + f_i, 0), \qquad \|K_i\|_2 \le C m_i^{-3/2}, \qquad \|f_i\|_2 \le C m_i^{1/2} e^{-S_i}.$$

The scalar coordinate is omitted in the unbordered case; in the bordered case the cutoff support is disjoint from the border. The residual is an ordinary L^2 commutator source, not a point source. Therefore

$$\boldsymbol{g}_i = (K_i, 0) - \mathcal{H}^{-1}(f_i, 0)$$

is the exact Green column at the cut.

Let $\Gamma_{ik} = (g_k)_{\mathrm{field}}(c_i)$. The correction is homogeneous in the positive buffer at cut i and satisfies the natural exterior condition. Propagating it from the inner cutoff layer to the cut costs a further e^{-S_i}; rescaling a layer of width m_i^{-1} converts its L^2 bound to a point bound with factor $C m_i^{1/2}$. Consequently

$$D^{1/2} \Gamma D^{1/2} = I + E, \qquad |E_{ik}| \le C B_0 m_i m_k e^{-S_i - S_k}.$$

This also follows directly by applying Green's identity to the two commutator residuals. The stated smallness makes this finite matrix invertible.

Let ι extend the retained source by zero and $\mathcal{R}$ restrict the field, preserving the scalar in both cases. Subtracting the cut values gives the exact formulas

$$\mathcal{H}_D^{-1} = \mathcal{R}\mathcal{H}^{-1}\iota - \boldsymbol{g}_I \Gamma^{-1} \boldsymbol{g}_I^*, \qquad T = \boldsymbol{g}_I \Gamma^{-1}.$$

There is no hidden kernel: extending a homogeneous Dirichlet field by zero leaves only cut delta sources, and the invertible cut matrix forces their coefficients to vanish. Moreover

$$\|\boldsymbol{g}_i D_{ii}^{1/2}\| \le C(m_i^{-1} + B_0 m_i e^{-S_i}) \le C m_i^{-1},$$

which proves the first two bounds. Interior and exterior derivatives of a Green column add to its delta coefficient, so $\Lambda = \Gamma^{-1} - \operatorname{diag} \alpha_i^{\mathrm{out}}$. This gives the final estimate and positivity. The compact symmetric border cancels in the same Green identities, so the proof covers it without an additional boundary term. □

We now translate this estimate to the radial problem. The exact norm and flux conversions are

$$\|u\|_{L^2(\varpi dr)}^2 = 12\lambda \int \upsilon^2 |\psi|^2 \, dX, \qquad w u_r = \lambda \sqrt{\mathcal{P}} \left(\psi_X - \frac{\mathcal{P}_X}{2\mathcal{P}} \psi \right). \tag{88}$$

Using $\sqrt{\lambda}\psi$ removes the absolute profile amplitude. For a Morse cell the short and long positive cut regions have, respectively,

$$(m_{X,j}, \Delta X, \upsilon) \asymp (j\sqrt{A_m}, 1, A_m), \qquad (j A_m^{9/2}, A_m^{-4}, A_m^5).$$

Each contains action $c\tau_{m,j}$ with $\tau_{m,j} = j\sqrt{A_m}$, and $\upsilon / m_{X,j} \le C/j$. The full-model inverse costs a fixed power of $\mathcal{E}_m$. For the central oscillator, far cuts have $m_{X,j} \asymp j|X|$ and action cjX^2; its two-sided weighted inverse was proved in the preceding section.

The Liouville factor also changes the leading conormal impedance. At a central cut with outward sign ϵ_i, the exact radial boundary matrix is

$$\Lambda_r = \lambda \operatorname{diag}(\sqrt{\mathcal{P}_i})\Lambda_X \operatorname{diag}(\sqrt{\mathcal{P}_i}) + \lambda a_c \operatorname{diag}(\epsilon_i \mathcal{P}_i). \tag{89}$$

Thus its positive diagonal is $\kappa_i = \lambda \mathcal{P}_i(\alpha_i^{\mathrm{in}} + \epsilon_i a_c)$, comparable to $\lambda \mathcal{P}_i m_{X,j}$ at the chosen far cuts. For a Morse model it is simply $\lambda \alpha_i^{\mathrm{in}}$. Retaining this gauge term is necessary for exact flux matching.

6.3 The inner region: retain the full positive interface energy

Fix $j \in \mathcal{I}_n$. The finite inner domain consists of the central model, the earlier cells $m < n$, and their positive connectors. Its two outer boundaries are the entrances to the selected transition joins. Impose zero values there. At frequency two use the auxiliary compact symmetric border. In radial Hilbert notation,

$$d\mu = \varpi\, dr, \qquad B_j = -\varpi^{-1} L_j, \qquad F = -\varpi^{-1} f, \qquad \langle u, F\rangle_{L^2(d\mu)} = -\int \overline{u} f\, dr. \tag{90}$$

For a border column $k = -\varpi^{-1} g_0$, its Hilbert adjoint is $k^* u = -\ell_0 u$; changing the sign of the scalar source row therefore makes the border symmetric. This convention will be used in all intermediate energy identities.

Let D_h be the direct sum of the cut-model Dirichlet inverses and the exact positive-connector inverses. On every positive connector,

$$Q_j(u) = \int \left(w|u_r|^2 + j^2 \frac{w'}{r} |u|^2 \right) dr \geq c j^2 \|u\|^2_{L^2(d\mu)}. \tag{91}$$

Here $c = 1/6$ on the short joins where $rw'/w \geq 2$, and a smaller fixed positive c covers the cone heads and central connectors. Its Poisson map P_C and full conormal matrix Λ_C consequently satisfy $\|P_C a\| \leq (C/j)(a^* \Lambda_C a)^{1/2}$.

A short connector can have a large mixed entry in Λ_C. We keep that entry instead of treating it as an error. At all internal interfaces set

$$S_0 = \operatorname{diag}(\kappa_i) + \bigoplus_C \Lambda_C, \qquad S = S_0 + E_h. \tag{92}$$

The positive matrix S_0 has blocks of size at most two and dominates $D_\kappa = \operatorname{diag}(\kappa_i)$. All errors E_h now come from crossing positive buffers inside the local models.

To see this uniformly in the number of cells, use a source weight W which is one on the finite middle and has cell size

$$W_m = (\mathcal{E}_m/\mathcal{E}_N)^{M_0} (H_m/H_N)^{-\gamma}. \tag{93}$$

Make it constant across each whole positive gap, with the geometric mean of the neighboring weights, and vary it inside the model cells. For $|\gamma| \leq 1$,

$$|\log(W_{m+1}/W_m)| \leq C m^2 + C M_0 m^{3/2} \log(e+m). \tag{94}$$

Thus this interpolation changes cell norms by fixed coefficient powers. Let $\Omega = \operatorname{diag}(W(c_i))$. It commutes with S_0, because the two ends of every positive connector have the same weight.

The buffer lemma and (88) give local normalized errors

$$\mathcal{E}_m^{C(M_0)} \langle j\rangle^C e^{-c\tau_{m,j}}, \qquad C\langle j, X_0\rangle^C e^{-cjX_0^2}, \qquad C_N \langle j\rangle^C e^{-cjL} \tag{95}$$

for earlier cells, central cuts and initial cone heads. The leading Poisson layer at a model cut has Hilbert norm at most $(C/j)\sqrt{\kappa_i}|a_i|$, because $\upsilon/m_{X,j} \le C/j$. Its remaining part has the first type of error in (95). The assertion of uniform smallness includes frequencies arbitrarily far above a cell's threshold. To see this, put $t = j\sqrt{A_m} \ge C_* q_m$. Since $\log\langle j\rangle \le \log(1+t) + Cm^2$, the logarithm of that local error is bounded, with fixed $a, b, c > 0$, by

$$aq_m + b\log(1+t) - ct \le aq_m - \tfrac{3}{4}ct + C_b \le -\tfrac{1}{2}ct + C_b \quad \text{if } C_* \ge 4a/c. \tag{96}$$

Here $m^2 = o(q_m)$ was included in a, and the middle inequality uses $b\log(1+t) \le ct/4 + C_b$. Increasing N now makes the original error smaller than any prescribed positive tolerance, simultaneously for all $m < n$ and all such j. Thus we choose M_0 for source conversion, then C_*, then N, and finally the central cuts and the fixed head lengths. Their errors in (95) have the same polynomial-versus-exponential comparison, on the finitely many central pieces.

Let T_h be the homogeneous interpolation of common interface values, with zero scalar row. The resulting two estimates are

$$\begin{aligned} \|S_0^{-1/2}\Omega^{\epsilon} E_h \Omega^{-\epsilon} S_0^{-1/2}\| &< 1/2, \\ \|W^{\epsilon} T_h \Omega^{-\epsilon} S_0^{-1/2}\| &\le C/j, \qquad \epsilon = \pm 1. \end{aligned} \tag{97}$$

For the first, normalize each model error by its two adjacent $\kappa_i^{1/2}$. Each interface belongs to at most two such blocks, so their sum has operator norm at most twice the largest local norm. Then use $\|S_0^{-1/2} D_\kappa^{1/2}\| \le 1$ on both sides. For the second, sum the squared local Poisson norms over disjoint model interiors and over the positive connectors, whose overlap is bounded. This proves both estimates independently of the number of cells.

The outgoing interface flux of a zero-boundary source solution is $-T_h^* F$. Matching values and fluxes therefore gives the exact inverse

$$G_h = D_h + T_h S^{-1} T_h^*. \tag{98}$$

The two signs of the weight in (97) control the second term by duality. Local Dirichlet costs are coefficient powers, so, for a sufficiently large fixed M_1,

$$\|W_1 G_h W^{-1}\| \le C, \qquad \|W^{\pm 1} G_h W^{\mp 1}\| \le \mathcal{E}_n^{C}, \qquad W_1 = W(\mathcal{E}/\mathcal{E}_N)^{-M_1}. \tag{99}$$

The first estimate is localized and will handle ordinary inner sources. Its net output coefficient exponent is $M_0 - M_1$; at the end we rename $M_1 - M_0 > 0$ as the output exponent in Proposition 4.2. The coarser second estimate is used only on boundary residuals which already contain an exponentially small factor. Invertibility of the Dirichlet blocks and of S proves both inverse identities in (98); its symmetry is the corresponding Green identity.

6.4 The transition region and an outer inverse without division by height

The outer core begins at a selected join, contains cell n, and ends where the deep tail starts. The preceding large positive cap belongs to the inner domain. On the right the deep entrance is a mass entrance; on the left it is the initial resistance entrance described earlier. Let r_o increase outward, use positive measure dr_o, and choose $\lambda > 0$ with $e^{-Cn^2} \le \lambda/\lambda_n \le e^{Cn^2}$. In the state (u, z), $z = wu_{r_o}/\lambda$, the core has the bounds

$$\begin{aligned} \|T\| + \|T^{-1}\| \le \mathcal{K}_n := e^{Cn^2 + C_o \tau_{n,j}}, \qquad & \int_{\text{core}} \varpi \, dr_o \le \lambda e^{Cn^2}, \\ \|P_d\|_{L^2(d\mu)} \le \sqrt{\lambda} e^{Cn^2}, \qquad & \|B_{d,D}^{-1}\| \le C j^{-4}. \end{aligned} \tag{100}$$

Here P_d is the deep natural solution with entrance value one.

For clarity, these four estimates follow from different parts of the construction. The last is (76). The core mass is $12\lambda_n \int (v/r)^2 dX$, plus the negligible joining masses, so it has the stated size by (55). The weighted node contraction (68) bounds the squared norm of P_d by the first deep mass. If an initial resistance edge comes first, its next mass height is at most $C/(\Lambda\mu R_0)$ and the edge mass is small; the same bound follows. For the transfer estimate, on a positive piece the off-diagonal coefficients in (u, z) are λ/w and $j^2 w_r/(\lambda r)$. Balancing their integrals bounds the transfer in either direction by

$$e^{Cn^2} \exp\left(2j\sqrt{\left(\int dr_o/w\right)\left(\int w_r\, dr_o/r\right)}\right).$$

The product inside the square root is $O(A_n)$ on either positive cap and smaller on the short join. On the negative component use $u_{XX} + j^2\rho u = 0$ and the energy $E = |u_X|^2 + (1+j^2\rho)|u|^2$. It satisfies $|E_X| \le [1 + j^2|\rho_X|/(1+j^2\rho)]E$. The interval has length $O(n^2)$ and ρ has one peak, so the integrated second term is at most $2\log(1 + j^2 \max \rho)$. Converting the flux back to (u, z) costs only e^{Cn^2}. Together these estimates prove (100).

The natural outer solution may vanish at the entrance. A logarithmic slope there is therefore the wrong variable. Normalize its full state:

$$v = (V(r_{o,-}), f_V(r_{o,-})/\lambda), \qquad |v| = 1, \qquad f_u = w u_{r_o}, \qquad v^\perp = (-v_2, v_1). \tag{101}$$

Let U have entrance state $v^\perp$. For $\mathscr{H}_j = -L_j$, the kernel

$$K_o(r_o, t_o) = \lambda^{-1} U(\min(r_o, t_o)) V(\max(r_o, t_o))$$

has the natural endpoint condition and the complementary Robin condition $v \cdot (u, f_u/\lambda) = 0$ at the entrance. Its normalization is exact: $w(UV_{r_o} - U_{r_o}V) = -\lambda$. Thus

$$R_o F = \int K_o(r_o, t_o) F(t_o)\, d\mu(t_o), \qquad G_o f = -\int K_o(r_o, t_o) f(t_o)\, dt_o \tag{102}$$

are respectively the Hilbert inverse of B_j and the original-source inverse of L_j. No entrance height has been divided out. Green's identity gives the remaining boundary equation

$$v^\perp \cdot (u, f_u/\lambda) = \lambda^{-1} \int V \varpi F\, dr_o, \qquad \|V\|_{L^2(d\mu)} \le \sqrt{\lambda}\, \mathcal{E}_n^C \mathcal{K}_n. \tag{103}$$

The natural endpoint term is zero by the deep-tail construction.

The factor λ^{-1} in the kernel causes no amplitude loss. On the core it is canceled by the mass bound in (100). On the deep region exact kernel restriction gives

$$(R_o)_{d,d} = B_{d,D}^{-1} + P_d K_o(r_{o,+}, r_{o,+}) P_d^*, \qquad |K_o(r_{o,+}, r_{o,+})| \le \lambda^{-1} \mathcal{E}_n^C \mathcal{K}_n^2.$$

The two Poisson norms cancel it there, and the cross blocks have one Poisson factor each. Consequently

$$\|R_o\| \le \mathcal{E}_n^C \mathcal{K}_n^2. \tag{104}$$

The localized deep kernel (75) and the outward homogeneous quotient (73) give more precisely

$$\|\vartheta_k R_o \vartheta_m\| \le \mathcal{E}_k^C \mathcal{E}_m^C \mathcal{K}_n^2 e^{-|S_k - S_m|}, \qquad k, m \ge n - O(1). \tag{105}$$

A product of two outward quotient factors is bounded by the same exponential, since both indices lie beyond the entrance. Converting the source and target to $L^2(dr_o)$ adds $\varpi^{-1/2}$ on each of their cells, bounded by coefficient powers. Hence (105) also holds for G_o in ordinary source units. These observations justify both inverse identities and reciprocity directly from the kernel.

6.5 The two boundary values which remain to be matched

At each truncation boundary use the scale $\lambda_i = j^2 W_{\mathrm{in},i}/r_{\mathrm{in},i}$. Its ratio to the transition-cell height is between e^{-Cn^2} and e^{Cn^2}, so the preceding outer estimates apply. Let κ_i be the impedance of the artificial incoming positive block used for height selection and put $d_i = \kappa_i/\lambda_i$, $D_0 = \mathrm{diag}(d_i)$. The incoming estimate gives $0 < d_i \le 1/2$; in the first band $d_i = 1/j$. Use normalized boundary coordinates $x_i = \sqrt{\lambda_i}u_i$, $z_i = f_{u,i}/\sqrt{\lambda_i}$.

Let $\mathcal{B}_h$ denote the inner Hilbert operator, including its symmetric scalar border, and let $\mathcal{Q}_h$ be its quadratic form. We need to know how much the true inner derivative matrix differs from this artificial incoming one. Take its positive-block Poisson function of boundary height $\lambda_i^{-1/2}$, cut it off near the inner end of that block, and extend by zero. Call the resulting field $\mathcal{L}_i$, and its ordinary Hilbert residual $\mathcal{F}_i = \mathcal{B}_h\mathcal{L}_i$, including zero scalar row. Allocate separate fixed fractions of the preceding cap to this cutoff and to the earlier model's buffer. Positive comparison and the commutator

$$\mathscr{H}_j(\chi u) - \chi\mathscr{H}_j u = -2w\chi' u' - (w\chi')' u$$

then give

$$\begin{gathered}\|W^{\pm1}\mathcal{L}_i\omega_i^{\mp1}\| \le C\sqrt{d_i}/j, \qquad \|W^{\pm1}\mathcal{F}_i\omega_i^{\mp1}\| \le \sqrt{d_i}\mathcal{E}_n^C e^{-S_{b,i}},\\ S_{b,i} = c\tau_{n-1,j} \quad \text{or, in the first band, } cjL.\end{gathered} \tag{106}$$

Here ω_i is the constant weight on this positive gap. The first bound uses the energy $\kappa_i/\lambda_i = d_i$ and (91); the second uses the extra attenuation on the cutoff layer. The two lifts have disjoint supports, and their energy matrix is diagonal with entries $d_i(1 + O(\mathcal{E}_n^C e^{-2S_{b,i}}))$.

The exact homogeneous Poisson map is $P_h = \mathcal{L} - G_h\mathcal{F}$. Completing the quadratic form gives the true normalized derivative matrix

$$D_b = \mathcal{Q}_h(\mathcal{L},\mathcal{L}) - \mathcal{F}^* G_h \mathcal{F} = D_0 + E_b, \qquad |(E_b)_{ik}| \le \mathcal{E}_n^C\sqrt{d_i d_k}e^{-S_{b,i}-S_{b,k}}. \tag{107}$$

The two cross terms cancel one correction term because $\mathcal{B}_h G_h\mathcal{F} = \mathcal{F}$ and the correction has zero boundary values. This identity is also why the error has *two* attenuation factors. The scalar border is symmetric and every lift has row zero, so there is no missing scalar term.

Use the outer normalized states (101) with these λ_i. Let $A = \mathrm{diag}(v_{1,i})$, $B = \mathrm{diag}(v_{2,i})$, and let $\mathcal{V}$ consist of the two outer columns $V_i/\sqrt{\lambda_i}$. The inner source flux is $-P_h^* F_h$, while (103) supplies the outer equation. Thus the complete matching problem is just

$$z = D_b x - P_h^* F_h, \qquad -Bx + Az = \mathcal{V}^* F_o, \qquad Kx = \mathcal{V}^* F_o + AP_h^* F_h, \quad K = A(D_0 + E_b) - B. \tag{108}$$

The angle selected in the preceding section is exactly what is needed to invert the diagonal part of K. Its state is $(u, -u \pm f_u/\lambda_i)$; the change to $(u, f_u/\lambda_i)$ has a fixed condition number. Transport across the nearly identity join preserves the angle lower bound up to a fixed factor. Hence

$$|v_{1,i}d_i - v_{2,i}| = \sqrt{1+d_i^2}\,|\sin\angle(v_i,(1,d_i))| \ge c\delta_n. \tag{109}$$

This argument remains valid when $v_{1,i} = 0$. Any later dependence of the central connectors on the selected tails is confined to E_b; the artificial incoming direction itself was fixed before the height parameter was varied.

Two estimates now decide the choice of scales:

$$\delta_n^{-1}\|E_b\| < c/4, \qquad \mathcal{E}_n^C\delta_n^{-1}e^{C_o\tau_{n,j} - c\tau_{n-1,j}} < 1/4. \tag{110}$$

The first makes K invertible. The second controls responses which return to the fixed interior. They are compatible because

$$\tau_{n-1,j}/\tau_{n,j} \asymp e^n, \qquad \tau_{n-1,j} \geq C_* q_{n-1}, \qquad q_n/q_{n-1} \to 1, \qquad n^2/q_{n-1} \to 0.$$

Choose C_* after the fixed coefficient exponent, then N large. The first relation absorbs $C_o\tau_{n,j}$ into a quarter of the preceding action, and the other relations absorb $Cq_n + n^2 + |\log c_0|$ into another quarter. The first estimate has two such attenuations by (107). At the finite initial band choose the fixed cone-head length L large enough instead. Therefore $\|K^{-1}\| \leq C\delta_n^{-1}$ uniformly in the band and frequency.

6.6 The global estimate and the order of choices

Set $K_1 = A + BD_b$. Solving (108) and reconstructing the outer homogeneous coefficient gives the exact matched inverse

$$G = \begin{pmatrix} G_h & 0 \\ 0 & R_o \end{pmatrix} + \begin{pmatrix} P_h & 0 \\ 0 & \mathcal{V} \end{pmatrix} \begin{pmatrix} K^{-1}A & K^{-1} \\ K^{-T} & K_1K^{-1} \end{pmatrix} \begin{pmatrix} P_h^* & 0 \\ 0 & \mathcal{V}^* \end{pmatrix}. \tag{111}$$

For example, the outer coefficient is $K_1x - BP_h^*F_h$. Since $D_b = D_b^T$, A, B commute and $A^2 + B^2 = I$, direct multiplication gives

$$KA = AK^T, \qquad K^TK_1 = K_1^TK, \qquad K_1K^{-1}A - B = K^{-T}.$$

These identities prove both the formula and the symmetry of its middle matrix, whose norm is $O(\delta_n^{-1})$. We choose equal boundary weights at the two ends by using their common band index. A bounded index shift changes their ratio only by a coefficient power and has the same estimates.

There are two ways to pay for a boundary correction. We give the weighted estimate explicitly for each, because this is where local inverse bounds become an estimate on the whole annulus.

An outer block and its coefficient weights. First consider terms involving only R_o, $\mathcal{V}$ and the leading lifts $\mathcal{L}$. Their source and output indices are at least $n - O(1)$. Combining (105) with $\|K^{-1}\| \leq C\delta_n^{-1}$ gives, after increasing fixed constants,

$$\|\vartheta_k G_{\text{out}} \vartheta_m\| \leq C \exp\{a(q_k + q_m) + dq_n - |S_k - S_m|\}.$$

Here d includes the finite-core cost: indeed $\delta_n^{-1}e^{C_o\tau_{n,j}} \leq c_0^{-1}\exp(n^2 + C_oC_*q_n)$. Since the output is outer, $q_n \leq c_1q_k + C_1$; write $d_1 = dc_1$. For $\Delta S = S_k - S_m$, conjugation by the source and output weights therefore gives the following exact exponent bound:

$$\begin{aligned} \log\|\vartheta_k W_1 G_{\text{out}} W^{-1}\vartheta_m\| \leq C &- (M_1 - 2a - d_1)q_k \\ &+ (a - M_0)(q_m - q_k) - |\Delta S| + \gamma\Delta S. \end{aligned} \tag{112}$$

Fixed neighboring-index and source-measure changes have been included in a, d_1, C. Choose $M_1 > \max(M_0, 2a + d_1)$, after C_*. For any $0 < \gamma \leq \gamma_+ < 1$, use

$$|q_m - q_k| \leq \varepsilon|S_m - S_k| + C_\varepsilon, \qquad \varepsilon = \frac{1 - \gamma_+}{2(1 + |a - M_0|)}.$$

Then (112) is at most $C_\gamma - \frac{1}{2}(1 - \gamma_+)|S_k - S_m|$. This is a summable block kernel in both indices. On either tail, with $s_0 = \inf_{m \geq N}(S_{m+1} - S_m) > 0$ and $\eta = \frac{1}{2}(1 - \gamma_+)$, we have

$$\sup_k \sum_m e^{-\eta|S_k - S_m|} \leq 1 + \frac{2}{e^{\eta s_0} - 1}.$$

The same estimate holds with k, m reversed. The two endpoint copies and the bounded overlap of the localizers change it only by a fixed factor. The block Schur estimate proves the outer contribution to the desired operator bound. Notice that increasing the output exponent works precisely because k is near or beyond the transition index; it gives no such estimate at a fixed interior output.

A boundary block returning to the interior. The upper-right block of (111) is $P_h K^{-1} \mathcal{V}^*$. Substituting $P_h = \mathcal{L} - G_h \mathcal{F}$, its part which can reach the fixed interior is

$$C_{ho} = -(G_h \mathcal{F}) K^{-1} \mathcal{V}^*.$$

Let Ω be the common boundary weight, and use W_h, W_o for the restrictions of W. The preceding estimates supply

$$\begin{aligned} \|W_h G_h \mathcal{F} \Omega^{-1}\| &\le \mathcal{E}_n^{C_1} e^{-c\tau_{n-1,j}}, \\ \|\Omega K^{-1} \Omega^{-1}\| &\le C\delta_n^{-1}, \\ \|\Omega \mathcal{V}^* W_o^{-1}\| = \|W_o^{-1} \mathcal{V} \Omega\| &\le C_\gamma \mathcal{E}_n^{C_2} e^{C_o \tau_{n,j}}. \end{aligned} \tag{113}$$

The first is (99) applied to the residual bound (106); $d_i \le 1/2$ absorbs its two boundary factors. The middle bound uses equal weights at the two ends. For the last bound, the localized column has decay $\mathcal{E}_n^C \mathcal{E}_k^C e^{C_o \tau_{n,j}} e^{-(S_k - S_n)}$. After either sign of the weight, its extra exponent is $(C \pm M_0)(q_k - q_n) - (1 \mp \gamma)(S_k - S_n)$. The same slow-variation inequality used above makes its squared sum finite, leaving a fixed coefficient power at n and a constant C_γ. This proves the displayed adjoint estimate as well as its reciprocal-weight version.

Multiplying the three factors in (113) now exhibits the two competing costs in one operator estimate:

$$\begin{aligned} \|W_h C_{ho} W_o^{-1}\| &\le C_\gamma \mathcal{E}_n^C \delta_n^{-1} e^{C_o \tau_{n,j} - c\tau_{n-1,j}} \\ &\le C_\gamma e^{-c\tau_{n-1,j}/2} \le C_\gamma. \end{aligned} \tag{114}$$

The second line is the explicit exponent comparison proving (110); enlarge the constants there before choosing C_*, N. For the initial finite band replace the preceding action by cjL. On the tails $\mathcal{E}/\mathcal{E}_N \ge 1$, while its smooth extension has a positive minimum on the fixed middle. Thus $W_1 \le C_{\mathrm{mid}} W$ with a fixed constant, which gives the required output-weight bound. The transpose block uses the opposite sign of the weight in (113). Terms with two residual columns have two preceding attenuations; terms with one residual and one leading lift have the same estimate. Thus every correction in (111) is covered, while the direct inner term is already bounded by (99). We conclude

$$\|W_1 G W^{-1}\| \le C \tag{115}$$

at angular order zero, with the original source-density conversions included by the coefficient powers already used above.

Lemma 6.2 (One geometry for all flat exponents)**.** *The profile, height choices, central cuts and cone-head lengths can be chosen once so that* (115) *holds for every compact range* $[\gamma_-, \gamma_+] \Subset (0, 1)$*. Only the norm constants and coefficient exponents may depend on that range.*

Proof. Every finite matching choice uses only adjacent weight ratios. Estimate (94) is uniform for $|\gamma| \le 1$. Fix a source exponent M_0 which absorbs the original-source conversion, then choose C_*, N, the central cuts and head lengths using that uniform bound. In particular, the invertibility of S and K is independent of a compact range of flat exponents.

After this geometry is fixed, the only new issue is the infinite summation of the deep kernel. Conjugation by $H^{-\gamma}$ changes its exponential factor to

$$e^{-|S_k-S_m|+\gamma(S_k-S_m)} \le e^{-(1-|\gamma|)|S_k-S_m|}.$$

Its row and column sums are finite when $|\gamma| \le \gamma_+ < 1$. Coefficient factors are absorbed using

$$|q_k - q_m| \le \varepsilon |S_k - S_m| + C_\varepsilon,$$

which follows from $q'(x)/S'(x) \to 0$. Choose ε after γ_+; this changes the summation constant and output exponent, not the geometry. The boundary terms which enter the interior have already acquired the attenuation in (110). Compact-range summation constants multiply an existing bound and do not enter the finite matrix inversions. Thus no later choice of endpoint exponent changes the profile. □

The same estimates with one radial derivative use the local equation rather than differentiating a spectral decomposition. On a slightly enlarged cell,

$$\begin{aligned} \|u_r\|_2 &\le \mathcal{E}_m^C(\|f\|_2 + (1+j)\|u\|_2), \\ \|u_{rr}\|_2 &\le \mathcal{E}_m^C(\|f\|_2 + (1+j^2)\|u\|_2). \end{aligned} \tag{116}$$

To derive the first, use interval interpolation $\|u_r\|_2^2 \le C\|u\|_2(\|u_{rr}\|_2 + \|u\|_2)$ and substitute $wu_{rr} = f - w'u_r + j^2 w'u/r$; absorb the resulting $C\|u\|_2\|u_r\|_2$ by Young's inequality. Substitution back gives the second. Rescaling and the reciprocal profile cost coefficient powers. A cutoff on the enlarged interval supplies the displayed local version without imposing a boundary condition. At mode two replace f by $f - \nu g_0$, a compact contribution already controlled by its scalar norm. Thus one normal derivative uses at most one extra angular derivative. On an outer transition region any fixed j power is itself bounded by a coefficient power, since $j < \mathcal{H}_n$.

6.7 The realization, the zero mode and the final border

We now identify the common domain and complete the statement of Proposition 4.2.

Definition 6.3 (The weighted graph domain)**.** Let $\mathcal{Y}_\gamma^{\mathrm{rad}}$ be the source space (33) at order zero with $\int \langle f \rangle \, dr = 0$, including its scalar row. Let $\mathcal{X}_\gamma^{\mathrm{rad}}$ consist of (u, ν) with $u = a_L \chi_L + v$ and squared norm

$$|a_L|^2 + |\nu|^2 + \sum_m \mathcal{E}_m^{-2M_1} H_m^{-2\gamma} \|\vartheta_m v\|^2_{L^2(dr\, d\theta)}.$$

The angular measure is normalized. Define

$$\begin{aligned} \mathcal{D}_\gamma^{\mathrm{rad}} &= \{(u,\nu) \in \mathcal{X}_\gamma^{\mathrm{rad}} : \mathbb{L}_w(u,\nu) \in \mathcal{Y}_\gamma^{\mathrm{rad}}\}, \\ \|U\|_{\mathcal{D}_\gamma^{\mathrm{rad}}} &= \|U\|_{\mathcal{X}_\gamma^{\mathrm{rad}}} + \|\mathbb{L}_w U\|_{\mathcal{Y}_\gamma^{\mathrm{rad}}}. \end{aligned}$$

The differential expression is interpreted distributionally. This definition applies first to the auxiliary border and then to the final one below.

The zero mode has not entered the spectral construction. It is solved exactly by two primitives. For $\int f_0 \, dr = 0$, set

$$p_0(r) = \int_{r_-}^{r} f_0(t)\, dt = -\int_{r}^{r_+} f_0(t)\, dt, \qquad u_0(r) = -\int_r^{r_+} \frac{p_0(t)}{w(t)}\, dt. \tag{117}$$

Then $(wu_0')' = f_0$, the right height is zero, and the left height is an allowed constant. Summing the cells beyond n gives $|p_0| \le \mathcal{E}_n^C H_n^\gamma \|f_0\|_{Y_{\gamma,0}}$. Division by w and one integration cost only coefficient powers, so the primitive converges and has the required endpoint estimate after subtracting $a_L \chi_L$. The coefficient a_L is a bounded source functional. A nonzero homogeneous flux would produce $\int dr/w$, excluded by its growth; a constant is excluded by the right normalization. The zero channel is therefore unique.

Lemma 6.4 (The two inverse identities and the graph closure)**.** *The matched nonzero-mode inverse* (111) *and the zero-mode primitive* (117) *define* $\mathbb{G}_w : \mathcal{Y}_\gamma^{\mathrm{rad}} \to \mathcal{D}_\gamma^{\mathrm{rad}}$*. They satisfy*

$$\mathbb{L}_w \mathbb{G}_w = I, \qquad \mathbb{G}_w \mathbb{L}_w = I. \tag{118}$$

The graph domain is the graph closure of smooth admissible fields. The Green identities have the ordinary dr source pairing, including the chosen scalar border.

Proof. On compact smooth finite-frequency sources, all blocks were constructed as genuine inverses, and their values and conormals match exactly. Their bounded estimates therefore give the first identity by density.

For uniqueness on the maximal graph domain, subtract this constructed solution from any other one. At each fixed nonzero frequency the homogeneous difference is locally H^2, because $w > 0$ in the interior. Its value and conormal agree at every finite interface. The endpoint weights put it in the natural Hilbert space, since $\gamma S_m - C q_m \to +\infty$ for every fixed C. The deep lemma therefore puts it on the unique natural line at each end. Its two remaining boundary values vanish by the invertible matrix K; the inner and outer inverses then make the whole difference zero. The zero mode is unique by the primitive calculation. This proves the second identity without imposing additional traces at the flat endpoints.

Distributional convergence of a graph-convergent sequence identifies its limiting image, so the maximal graph domain is closed. Smooth compact finite-frequency sources are dense in the compatible source space: truncate and mollify, then restore the integral using a fixed compact bump of integral one. Their images are smooth admissible fields, flat in the nonzero modes outside the support of the source. The output bound and the first identity imply convergence in graph norm; the second identity shows that every graph-domain field is reached this way. Finally, the local Green identities and the symmetric middle matrix in (111) give the stated reciprocity, with the source conversion in (90). □

The explicitly joined second-mode solution spans the unbordered kernel: otherwise its bordered inverse could not be injective. Its auxiliary row is nonzero, and we normalize that row to one. Choose a nonzero $\chi \ge 0$ supported on a right positive connector where $\phi_*'/\phi_* < 0$. Integration by parts gives

$$a_* := \int \phi_* L_{2,w}'[w\chi]\phi_*\, dr = \int w\chi \left(-\phi_*'^2 + \frac{8\phi_*\phi_*'}{r} - \frac{4\phi_*^2}{r^2} \right) dr < 0. \tag{119}$$

Put

$$\chi_* = \chi/a_*, \qquad g_* = L_{2,w}'[w\chi_*]\phi_*, \qquad \ell_* u = \int g_* u_2\, dr. \tag{120}$$

Then $\int g_*\phi_*\, dr = 1$, or equivalently $\ell_*(\phi_* e_2) = 1$. For a source f_2, its scalar obstruction is $\nu = \int \phi_* f_2\, dr$. Solve the compatible source $f_2 - \nu g_*$ with the auxiliary border, and add a multiple of ϕ_* to impose the new row. This finite-rank change is bounded in all the established endpoint spaces and preserves both inverse identities and the graph closure. From now on $\mathbb{L}_{w_*}, \mathbb{G}_{w_*}$ refer to this final border. All claims in Proposition 4.2 now follow.

The interface with the nonlinear argument. The result just proved is an angular-order-zero inverse with no angular loss, together with its first-normal estimate and its maximal graph domain. These are the inputs to the subsequent two-weight argument. For higher normal derivatives, the equation recursively replaces two normal derivatives by at most two angular derivatives and lower normal orders. At each fixed total order s, the *complete* coefficient product in that recursion is bounded by $C_s\mathcal{E}_m^{C_s}$. The enlarged discrete weight

$$M_m = \exp\{1 + m^{5/2}\log^2(e+m)\}$$

satisfies $\mathcal{E}_m^{C_s} \le C_s M_m$ for every fixed s, and

$$|\log M_k - \log M_m| \le \varepsilon|S_k - S_m| + C_\varepsilon.$$

These follow respectively from $\log(e+m) \to \infty$ and from the ratio of the derivatives of $\log M$ and S tending to zero. Thus complete coefficient products cost one extra M weight, independent of output order. Combining the base estimates at two neighboring exponents γ and the normal equation will give the fixed total-order loss. This final upgrade uses the one geometry already fixed here; it makes no new spectral or endpoint choice.

7 A geometry that stays fixed during iteration

The reference inverse is useful only if a nearby equation can be compared with it. Two features would normally move: the circles where the equation changes type, and the angular positions at which characteristics arrive. Section 3 fixes the first by a correction proportional to the curvature residual. We shall fix the second in the same way. The resulting coordinates will put all the operators on one set of cylinders with one set of transmission conditions.

This construction involves infinitely many components. Before moving coordinates, we explain the quantitative room available on them. The point is that the deformation dies out much faster than the reference slope. This room pays for every fixed number of coordinate derivatives.

7.1 The three scales and the function spaces

Let d_∂ be distance to the nearest endpoint. With $d_0 > 0$ fixed, put, sufficiently near that endpoint,

$$x_\partial = \sqrt{\log(d_0/d_\partial)}, \quad H = e^{-2x_\partial^3/3}, \quad E_0 = e^{1+x_\partial^{5/2}\log(e+x_\partial)}, \quad M = e^{1+x_\partial^{5/2}\log^2(e+x_\partial)}. \tag{121}$$

Use smooth positive extensions through the middle. The endpoint construction gives $x_\partial = n + O(1)$ on the nth component. Thus H agrees with H_n, and E_0 with $\mathcal{E}_n$, up to powers of $\mathcal{E}_n$. The slope itself has the slower decay $w_* = \exp(-n^{5/2} + O(n^2))$.

There are two useful inequalities, for every fixed A and $\epsilon > 0$:

$$E_0^A \le C_A M, \qquad H^\epsilon M^A \le C_{A,\epsilon}. \tag{122}$$

They follow by comparing logarithms. The additional logarithm in M is deliberate: however many fixed coefficients appear after differentiation, their *whole product* costs one M order. That order does not grow with the number of derivatives. The cubic exponent in H still dominates every fixed power of M.

Choose

$$\ell = \frac{d_\partial}{1 + x_\partial}, \qquad d\varrho = \frac{dr}{\ell}, \qquad \mathsf{D} = \partial_\varrho = \ell\partial_r$$

near the ends, with a positive extension in between. Then ϱ covers $\mathbb{R}$ and

$$|\mathsf{D}x_\partial| = \frac{1}{2x_\partial(1+x_\partial)}, \qquad |\mathsf{D}\log H| = \frac{x_\partial}{1+x_\partial}.$$

All positive D derivatives of $\log H, \log M, \log \ell$ are bounded. Ordinary radial differentiation costs factors bounded by $E_0^{C_s}$.

We take $\beta = 3/4$, within the convenient interval

$$\frac{1}{2} < \beta < 1. \tag{123}$$

The upper inequality leaves two neighboring exponents for the reference inverse. The lower one makes a quadratic correction decay faster than H, giving the strengthened metric estimate in Theorem 10.2. The words weak and strong below compare endpoint weights: both scales consist of smooth functions with flat endpoint extensions.

Definition 7.1 (Height, source and circle spaces)**.** For real even π-periodic functions set

$$\|u\|_{\gamma,s}^2 = \sum_{a+b+c\le s} \|M^c \mathsf{D}^a \partial_\theta^b (H^{-\gamma}u)\|_{L^2(d\varrho\, d\theta)}^2. \tag{124}$$

The intersection over $s \ge 0$ is $\mathcal{X}_\gamma^0$. Choose χ_L equal to one at the left end and zero at the right, with derivative supported in a compact positive component, and set $\mathcal{X}_\gamma = \mathcal{X}_\gamma^0 \oplus \mathbb{R}\chi_L$. Its extra scalar records the difference of the two endpoint heights. The compatible source space is

$$\mathcal{Y}_\beta = \{f \in \mathcal{X}_\beta^0 : \mathsf{Q}f = 0\}, \qquad \mathsf{Q}f = \int_{r_-}^{r_+} \langle f \rangle \, dr.$$

For circle sequences define

$$\|a\|_{\mathcal{Z}_{\gamma,s}}^2 = \sum_m \sum_{b+c\le s} M_m^{2c} H_m^{-2\gamma} \|\partial_\theta^b a_m\|_{L^2(\mathbb{T})}^2. \tag{125}$$

The angularly constant sector is denoted by $\mathcal{K}_\gamma$. Odd and even sectors will be indicated explicitly. Without a displayed γ, a height or source norm uses $\gamma = \beta$.

Lemma 7.2 (Calculus and flat extension)**.** *These spaces are tame Fréchet spaces. Products, finite-order differential expressions, and coefficient reciprocals after separation of their reference factors are smooth tame on neighborhoods where the remaining denominators have positive lower bounds. The loss is fixed independently of the output order. Every member of $\mathcal{X}_\gamma^0$, $\gamma > 0$, extends smoothly and flatly in r at both endpoints; this also holds for its parameter derivatives.*

Proof. Here the noncompact weighted scale, rather than the local Sobolev calculus, needs explanation. Write $v = H^{-\gamma}u$. Choose unit-scale functions $\phi_i(\varrho)$ with bounded derivatives, finite overlap and $\sum_i \phi_i^2 = 1$. Bounded logarithmic derivatives imply $M \asymp M_i := M(i)$ on their supports. The norm of v is equivalent to

$$\sum_{i,j} \int (M_i^2 + \xi^2 + j^2)^s |\widehat{\phi_i v}(\xi, j)|^2 \, d\xi.$$

Let $\omega_i = (M_i^2 + \xi^2 + j^2)^{1/2}$. A smooth partition in $\log \omega_i$ gives maps J, P with $PJ = I$ to and from the exponentially decreasing sequence space over $B = \ell_i^2 L_\xi^2 \ell_j^2$. Concretely, $(Jv)_{k,i} = \zeta_k(\omega_i)\widehat{\phi_i v}$ and $Pb = \sum_i \phi_i \mathcal{F}^{-1} \sum_k \widetilde{\zeta}_k(\omega_i) b_{k,i}$, where the enlarged cutoff is one

on the support of ζ_k. Cauchy–Schwarz in k gives $\|Jv\|_s \le C_s\|v\|_{s+1}$; finite overlap gives $\|Pb\|_s \le C_s\|b\|_s$. This is a tame retraction and supplies smoothing. Parity projection preserves it. The endpoint scalar is a finite-dimensional summand. If ψ is compactly supported with $\mathsf{Q}\psi = 1$, then $f \mapsto f - \psi\mathsf{Q}f$ gives the source projection: its continuity follows by pairing $H^{-\beta}f$ with the square-integrable fixed function $H^{\beta}\ell$.

On each unit cylinder the usual Sobolev product bound, at a fixed base order $b > 1$, puts only one factor at the high order. Summing with finite overlap gives the same bound globally. Derivatives of a reciprocal are products divided by a positive power of its denominator; interpolation puts intermediate derivatives into the same one-high-factor form. After expanding a complete expression, all its reference coefficients have a bound $E_0^{C_s}$; the first inequality in (122) costs one additional weight order. The differential expression has fixed order, so the total loss is fixed. The identical Fourier construction proves the assertion for circle sequences. A restriction to a transverse circle costs one normal Sobolev derivative, and a fixed cutoff supplies its right inverse.

Finally ∂_r^a is a finite combination of D derivatives with coefficients bounded by $E_0^{C_a}$. Local Sobolev embedding bounds $d_\partial^{-k}\partial_r^a\partial_\theta^b u$ by a seminorm times $H^\gamma E_0^{C_{a,b,k}}$, which tends to zero. This proves the flat extension, also after any fixed parameter differentiation. □

7.2 Quantitative root geometry

Write $\mathfrak{r}_n$ for the radius scale of a model, bounded above and below, and $v = r_X$. The radial construction gives

$$A_n \asymp e^{-n^2}, \quad B_n \asymp A_n^9, \quad c\mathfrak{r}_n A_n^5 \le v \le C\mathfrak{r}_n A_n, \quad w_* = \lambda_n v, \quad \log\lambda_n = -n^{5/2} + O(n).$$

The long positive portion and the negative portion have radial size comparable to A_n. Summing the remaining lengths gives $cA_{n+1} \le d_\partial \le CA_n$, which proves the comparison used above. On a bounded enlargement the index changes by $O(1)$, so the same comparisons hold. Every interval length and reciprocal is bounded by e^{Cn^2}.

The derivative and reciprocal bounds were proved in (60). They imply that every fixed jet of $w_*^{\pm1}$ costs only $\mathcal{E}_n^{C_s}$. We shall also use the model identities $r_X = v$, $v_X = -\rho r$, with $\rho = A_n(e^{-X} - e^{-2X}) - B_n$, to control the roots quantitatively.

Let $r_{n,L}, r_{n,R}$ be the roots of w_*' and let x measure distance into a negative component. Its reference characteristic speed is

$$v_0 = \sqrt{-w_*'/(rw_*)} = \sqrt{x}\, c_i(x).$$

At a root, $c_i(0)^2 = |\rho_X(X_i)|/v(X_i)^3$. Since the left and right values of $|\rho_X|$ are respectively comparable to A_n, A_n^9, while v there is comparable to $\mathfrak{r}_nA_n, \mathfrak{r}_nA_n^5$, we obtain

$$c_L(0) \asymp \mathfrak{r}_n^{-3/2}A_n^{-1}, \qquad c_R(0) \asymp \mathfrak{r}_n^{-3/2}A_n^{-3}. \tag{126}$$

Dividing a simple zero by its signed distance uses $f(x)/x = \int_0^1 f'(tx)\,dt$. It follows from the model equations that the normalized jets of c_i and c_i^{-1} on a collar of size $c\mathfrak{r}_nA_n^5$ cost at most $C_k(\mathfrak{r}_nA_n^5)^{-k}$; at the left root A_n suffices. Choose once and for all

$$\delta_n^{\mathrm{root}} = \epsilon\mathfrak{r}_nA_n^{20}.$$

On $|x| < 8\delta_n^{\mathrm{root}}$ the ratio $c_i(x)/c_i(0)$ lies in $[1/2, 2]$. Higher derivatives change constants, not the collar. The fixed central pieces are handled by their positive minima.

Lemma 7.3 (Travel-angle controls)**.** *There are profiles χ_n supported in $\delta_n^{\mathrm{root}} < x < 2\delta_n^{\mathrm{root}}$ at the left root, with*

$$DT_n[\chi_n] = 1, \qquad \|\partial_r^k \chi_n\|_\infty \le C_k \mathfrak{r}_n^{-k} A_n^{9-20k}, \qquad T_n(w) = \int_{w'<0} \sqrt{-w'/(rw)}\, dr.$$

The differential uses the variation $w_* e^{t\chi_n}$. A separate correction cutoff can rise from zero to one on $3\delta_n^{\mathrm{root}} < x < 4\delta_n^{\mathrm{root}}$. On both supports, $v_0 \asymp \mathfrak{r}_n^{-1} A_n^9$ and $|w_*'/w_*| \asymp \mathfrak{r}_n^{-1} A_n^{18}$.

Proof. For nonnegative nonzero $\phi \in C_c^\infty((1,2))$ take $f_n = \phi(x/\delta_n^{\mathrm{root}})$. Logarithmic variation gives $\delta v_0 = -f_n'/(2rv_0)$, hence

$$DT_n[f_n] = \frac{1}{2}\int ((rv_0)^{-1})' f_n\, dr = -\frac{(\delta_n^{\mathrm{root}})^{-1/2}}{4r_{n,L}c_L(0)} \int_1^2 t^{-3/2}\phi(t)\, dt\, (1 + O(A_n^{19})).$$

This is nonzero and comparable to $-A_n^{-9}$. Set $\chi_n = f_n/DT_n[f_n]$. The derivative bound follows by rescaling. The speed bounds follow from (126); the second is $|w_*'/w_*| = rv_0^2$. The disjoint cutoff is an integrated rescaled bump. The same argument at each fixed central root completes the finite part. □

7.3 One neighborhood for all components

The unknown metric has the form

$$w = w_* \exp\left(\nu\chi_* + \sum_m \kappa_m \chi_m\right). \tag{127}$$

Here χ_* is the compact positive control of the second mode. The remaining controls have disjoint negative supports. Put $\Phi_* = \phi_*\sqrt{2}\cos 2\theta$, with $\ell_*\Phi_* = 1$. Our state is

$$\mathbf{x} = (u, \nu, \kappa, h_L), \quad u \in \Phi_* + \ker \ell_* \subset \mathcal{X}_\beta, \quad \kappa \in \mathcal{K}_\beta, \quad h_L - \mathrm{id} \in \mathcal{Z}_{\beta,\mathrm{odd}}. \tag{128}$$

Each chart is an increasing lift commuting with translation by π. The full displacement seminorm is

$$\mathfrak{n}_s = |q| + \|u - \Phi_*\|_s + |\nu| + \|\kappa\|_{\mathcal{K}_{\beta,s}} + \|h_L - \mathrm{id}\|_{\mathcal{Z}_{\beta,s}}. \tag{129}$$

The unscaled u must appear here even at $q = 0$, since the metric derivative of $\mathcal{F}_0(w,u) = \mathscr{L}_w u$ depends on u.

For local norms include the component and all neighboring components used by its charts; for a positive component include both adjacent negative ones. These neighborhoods have bounded overlap. Compact scalars are included only where their supports meet a neighborhood. Let $\mathfrak{g}_{s,m}$ be the corresponding geometric norm of $(qu, \nu, \kappa, h_L - \mathrm{id})$, using only positive height jets at an end. Then $(\sum_m \mathfrak{g}_{s,m}^2)^{1/2} \le C_s \mathfrak{n}_s$.

Proposition 7.4 (Uniform geometric calculus)**.** *On a single neighborhood $\mathfrak{n}_{b_0} < \epsilon_0$, for fixed finite b_0, all metric type intervals are unchanged. With $Z = qu$, $E = 1 + w^2$,*

$$Q = w + Z_r + EZ_{\theta\theta}/r, \qquad D = 1 - 2wZ_r - Z_r^2 - E(Z_\theta/r)^2,$$

one has $w/w_* \in [1/2, 2]$, $Q/w \in [7/8, 9/8]$, $D \in [1/2, 3/2]$ and $1 - (Z_\theta/r)^2 \ge 3/4$. The regularized coordinates, inverses, height and source factors constructed below have reference jets bounded by $C_s\mathcal{E}_m^{C_s}$. Their differences satisfy

$$\|\Delta c_m\|_{C^s} \le C_s \mathcal{E}_m^{C_s} H_m^\beta \mathfrak{g}_{s+d_*,m} \tag{130}$$

with fixed d_*. The assertion includes the positive principal differences divided by $y^3(1-y)^3$ and every fixed number of state variations, with the corresponding one-high-factor tame estimates.

Proof. We prove the low comparisons now and the coordinate assertion as its formulas are constructed below. Local Sobolev embedding and (60) bound every required low positive height jet, control jet and chart displacement by $C\epsilon_0 H_m^\beta \mathcal{E}_m^C$. For each fixed A, $H_m^\beta \mathcal{E}_m^A$ is bounded and tends to zero. Thus one ϵ_0 suffices for every component and every inequality in a finite list. On a control support the relative change of w'/w is at most $C|\kappa_m|A_n^{-29}$ by the preceding lemma; near each root it is zero. The compact control has a positive reference minimum. This preserves the signs and all root germs. The formulas for Q, D and the reciprocal bound for w_* give the remaining comparisons.

Here is the uniform derivative argument used for the forthcoming coordinate constructions. In a characteristic equation $\dot\Theta = G(r,\Theta)$ the angular Jacobian is $\exp\int G_\theta(r,\Theta)\,dr$. The reference has $G_\theta = 0$; the perturbed integral is bounded by $C\epsilon_0 H_m^\beta \mathcal{E}_m^C$. It therefore stays uniformly bounded above and below. Highest flow derivatives satisfy a linear equation with this same integrating factor. Subtracting the reference equation places a difference in each source term. Product estimates give (130), with one high factor. Inversion follows by differentiating $\Psi^{-1}\Psi = \mathrm{id}$, since the low differential is relatively close to the reference. Products, positive powers, the finite boundary jets and the divided differences below have a fixed derivative cost. They preserve the difference factor H_m^β. At the end of each calculation absorb the full $\mathcal{E}_m^{C_s}$ product into one M order.

All parameter variations use the same formulas. Pulling back by $Z = qu$ uses $DZ = \dot q\, u + q\dot u$ and $D^2 Z = q_1 u_2 + q_2 u_1$, with higher derivatives zero; thus the full seminorms (129) apply also at $q = 0$. The explicit nondegeneracy and finite derivative counts needed in this argument are verified below, completing the proposition without an additional smallness condition at higher orders. □

7.4 Fixing characteristic returns by their own residual

Start with the type-preserving operator $\bar L$ of Lemma 3.3. On a negative component write its coefficients as (a,b,c) and put

$$d = b/a, \qquad v = \sqrt{b^2 - ac}/a, \qquad c_\pm = d \pm v.$$

Reflection $S\theta = -\theta$ gives $c_+ = -c_- \circ S$. Let f be the left-to-right minus flow and T its reference travel angle. Write $R_T\theta = \theta + T$. Define

$$\widetilde h_R = R_{-T} h_L f^{-1}, \quad h_R = \mathsf{O}\widetilde h_R, \quad \mathfrak{e} = (\mathsf{E}\widetilde h_R)\circ h_R^{-1}, \tag{131}$$

where O and E are odd and even projections. The derivative of h_R is an average of positive derivatives, so h_R is increasing. Direct multiplication gives

$$(\mathrm{id} + \mathfrak{e}) h_R f h_L^{-1} = R_{-T}.$$

Thus one even equation $\mathfrak{e} = 0$ imposes both returns, the plus one by reflection. Its mean controls the travel angle; its nonconstant part controls the boundary parametrization.

For an odd variation $\delta h_L = \psi \circ h_L$, at fixed f, the chain rule yields

$$\begin{aligned} D_h\mathfrak{e}[\psi] &= \mathcal{M}_T\psi + \mathcal{Q}_h(\mathfrak{e},\psi), \qquad \mathcal{M}_T\psi = \mathsf{E}[\psi(\theta+T)], \\ \mathcal{Q}_h(\mathfrak{e},\psi) &= \mathsf{E}\left[\mathfrak{e}\int_0^1 \psi'(\theta+T+t\mathfrak{e})\,dt\right] - \mathfrak{e}'\mathsf{O}[\psi(\theta+T+\mathfrak{e})]. \end{aligned} \tag{132}$$

Indeed the variation of $\widetilde h_R$, after composition with h_R^{-1}, is $\psi(\theta+T+\mathfrak{e})$; its odd part is the variation of h_R in these coordinates. This proves the formula. At the reference,

$$\mathcal{M}_T^{-1}(g_j \cos j\theta) = \frac{g_j}{\sin(jT)}\sin j\theta \quad (j > 0 \text{ even}). \tag{133}$$

The selected angle bound loses one angular derivative and a fixed coefficient weight. There is no inverse on the constant mode; this is why the radial controls were added.

Proposition 7.5 (Return correction)**.** *The operator $\bar{L}$ can be replaced, on the chosen neighborhood, by a symmetric divergence operator $\widehat{L}$ with the same normal coefficient, the same root germs, and exact returns $h_R \widehat{f}_{\pm} h_L^{-1} = R_{\pm T}$. It is unchanged on positive components. Its difference has the form*

$$\bar{L} - \widehat{L} = \mathscr{E}_{\mathbf{x}}(\mathfrak{e}), \tag{134}$$

where $\mathscr{E}_{\mathbf{x}}(e)\eta$ is smooth tame and bilinear in (e, η), with values in $\mathcal{Y}_\beta$.

Proof. Let Φ_r be the minus flow. Its required endpoint is $f_* = h_R^{-1} R_{-T} h_L$. With $H_c = h_R f$, set

$$K = f^{-1} f_* = H_c^{-1}(\mathrm{id} + \mathfrak{e}) H_c, \quad \Delta = K - \mathrm{id}, \quad J_r = \mathrm{id} + \chi(r)\Delta, \quad \widehat{\Phi}_r = \Phi_r J_r.$$

Use the interior cutoff of the control lemma. Each J_r is increasing, as a convex combination of increasing lifts. The new velocity is

$$\widehat{c}_- = c_- + \Phi_r'(\Phi_r^{-1}\theta)\chi'(r)\Delta(J_r^{-1}\Phi_r^{-1}\theta). \tag{135}$$

Define $\widehat{c}_+ = -\widehat{c}_- \circ S$, then $\widehat{d} = (\widehat{c}_+ + \widehat{c}_-)/2$, $\widehat{v} = (\widehat{c}_+ - \widehat{c}_-)/2$, $\widehat{a} = a$, $\widehat{b} = a\widehat{d}$ and $\widehat{c} = a(\widehat{d}^2 - \widehat{v}^2)$. These formulas give the asserted returns and symmetric principal matrix. The speed lower bound on the correction support and the factor $H_m^\beta \mathcal{E}_m^C$ make the correction less than $v/2$ on a single low-order neighborhood. Near the roots it is zero.

For the error factorization use

$$\Delta(\theta) = \mathfrak{e}(H_c\theta) \int_0^1 (H_c^{-1})'(H_c\theta + t\mathfrak{e}(H_c\theta))\, dt.$$

Freeze the state and these composition paths. Replace only the leading residual by an independent e. This defines linear increments $d_1[e], v_1[e]$ with actual values d_1, v_1 at $e = \mathfrak{e}$. Then the coefficients of $\bar{L} - \widehat{L}$ are

$$\delta b[e] = -a d_1[e], \qquad \delta c[e] = -a(2d + d_1)d_1[e] + a(2v + v_1)v_1[e].$$

The associated divergence expression is bilinear in (e, η). Its support is interior to each component and all its endpoint jets are flat. Its integral is zero. Each term contains a residual factor and a positive height jet, both of weight H_m^β; dividing by the source weight leaves enough decay for every fixed coefficient product. The flow estimates above and the product calculus prove the claimed smooth tame bounds. □

Combining the two residual corrections gives the exact identity

$$D_u \mathcal{F}_q \eta = \widehat{L}\eta + q\partial_\theta(B\mathcal{A}\eta_\theta) + \mathscr{E}_{\mathbf{x}}(\mathfrak{e})\eta, \qquad \mathcal{A} = \mathcal{F}_q(w, u), \quad B = E/(rQ). \tag{136}$$

Both added terms vanish at a solution. We shall invert $\widehat{L}$ while retaining these errors in the nonlinear theorem.

7.5 The root coordinate and the fixed transmission law

Let $S_\pm$ be inverse characteristics of $\widehat{L}$, equal to θ at the left boundary. Set

$$U = h_L S_+, \quad V = h_L S_-, \quad \sigma = (V - U)/2, \quad \varphi = (U + V)/2. \tag{137}$$

Their endpoint values are $(0, h_L)$ and (T, h_R). Moreover $U_\theta, V_\theta > 0$ and

$$\det D(\sigma, \varphi) = vU_\theta V_\theta > 0.$$

At fixed φ, σ increases strictly from 0 to T; this proves global invertibility onto the cylinder. Substitution in the principal quadratic form, followed by division by this Jacobian, gives the principal divergence density $\delta \operatorname{diag}(1, -1)$, where $\delta = \sqrt{b^2 - ac}$.

At a root $v = \sqrt{x}\, c_0(x, \theta)$. The travel coordinate is of order $x^{3/2}$, so use $\xi_L = (3\sigma/2)^{2/3}$ and $\xi_R = (3(T - \sigma)/2)^{2/3}$ on the two halves. To justify smoothness when the coefficients depend on angle, solve the fixed-time equation in independent variables x, λ:

$$\Theta_t = 2xt\, d(xt^2, \Theta) + 2\lambda t^2 c_0(xt^2, \Theta), \qquad 0 \le t \le 1. \tag{138}$$

The inverse time-one flow followed by the boundary chart is one smooth map $G(x, \lambda, \theta)$. The two characteristic families are obtained from this same map by $\lambda = \pm x^{3/2}$. Their common boundary chart is essential for the cancellation in the following estimate.

Lemma 7.6 (Finite-order control at a root)**.** *On a fixed rectangle $0 \le x \le 1$, $|\lambda| \le 1$, define, for a smooth $G(x, \lambda, \theta)$,*

$$\mathcal{E}G = \frac{G(x, x^{3/2}, \theta) + G(x, -x^{3/2}, \theta)}{2}, \qquad \mathcal{O}G = \frac{G(x, x^{3/2}, \theta) - G(x, -x^{3/2}, \theta)}{2x^{3/2}}.$$

Both extend smoothly to $x = 0$, and for every integer $s \ge 0$,

$$\|\mathcal{E}G\|_{C^s_{x,\theta}} \le C_s \|G\|_{C^s_{x,\lambda,\theta}}, \qquad \|\mathcal{O}G\|_{C^s_{x,\theta}} \le C_s \|G\|_{C^{s+1}_{x,\lambda,\theta}}. \tag{139}$$

The quotient at zero is $\partial_\lambda G(0, 0, \theta)$. The estimates also hold after any fixed number of parameter differentiations.

Proof. First suppose G is even in λ. Fix $0 < x_0 \le 1$ and take its degree-s Taylor polynomial at $(x_0, 0)$. Its monomials $(x - x_0)^i \lambda^{2j}$ become $(x - x_0)^i x^{3j}$. Their ath x derivative at x_0, $a \le s$, either vanishes or has a factor x_0^{3j-a+i} with nonnegative exponent. Their coefficients use at most s derivatives of G.

For the Taylor remainder, a chain-rule term using i direct x derivatives and k λ derivatives is bounded by

$$C_s \|G\|_{C^s} x_0^{3(s-i-k)/2} x_0^{3k/2-(a-i)} = C_s \|G\|_{C^s} x_0^{3s/2-a-i/2}.$$

Again the exponent is nonnegative. At exponent zero the remainder tends to zero by continuity of the top Taylor derivatives. The polynomial terms tend to the appropriate endpoint Taylor jet, so these are bounds for continuous derivatives up to $x = 0$. Apply this to the even part of G to prove the first estimate. Tangential derivatives commute with the construction and reduce the remaining derivative budget by their order.

For the second estimate the smooth even function

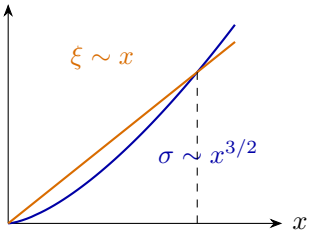
$$K(x, \lambda, \theta) = \frac{1}{2} \int_{-1}^{1} \partial_\lambda G(x, t\lambda, \theta)\, dt$$

satisfies $\|K\|_{C^s} \le C_s \|G\|_{C^{s+1}}$ and $\mathcal{O}G = K(x, x^{3/2}, \theta)$. Apply the even estimate to K. Both operations are linear in G, so the same bounds commute with parameter differentiation. □

Applying the lemma to the common flow gives

$$\sigma = x^{3/2} \chi_0(x, \theta), \quad \chi_0(0, \theta) = \tfrac{2}{3} c_0(0, \theta) h_L'(\theta) > 0, \quad \xi_L = x(3\chi_0/2)^{2/3}.$$

Thus the root coordinate is smooth with nonzero radial derivative. The right root uses its common boundary chart h_R in the same way. Rescaling from the unit rectangle to a root collar introduces powers of its length and reciprocal, all bounded by $\mathcal{E}_m^{C_s}$. Their complete product costs one M order. The even substitution uses s input derivatives and the odd divided difference uses $s+1$; these costs remain fixed as the output order increases.



The travel angle has zero derivative.
Its 2/3 power has a nonzero derivative.

Figure 10: Regularization at a turning circle. The graph displays the leading powers with unit coefficients. The positive coefficient of x in ξ is controlled uniformly at every turning circle.

Write $\Psi_i = (\xi_i, \varphi)$ and $\alpha_i = (\delta \circ \Psi_i^{-1})/\sqrt{\xi_i}$. The factor is smooth and positive, also at the root, since

$$\alpha_i = \frac{a|\det D\Psi_i|}{U_\theta V_\theta}.$$

For $u = (h_i W_i) \circ \Psi_i$, $h_i = \alpha_i^{-1/2}$, the equation is

$$\begin{aligned}(\partial_{\xi_i}^2 - \xi_i \partial_\varphi^2 + V_i) W_i &= F_i,\\ V_i &= -\alpha_i^{-1/2}(\partial_{\xi_i}^2 - \xi_i \partial_\varphi^2)\alpha_i^{1/2},\\ F_i &= h_i((f/|\det D\Psi_i|) \circ \Psi_i^{-1}).\end{aligned} \tag{140}$$

Indeed the density in these coordinates is $\alpha_i \operatorname{diag}(1, -\xi_i)$; the positive conjugation cancels its first derivatives. The source formula preserves $\int u f \, dr \, d\theta = \int W_i F_i \, d\xi_i \, d\varphi$.

The middle value is $\xi_c = (3T/4)^{2/3}$. There $\alpha_L = \alpha_R$ and $\partial_\sigma \log \alpha_R - \partial_\sigma \log \alpha_L = \xi_c^{-3/2}$. Consequently value and conormal matching become

$$\begin{pmatrix} W_R \\ (W_R)_{\xi_R} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -1/(2\xi_c) & -1 \end{pmatrix} \begin{pmatrix} W_L \\ (W_L)_{\xi_L} \end{pmatrix}. \tag{141}$$

This matrix is independent of the state. At a type root, with boundary chart h_{bd}, the physical pair is

$$u = (h_i W_i)(0, h_{\mathrm{bd}}), \qquad p_{\mathrm{in}} = \frac{h_{\mathrm{bd}}'}{h_i(0, h_{\mathrm{bd}})}(\partial_{\xi_i} W_i + \partial_{\xi_i} \log h_i \, W_i)(0, h_{\mathrm{bd}}). \tag{142}$$

The tangential shear cancels because $a\varphi_r + b\varphi_\theta = 0$ at the root. This identity keeps the zero mode and the full Green pairing, not just the principal characteristics.

For completeness, all reference coordinate reciprocals used here have the promised bounds. At a root

$$\xi = x\left[\tfrac{3}{2}\int_0^1 \sqrt{t}\, c_i(xt)\, dt\right]^{2/3}.$$

Outside its collar the Morse potential has a lower bound cA_n^{24}; the upper and lower bounds for v then bound $|\xi_r| = v_0/\sqrt{\xi}$ above and below by powers of A_n. Also $\xi_c \asymp A_n^{1/3}$. Differentiating these formulas and the inverse identity gives $\mathcal{E}_n^{C_s}$ bounds for charts, inverses, $\alpha_i = w_*|\xi_r|$, h_i, potentials and source factors. The perturbed bounds follow from the flow and inverse argument in Proposition 7.4.

7.6 Crossing the positive components

On a positive component continue the signed root coordinate and put $y_i = -\xi_i$. Build a radial reference coordinate $y \in [0,1]$ equal to y_L near the left root and $1 - y_R$ near the right. This can be done explicitly by choosing positive y_r with integral one: use the prescribed derivatives on disjoint root collars and a positive constant on their complement, with cutoff interpolation. The collars have total prescribed coordinate increment less than $1/4$, so that constant and its reciprocal are bounded by $\mathcal{E}_m^C$. Interpolate the logarithms of the two root height factors, choosing height factor one in the middle. All reference coordinate, height and source factors retain the same jet bounds.

For a perturbed state extend the *differences* of the negative coordinate jets through degree four and of the logarithmic height jets through degree three, using cutoff Taylor polynomials. This leaves the reference map unchanged on a compact interval J_* containing the second-mode normalization and metric control. Relative closeness of the differential gives a global diffeomorphism: it is proper, locally invertible, and of boundary degree one. The map and its inverse have the estimates already proved.

Writing the positive height factor in physical coordinates as $h_{\mathbf{x}}$ and its chart as $\Psi_{\mathbf{x}}$, the positive canonical equation for $-\widehat{L}u = F_{\mathrm{ph}}$ is

$$P_{\mathbf{x}}U = -\left(\frac{h_{\mathbf{x}}}{|J_{\mathbf{x}}|}\widehat{L}[h_{\mathbf{x}}(U \circ \Psi_{\mathbf{x}})]\right) \circ \Psi_{\mathbf{x}}^{-1}, \qquad F = \left(\frac{h_{\mathbf{x}} F_{\mathrm{ph}}}{|J_{\mathbf{x}}|}\right) \circ \Psi_{\mathbf{x}}^{-1}. \tag{143}$$

These formulas again preserve the pairing. The continued negative principal part is $-\partial_{y_i}^2 - y_i\partial_\varphi^2$. The specified jets match principal densities through order two. Thus, with $d = y(1-y)$,

$$P_{\mathbf{x}} - P_* = -\partial_a(d^3 e_{\mathbf{x}}^{ab}\partial_b) + \delta V_{\mathbf{x}}. \tag{144}$$

There is no boundary condition on the potential. Division by d^3 has a fixed cost: if F, F', F'' vanish at zero, then

$$F(y)/y^3 = \tfrac{1}{2}\int_0^1 (1-t)^2 F'''(ty)\,dt.$$

Each further derivative differentiates the integrand; the L^2 rescaling factor $t^{-1/2}$ is integrable. The right end is identical. This verifies the final divided-coefficient assertion of the uniform geometric calculus. All operations use at most two height derivatives, one root divided difference, four boundary jets, three divisions and fixed trace and embedding costs; their sum fixes d_*.

The reference positive operator is

$$P_* = -\partial_y(a\partial_y) - c\partial_\varphi^2 + V, \qquad a = h_*^2 w_* y_r, \quad c = \frac{h_*^2 w_*'}{r y_r}.$$

It obeys $a \geq \mathcal{E}_m^{-C}$ and $c \geq \mathcal{E}_m^{-C} d$, with all fixed coefficient jets bounded by $\mathcal{E}_m^{C_s}$. Near a root it is exactly $a = 1, c = y_i$; on the rest this follows from the positive-model and transition lower bounds for w_*'/w_*. These are the coefficients of the next energy estimate.

At a type root the adjacent charts and height factors have the same value and first physical derivative. Formula (142) therefore becomes equality of canonical values and opposite inward derivatives. Together with (141) this is one fixed transmission domain. Its smooth members really are smooth in the original variables: the value matches tangential derivatives, the conormal matches the first normal derivative, and the equation solves for au_{rr} since $a > 0$. Repeated normal differentiation then matches every higher jet. This observation is what permits us to use a common closed operator domain in the next section.

8 Inverting on one common domain

A tame estimate with derivative loss is not a bounded inverse on a single Banach space. It cannot by itself justify a Neumann series. We first identify a source space on which the *relative* operator is bounded. The extra endpoint factor in its coefficients will compensate for the coefficient-weight loss of the reference inverse. Only after that Banach-space inversion do we recover all derivatives. This order of argument keeps one neighborhood for the entire nonlinear problem.

8.1 Sources, scalars and the domain

Use the canonical cylinders of Section 7 and retain the physical sign of the equation: on a positive cylinder its canonical operator is $-P_{\mathbf{x}}$. Let $h_{\mathbf{x}}$ denote the height factor in physical coordinates, $\Psi_{\mathbf{x}}$ the chart, and $J_{\mathbf{x}}$ its Jacobian. Define

$$\mathscr{C}_{\mathbf{x}}u = (u/h_{\mathbf{x}}) \circ \Psi_{\mathbf{x}}^{-1}, \qquad \mathscr{S}_{\mathbf{x}}f = (h_{\mathbf{x}}f/|J_{\mathbf{x}}|) \circ \Psi_{\mathbf{x}}^{-1}, \qquad \mathbb{L}_{\mathbf{x}} = \mathscr{S}_{\mathbf{x}}\widehat{L}_{\mathbf{x}}\mathscr{C}_{\mathbf{x}}^{-1}.$$

These preserve the pairing with $dr\,d\theta$. For real t and angular order $s \geq 0$ use the volume-source norms

$$\|F\|_{t,s}^2 = \sum_m M_m^{2t} H_m^{-2\beta} \|F_m\|_{L^2_{\mathrm{rad}} H^s_\varphi}^2, \qquad F \in \mathbb{V}_{t,s}. \tag{145}$$

The index includes the two halves of every negative component and the positive components. No matching or integral condition is imposed on these sources. For inverse outputs, the corresponding volume norm is supplemented by the absolute values of the three scalar unknowns introduced below; these scalar coordinates are assigned to the finite middle.

The charts agree with the reference on J_*, so the second-mode normalization has the fixed form $\ell_*^c W$. Choose a compact radial ψ in J_* with $\mathsf{Q}\psi = 1$ and put $c_* = \mathscr{S}_*\psi = \mathscr{S}_{\mathbf{x}}\psi$. A physical height is represented by

$$\eta = \mathscr{C}_{\mathbf{x}}^{-1}W + a_L\chi_L, \qquad \ell_*^c W = 0.$$

Its left endpoint constant is a_L; the right one is zero. Define

$$k_{\mathbf{x}} = D_\nu \mathcal{F}_q(w,u), \quad k_{\mathbf{x}}^c = \mathscr{S}_{\mathbf{x}}k_{\mathbf{x}}, \quad s_{\mathbf{x}} = \mathscr{S}_{\mathbf{x}}\widehat{L}_{\mathbf{x}}\chi_L.$$

Here D_ν differentiates the true residual with u, q fixed, including both its metric and reference height. The augmented operator is

$$\mathbb{P}_{\mathbf{x}}(W, a_L, \mu, \lambda) = \mathbb{L}_{\mathbf{x}}W + a_L s_{\mathbf{x}} + \mu k_{\mathbf{x}}^c + \lambda c_*. \tag{146}$$

The parameter μ removes the second-mode obstruction. The additional λ removes the integral obstruction for arbitrary sources; it will disappear for the original compatible problem.

At the reference let $\mathfrak{g}_* = \mathscr{C}_* 1$ and $\mathsf{Q}_* F = \langle \mathfrak{g}_*, F \rangle$. The coefficient bounds make this continuous and $\mathsf{Q}_* c_* = 1$. Set $\lambda = \mathsf{Q}_* F$ and apply the compatible reference inverse to $F - \lambda c_*$. This constructs a two-sided augmented inverse $\mathbb{G}_*$. In the zero angular mode the construction can be seen directly: with $f_* = \mathscr{S}_*^{-1}F$,

$$p_0(r) = \int_{r_-}^{r} (\langle f_* \rangle - \lambda\psi)(t)\,dt, \qquad u_0(r) = -\int_r^{r_+} \frac{p_0(t)}{w_*(t)}\,dt.$$

The first integrand has total integral zero. At either endpoint use the primitive starting there; division by w_* costs a coefficient factor, not an endpoint exponent. Subtract the left endpoint value times χ_L. This yields the required field, endpoint scalar and normalization. It also proves uniqueness in the zero mode.

8.2 Why two endpoint exponents are needed

The reference construction gives one natural endpoint inverse for every exponent in a compact subinterval of $(0,1)$ (Lemma 6.2). Choose $0 < \epsilon < \min(\beta, 1-\beta)$. The inverses at $\beta - \epsilon$ and $\beta + \epsilon$ agree on compact smooth sources: their difference satisfies the homogeneous equation, the normalization and the natural endpoint conditions. The reference uniqueness, together with the zero-mode formula above, makes it zero. Hence both estimates apply to the same $\mathbb{G}_*$.

Lemma 8.1 (A fixed coefficient loss)**.** *For a fixed L, every real t and every $s \geq 0$,*

$$\|\mathbb{G}_* F\|^{\mathrm{vol}}_{t-L,s} \leq C_{t,s} \|F\|_{t,s}. \tag{147}$$

The first normal derivative has the same bound with at most one additional angular source derivative. For a fixed $b \geq 2$,

$$\mathbb{D}_* := \mathbb{G}_*(\mathbb{V}_{0,b}), \qquad \|v\|_{\mathbb{D}_*} = \|\mathbb{P}_* v\|_{0,b} \tag{148}$$

is a complete closed graph domain. It includes every smooth admissible field with the stated transmissions, endpoint conditions and slice, whose reference source belongs to $\mathbb{V}_{0,b}$.

Proof. Write $\bar{S}_m = S_m - S_N$ on the tails and $\bar{S}_m = 0$ on the finite middle. Let π_k, ι_m be restriction to an output piece and extension of a source on a piece by zero. The reference estimate at exponent γ gives

$$\|\pi_k \mathbb{G}_* \iota_m\|_{s\to s} \leq C_s \mathcal{E}_k^{C_s} \mathcal{E}_m^{C_s} e^{-\gamma(\bar{S}_k - \bar{S}_m)}.$$

Conjugate by the β weight. Choose $\gamma = \beta + \epsilon$ when $\bar{S}_k \geq \bar{S}_m$ and $\gamma = \beta - \epsilon$ otherwise. The resulting block K_{km} satisfies

$$\|K_{km}\|_{s\to s} \leq C_s \mathcal{E}_k^{C_s} \mathcal{E}_m^{C_s} e^{-\epsilon|\bar{S}_k - \bar{S}_m|}. \tag{149}$$

There is no angular-order loss in this volume estimate. The first normal estimate of the radial construction adds at most one angular derivative and otherwise has the same block argument.

For each $\delta > 0$ the enlarged weight satisfies

$$|\log M_k - \log M_m| \leq \delta |\bar{S}_k - \bar{S}_m| + C_\delta.$$

This follows from the derivative of $n^{5/2}\log^2(e+n)$ with respect to $2n^3/3$ tending to zero. Using $\mathcal{E}^{C_s} \leq C_s M$, we find, for $L \geq 2$,

$$M_k^{t-L} \|K_{km}\| M_m^{-t} \leq C_s M_k^{2-L} (M_k/M_m)^{t-1} e^{-\epsilon|\bar{S}_k - \bar{S}_m|} \leq C_{t,s} e^{-(\epsilon/2)|\bar{S}_k - \bar{S}_m|}.$$

Choose δ so that $|t-1|\delta \leq \epsilon/2$. The last matrix has bounded row and column sums: the S_n are uniformly separated and there are only finitely many pieces at each index. The ℓ^2 Schur estimate proves (147).

The definition of $\mathbb{D}_*$ makes it complete. Approximate a source by compact smooth sources. The estimate gives local volume convergence and the equation passes to distributions, with weak transmissions. If $v_j \to v$ in the ambient volume space and $\mathbb{P}_* v_j \to F$ in the source space, then $v_j = \mathbb{G}_* \mathbb{P}_* v_j \to \mathbb{G}_* F$, so $v = \mathbb{G}_* F$; the graph is closed. First-normal estimates identify the local weak traces. Finally, any smooth admissible y with source F differs from $\mathbb{G}_* F$ by a homogeneous normalized natural endpoint solution, hence is equal to it. This proves the stated exhaustion of admissible smooth fields. □

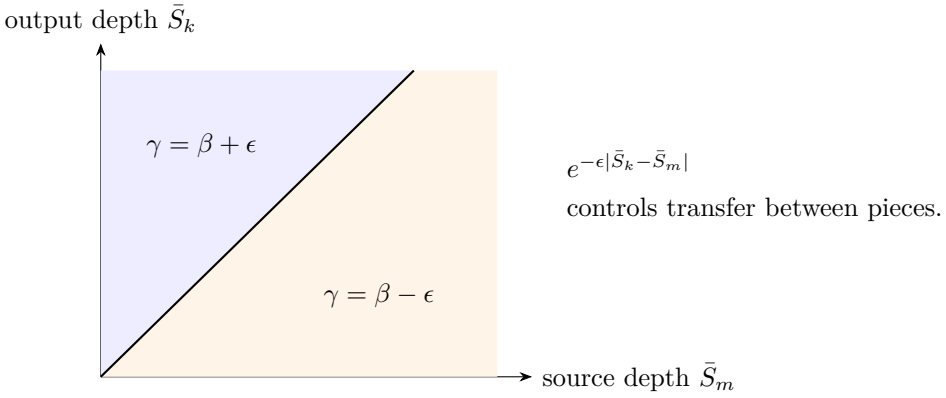


Figure 11: The same reference inverse is estimated with two endpoint weights. Each weight is used on one side of the diagonal. Their minimum gives decay in the distance between source and output depths.

8.3 The estimate that explains cubic matching

On a normalized positive cylinder consider

$$P_j u = -(au')' + j^2 cu + Vu = F, \qquad a \geq \mathcal{E}_m^{-C}, \quad c \geq \mathcal{E}_m^{-C} d, \quad d = y(1-y).$$

Coefficient derivatives through order two, and a^{-1}, are bounded by $\mathcal{E}_m^{C_1}$. Boundary values are arbitrary. We claim

$$\begin{aligned} \|du'\|_2 + \|\langle j\rangle d^{3/2} u\|_2 + \|d^2 u''\|_2 + \|\langle j\rangle^2 d^3 u\|_2 + \|\langle j\rangle d^{5/2} u'\|_2 \\ \leq \mathcal{E}_m^C (\|F\|_2 + \|u\|_2), \end{aligned} \tag{150}$$

where $\langle j\rangle = (1+j^2)^{1/2}$ in this display. For fixed j, a distributional L^2 solution with $F \in L^2$ is H^2 up to the endpoints by the regular ordinary differential equation. Uniform bounds come from the following calculation, not from that preliminary regularity statement.

Multiply by $d^2 \bar{u}$ and integrate. Since $d = 0$ at the ends,

$$\int a|(du)'|^2 + j^2 \int cd^2 |u|^2 = \operatorname{Re}\langle F, d^2 u\rangle + \int a|d'|^2 |u|^2 - \int V d^2 |u|^2.$$

This bounds du' and $jd^{3/2}u$. Now put $v = d^2 u$; it has zero trace and

$$P_j v = d^2 F - 2a(d^2)' u' - (a(d^2)')' u.$$

The right side is controlled by the first estimate. With $q_0 = c/a$, the same is true of $F_1 = -v'' + j^2 q_0 v$. Squaring and integrating the cross term twice gives

$$\|v''\|_2^2 + j^4 \|q_0 v\|_2^2 + 2j^2 \int q_0 |v'|^2 = \|F_1\|_2^2 + j^2 \int q_0'' |v|^2.$$

The final integral is controlled because $d^4 \leq Cd^3$. Use $q_0 \geq \mathcal{E}_m^{-C} d$ and expand v', v'' to obtain (150).

Every second-order term in the perturbation (144) is now controlled: $d^3 u_{yy}$ by $d^2 u''$, $d^3 u_{y\varphi}$ by $\langle j\rangle d^{5/2} u'$, and $d^3 u_{\varphi\varphi}$ by $\langle j\rangle^2 d^3 u$. A derivative on the coefficient leaves only terms controlled by the first two norms. Fourier convolution in φ therefore gives, for $s \geq b$,

$$\|\delta P_{\mathbf{x}} u\|_s \leq C_s H_m^\beta \mathcal{E}_m^{C_s} \left[\mathfrak{n}_{b_0,m}(\|F\|_s + \|u\|_s) + \mathfrak{n}_{s+d_*,m}(\|F\|_b + \|u\|_b)\right]. \tag{151}$$

This is the reason for matching the principal density to order two. It controls a second-order perturbation without imposing artificial boundary conditions on the reference solution.

We also use an elementary first-normal estimate on every normalized piece. Its radial equation has the form $Au'' + Bu' + (j^2C + D_0)u = F$ with A bounded away from zero and all coefficients bounded by fixed powers of $\mathcal{E}_m$. The equation and $\|u'\|_2^2 \le C\|u\|_2(\|u''\|_2 + \|u\|_2)$ give

$$\|u'\|_2 \le \mathcal{E}_m^C(\|F\|_2 + \langle j\rangle\|u\|_2), \qquad \|u''\|_2 \le \mathcal{E}_m^C(\|F\|_2 + \langle j\rangle^2\|u\|_2). \tag{152}$$

Young's inequality absorbs the intermediate u' term. Density extends the estimates to volume solutions. Neither estimate assumes zero boundary data.

8.4 The bounded relative operator

Define on compact smooth sources

$$\mathscr{R}_{\mathbf{x}} = (\mathbb{P}_{\mathbf{x}} - \mathbb{P}_*)\mathbb{G}_*.$$

If $\mathbb{G}_*F = (W, a_L, \mu, \lambda)$, then

$$\mathscr{R}_{\mathbf{x}}F = \delta\mathbb{L}_{\mathbf{x}}W + a_L(s_{\mathbf{x}} - s_*) + \mu(k^c_{\mathbf{x}} - k^c_*). \tag{153}$$

The last scalar has no difference. On a negative piece only the canonical potential varies. On a positive piece use (151), putting its compact scalar sources into F. Every coefficient difference at the output piece k contains H^β_k.

Here is the actual base-space calculation. Combine that factor with (149). For low coefficients the norm of the block from m to k is bounded by

$$C\mathfrak{n}_{b_0}\, M_k^C M_m^C e^{-\beta\bar{S}_k - \epsilon|\bar{S}_k - \bar{S}_m|}. \tag{154}$$

For every fixed C and $\delta > 0$, $M_j^C \le C_\delta e^{\delta\bar{S}_j}$, while $\bar{S}_m \le \bar{S}_k + |\bar{S}_k - \bar{S}_m|$. Choose $\delta < \min(\beta/4, \epsilon/2)$. The matrix in (154) is bounded by

$$C'\mathfrak{n}_{b_0} e^{-(\beta/2)\bar{S}_k - (\epsilon/2)|\bar{S}_k - \bar{S}_m|}.$$

Its row and column sums are uniformly bounded. Thus

$$\|\mathscr{R}_{\mathbf{x}}\|_{\mathbb{V}_{0,b}\to\mathbb{V}_{0,b}} \le C\mathfrak{n}_{b_0} < \tfrac{1}{2} \tag{155}$$

after increasing b_0 to cover the finite low derivatives and then decreasing ϵ_0. This is a bounded operator on precisely one source space; no identification of a derivative-losing inverse with a bounded endomorphism has been made.

For higher orders keep one coefficient factor at high order in the product estimate. The same summation proves, for $t \ge 0$, $s \ge b$,

$$\|\mathscr{R}_{\mathbf{x}}F\|_{t,s} + \|[\Lambda, \mathscr{R}_{\mathbf{x}}]F\|_{t,s} \le C_{t,s}\big[\epsilon_0\|F\|_{0,s} + \mathfrak{n}_{t+s+d_*}\|F\|_{0,b}\big], \tag{156}$$

where $\Lambda = |D_\varphi|$. Assign frequency zero to a_L, λ and frequency two to μ. For multiplication by a coefficient, the commutator Fourier matrix gains $|k| - |j|$, bounded by $|k - j|$; one additional coefficient derivative therefore proves the same bound. The scalar commutators are retained: $\Lambda(s_{\mathbf{x}} - s_*)a_L$ and $(\Lambda - 2)(k^c_{\mathbf{x}} - k^c_*)\mu$.

Approximation by compact smooth sources identifies this extension with the differential difference in distributions on $\mathbb{D}_*$. The canonical transmissions are fixed, hence

$$\mathbb{P}_{\mathbf{x}} = (I + \mathscr{R}_{\mathbf{x}})\mathbb{P}_* \quad \text{on } \mathbb{D}_*. \tag{157}$$

This is the desired comparison on a common domain.

8.5 Higher derivatives from the one base inverse

By (155), set

$$\widetilde{F} = (I + \mathscr{R}_{\mathbf{x}})^{-1} F, \qquad v = \mathbb{G}_* \widetilde{F}.$$

The inverse in the first formula is taken only on $\mathbb{V}_{0,b}$. We now explain why this same solution has all the required regularity.

First recover angular derivatives. If $C = [\Lambda, \mathscr{R}]$, then for $U_\delta = e^{i\delta\Lambda}$,

$$[U_\delta, \mathscr{R}] = i \int_0^\delta U_{\delta - t} C U_t \, dt.$$

Apply this to $(I + \mathscr{R})\widetilde{F} = F$. The base inverse bounds the difference quotients of $\widetilde{F}$ whenever those of F are bounded. Fatou's lemma on Fourier coefficients gives the first derivative; iteration gives all integer orders. Quantitatively, the commuted equation at order N contains $\sum_{k=0}^{N-1} \Lambda^k C \Lambda^{N-1-k} \widetilde{F}$. Interpolate the bound for C at the base and at order $N-1$, treating the high coefficient size as a weight in the source Hilbert norm. Each term is then bounded by

$$C_N(\|\widetilde{F}\|_{N-1} + (1 + \mathfrak{n}_{N+d_*})\|\widetilde{F}\|_b).$$

Interpolation and Young's inequality absorb the first term into half the order-N norm plus a base norm. This proves the one-high-factor estimate without smallness of high coefficient norms. Real orders follow by Hilbert interpolation, or may be replaced by their ceilings.

Next use the already solved identity $\widetilde{F} = F - \mathscr{R}_{\mathbf{x}} \widetilde{F}$. Estimate its last term by (156). It belongs to every coefficient-weight order once angular regularity is known. Thus

$$\|\widetilde{F}\|_{t,s} \le C_{t,s}\big[\|F\|_{t,s} + (1 + \mathfrak{n}_{t+s+d_*})\|F\|_{0,b}\big]. \tag{158}$$

No second inversion is required.

The normal recursion uses the original source. Estimate (158) controls angular derivatives of $\widetilde{F}$. We do not assume normal derivatives of that corrected source. Use instead the current equation $\mathbb{P}_{\mathbf{x}} v = F$, and subtract its three scalar columns:

$$F^\circ = F - a_L s_{\mathbf{x}} - \mu k^c_{\mathbf{x}} - \lambda c_*.$$

Rescale each negative half to a unit radial interval, denoted by y; the positive intervals already have this normalization. All rescaling factors and their reciprocals are reference coefficient factors. Solving the equation for its nonvanishing normal coefficient gives

$$\partial_y^2 W = \sum_{(i,j)\in I} A_{ij} \partial_y^i \partial_\varphi^j W + B F^\circ, \qquad I = \{(1,1),(0,2),(1,0),(0,1),(0,0)\}. \tag{159}$$

On a negative half the unscaled equation is $W_{\xi\xi} = \xi W_{\varphi\varphi} - VW + F^\circ$. On a positive piece the normal coefficient has a lower bound $\mathcal{E}_m^{-C}$ on the common neighborhood. Thus all the A_{ij}, B obey the coefficient calculus of Proposition 7.4, with a fixed state-derivative loss d_{geo}.

For $a \ge 2$ the actual mixed recursion is

$$\begin{aligned} \partial_y^a \partial_\varphi^b W = {} & \sum_{(i,j)\in I} \sum_{p=0}^{a-2} \sum_{q=0}^{b} \binom{a-2}{p} \binom{b}{q} (\partial_y^p \partial_\varphi^q A_{ij}) \partial_y^{a-2-p+i} \partial_\varphi^{b-q+j} W \\ & + \partial_y^{a-2} \partial_\varphi^b (B F^\circ). \end{aligned} \tag{160}$$

Every field on the right has at most $a-1$ normal derivatives and total order at most $a+b$. A derivative falling on a coefficient lowers the remaining field order by that amount.

Consequently the expansion terminates in angular derivatives of W, angular derivatives of W_y, and mixed derivatives of F°. At total order s their respective orders are at most s, $s-1$, and $s-2$. Starting with the already established angular and first-normal regularity, this identity in distributions proves the existence of the mixed derivatives successively.

Here is the quantitative form of this recursion. Fix a base order $b_1 \geq 4$, and let $\mathfrak{n}_{r,m}$ be the localized displacement seminorm on the bounded-overlap neighborhoods used in Section 7. For $s \geq b_1 + 2$ put

$$\mathcal{A}_{s,m} = \|W_m\|_{L^2_y H^s_\varphi} + \|\partial_y W_m\|_{L^2_y H^{s-1}_\varphi} + \|F^\circ_m\|_{H^{s-2}_{y,\varphi}}.$$

The finite expansion of (160) gives

$$\|W_m\|_{H^s_{y,\varphi}} \leq C_s \mathcal{E}_m^{C_s} \left[\mathcal{A}_{s,m} + (1 + \mathfrak{n}_{s+d_{\mathrm{geo}},m})\mathcal{A}_{b_1,m}\right]. \tag{161}$$

To justify the tame form, keep the derivative count in each expanded term: the sum of coefficient derivative orders and the remaining field or source order is at most s. Interpolate each intermediate coefficient norm between its fixed low norm and its order-$s + d_{\mathrm{geo}}$ norm, and each remaining field norm between the corresponding entries of $\mathcal{A}_{b_1,m}$ and $\mathcal{A}_{s,m}$. The interpolation exponents have sum at most one. The weighted arithmetic–geometric mean inequality therefore places at most one factor at high order. Sobolev embedding uses only the fixed base order b_1. Reciprocal normal coefficients have the same property by their positive lower bound. This proves (161); it does not require smallness of any order-s state norm.

The reference factors in that finite expansion may have an exponent C_s growing with s. Absorb their *complete product* only after the expansion: $\mathcal{E}_m^{C_s} \leq C_s M_m$ costs one coefficient-weight order. Paying for each recursion separately would obscure this fixed loss. For precision, write

$$\left(\|Z\|^{\mathrm{mix}}_{t,s}\right)^2 = \sum_m M_m^{2t} H_m^{-2\beta} \sum_{a+b\leq s} \|\partial_y^a \partial_\varphi^b Z_m\|_2^2.$$

The zeroth-normal entry of $\mathcal{A}_{s,m}$ is controlled by (147). For its first-normal entry, use the first-normal reference bound at angular order $s-1$: its one extra angular derivative asks for precisely order s of $\widetilde{F}$. Summing (161) therefore yields

$$\begin{aligned}\|W\|^{\mathrm{mix}}_{t,s} \leq C_{t,s}\big[&\|\widetilde{F}\|_{t+L+1,s} + \|F\|^{\mathrm{mix}}_{t+1,s-2} \\ &+ (1 + \mathfrak{n}_{t+s+D})\|F\|^{\mathrm{mix}}_{0,b_1}\big].\end{aligned} \tag{162}$$

Here D is fixed: it covers $L+1$, d_{geo} and the fixed base and embedding orders. The scalar columns are compactly supported and satisfy the same tame estimate; their scalars are bounded by the base inverse. In the high-coefficient term the M^{t+L+1} weight is placed on the localized state norm, leaving the low field in the reference output space of weight $-L$. Choose the original b_0 large enough to control the finitely many base norms used here. The finitely many orders below $b_1 + 2$ follow from this same base calculation.

Apply (158) to the first term in (162). The weight and derivative indices there have sum $t + s + L + 1$; the source term has sum $t + s - 1$. Thus taking $t + s \leq r$ costs a fixed number of grading orders, independent of r. The physical coordinate and source transformations have their already proved fixed tame losses. Increasing d_* once to include those losses gives (163). Physical seam jets match by the argument at the end of Section 7.

Undoing the transformations gives

$$\widehat{L}_{\mathbf{x}}\eta + k_{\mathbf{x}}\mu + \lambda\psi = f, \qquad \ell_*\eta = 0.$$

Endpoint fluxes vanish. Moreover $\mathsf{Q}k_{\mathbf{x}} = 0$ by differentiating the integral compatibility of the original residual. Integration therefore gives $\lambda = \mathsf{Q}f$. For $f \in \mathcal{Y}_\beta$ it is zero.

Proposition 8.2 (The perturbed scalar inverse)**.** *On the single neighborhood fixed above there is a unique normalized pair $G_{\mathbf{x}} f = (\eta, \mu)$ solving $\widehat{L}_{\mathbf{x}}\eta + k_{\mathbf{x}}\mu = f$ for every $f \in \mathcal{Y}_\beta$. The height belongs to $\mathcal{X}_\beta \cap \ker \ell_*$ and has right endpoint value zero. Its left endpoint value is free. For a fixed d_*,*

$$\|G_{\mathbf{x}} f\|_s \le C_s\big[\|f\|_{s+d_*} + (1 + \mathfrak{n}_{s+d_*})\|f\|_{b+d_*}\big]. \tag{163}$$

The inverse is smooth tame jointly in state and source.

Proof. Existence and the estimate were proved above. Factorization on $\mathbb{D}_*$ gives both inverse identities, and its exhaustion property gives uniqueness among smooth admissible fields. The resolvent identity first proves continuity in the base space and then, using the higher estimates, at every stronger order. The coefficient family is smooth tame. Hamilton's inverse-family theorem [8, II, Theorem 3.1.1, p. 150] now applies: a smooth tame linear family with unique solutions and a continuous tame inverse has a smooth tame inverse. Its uniqueness and continuity hypotheses have thus been checked before invoking it. □

9 From the linear seed to a nonlinear height

We can now see why the unknowns were chosen as they were. The height solves the curvature equation. One compact metric parameter removes its second-mode obstruction. One radial control per negative component adjusts the mean travel angle. Finally the odd boundary charts adjust the nonconstant return maps. The last two tasks are separated by taking the angular mean. Their feedback through the height contains the small amplitude q, which makes the coupled linear problem invertible.

9.1 The residual and its four unknowns

Use the state (128), with $q \in \mathbb{R}$ a parameter. Set

$$\mathcal{R}(\mathbf{x}, q) = (\mathcal{A}, \mathfrak{e}), \qquad \mathcal{A} = \mathcal{F}_q(w, u), \qquad \mathfrak{e} = \mathfrak{e}(qu, \kappa, h_L). \tag{164}$$

The tangent and residual spaces are

$$\mathfrak{X} = (\mathcal{X}_\beta \cap \ker \ell_*) \times \mathbb{R} \times \mathcal{K}_\beta \times \mathcal{Z}_{\beta,\mathrm{odd}}, \qquad \mathfrak{Y} = \mathcal{Y}_\beta \times \mathcal{Z}_{\beta,\mathrm{even}}.$$

The second residual space includes its angular means. The scalar space uses the integral compatibility from the connection form. The normalized scalar equation is smooth through $q = 0$ by (25). The coefficient and flow calculus of Section 7 makes $\mathcal{R}$ smooth tame on one open neighborhood $\mathfrak{n}_{b_0} < \epsilon_0$. At

$$q = 0, \qquad \mathbf{x}_0 = (\Phi_*, 0, 0, \mathrm{id})$$

it is zero: $\mathscr{L}_{w_*}\Phi_* = 0$ and the reference return maps are the prescribed rotations.

Write the differentiated circle rows as

$$C^\circ_{\mathbf{x}} = D_Z\mathfrak{e}, \quad R^\circ_{\mathbf{x}} = \langle C^\circ_{\mathbf{x}}\rangle, \quad \mathcal{D}_{\mathbf{x}} = D_\kappa\mathfrak{e}, \quad J_{\mathbf{x}} = \langle \mathcal{D}_{\mathbf{x}}\rangle. \tag{165}$$

The map on component m depends on its height jets, its chart and κ_m. It does not depend on ν or the endpoint scalar, whose positive jets are supported in positive components. Thus $J_{\mathbf{x}}$ is diagonal. At a radial state $\mathfrak{e} = T(w) - T(w_*)$, so the normalization of the controls gives $J_0 = I$. On one small low-order neighborhood its entries are uniformly separated from zero and its inverse is smooth tame.

The reference mean derivative annihilates nonzero height frequencies and odd chart directions. For a radial increment Z it is

$$(R_0^\circ Z)_m = -\int_{I_m} v_0 \frac{1+w_*^2}{w_*} Z_r \, dr. \tag{166}$$

To verify this, at a radial state the speed is $v_0 w_* \sqrt{D}/(w_* + Z_r)$, with $D = 1 - 2w_* Z_r - Z_r^2$. Its logarithmic variation is $-(1+w_*^2)Z_r/w_*$. Angular translation invariance gives the nonzero-mode assertion. In particular $R_0^\circ \Phi_* = 0$.

With $K_{\mathbf{x}} = D_\kappa \mathcal{A}$ and $k_{\mathbf{x}} = D_\nu \mathcal{A}$, the corrected linear equations are

$$\begin{aligned} \widehat{L}_{\mathbf{x}} \eta + k_{\mathbf{x}} \mu + K_{\mathbf{x}} \xi &= f, \qquad \ell_* \eta = 0, \\ q C_{\mathbf{x}}^\circ \eta + \mathcal{D}_{\mathbf{x}} \xi + \mathcal{M}_T v &= g. \end{aligned} \tag{167}$$

The actual chart variation is $v \circ h_L$. The equation differs from the true derivative only by the two residual terms already identified and the chart error in (132).

	η	μ	ξ	v	
curvature	$\widehat{L}$	k	K	0	
returns	qC°	0	$\mathcal{D}$	$\mathcal{M}_T$	$\langle \mathcal{D} \rangle = J, \quad \langle \mathcal{M}_T \rangle = 0$

Figure 12: The corrected derivative and the roles of its four columns. Taking the mean of the second row removes the chart column. Feedback from that row to the height is multiplied by q.

9.2 The control contribution gains all component weights

Let $G_{\mathbf{x}}$ be the scalar inverse of Proposition 8.2 and set $U_{\mathbf{x}} = G_{\mathbf{x}} K_{\mathbf{x}}$. The useful base space for control increments is

$$B_\beta = \left\{ \xi : \sum_m H_m^{-2\beta} |\xi_m|^2 < \infty \right\}.$$

For every s, with a fixed loss d_*,

$$\|K_{\mathbf{x}} \xi\|_{\mathcal{Y}_\beta, s} + \|U_{\mathbf{x}} \xi\|_{\mathcal{X}_\beta, s} + |(U_{\mathbf{x}} \xi)_\mu| \le C_s (1 + \mathfrak{n}_{s+d_*}) \|\xi\|_{B_\beta}. \tag{168}$$

In the middle term the norm denotes the height component of $U\xi$.

Here is the reason for this strong smoothing in the component index. Differentiate the removable formula for $\mathcal{F}_q$ with respect to a control. Every term contains that control increment and a positive height jet: the curvature expression is unchanged by adding a constant to the height. This remains true at $q = 0$, where the derivative is that of $\mathscr{L}_w u$. On component m write $\xi_m = H_m^\beta b_m$. The height jet contributes another H_m^β. After dividing by the source weight there remains a factor H_m^β, which absorbs every fixed power of M_m and every complete coefficient product. The endpoint scalar has no height jet on these control supports. The local product bound puts one state factor at high order, and summation of $|b_m|^2$ gives the estimate for K. Composing with G adds only its fixed loss.

9.3 Solving the coupled equations

Put $g_0 = \langle g \rangle$. Solving the first row of (167) gives $y = (\eta, \mu) = Gf - U\xi$. Taking the mean of the second then gives

$$(I - qJ^{-1}R^{\circ}U)\xi = J^{-1}(g_0 - qR^{\circ}Gf).$$

The rows act on the height component of G and U. By (168), $R^{\circ}U$ is bounded on the *same* B_β, with a constant controlled by finitely many state norms. Choose the low neighborhood to control those norms, then choose $|q|$ so that $\|qJ^{-1}R^{\circ}U\|_{B_\beta \to B_\beta} < 1/2$. The solution is therefore

$$\begin{aligned} \xi &= (I - qJ^{-1}R^{\circ}U)^{-1}J^{-1}(g_0 - qR^{\circ}Gf), \\ (\eta, \mu) &= Gf - U\xi, \\ v &= \mathcal{M}_T^{-1}(g - qC^{\circ}\eta - \mathcal{D}\xi). \end{aligned} \tag{169}$$

The last argument has zero mean by the first equation, so the inverse in (133) applies.

Higher control norms come from the already solved identity

$$\xi = J^{-1}(g_0 - qR^{\circ}Gf) + qJ^{-1}R^{\circ}U\xi.$$

The final term has every higher norm using only $\|\xi\|_{B_\beta}$ and one high state norm, by the smoothing estimate. The first has the requisite source norms with fixed loss. Consequently the same solution lies in all control spaces, with a tame bound. Substitution gives the height and chart estimates. The resolvent identity proves base continuity; this identity and its higher estimates prove continuity in every order. Hamilton's inverse-family theorem applies to the control inverse, making all three formulas smooth tame. Thus

$$V(\mathbf{x}, q)(f, g) = (\eta, \mu, \xi, v \circ h_L)$$

is a smooth tame map, linear in the residual pair.

9.4 The exact quadratic-error identity

We invoke Hamilton's implicit theorem with quadratic error [8, III, Theorem 3.3.1, pp. 211–212]. In the notation here its hypothesis is a smooth tame $\mathcal{R}$ on an open subset of tame spaces, together with a linear smooth tame V and a bilinear smooth tame $\mathcal{Q}$ such that

$$D_{\mathbf{x}}\mathcal{R}\, Vy = y + \mathcal{Q}(\mathbf{x}, q)[\mathcal{R}, y]. \tag{170}$$

At an initial zero this gives a local smooth tame solution branch in the parameter. We check the identity with two *independent* residual arguments; an informal quadratic bound would not suffice.

For the increments in (169), the actual derivative is

$$D_{\mathbf{x}}\mathcal{R}(\eta, \mu, \xi, v \circ h_L) = \big(f + q\partial_\theta(B\mathcal{A}\eta_\theta) + \mathscr{E}_{\mathbf{x}}(\mathfrak{e})\eta,\ \ g + \mathcal{Q}_h(\mathfrak{e}, v)\big). \tag{171}$$

For $r = (r_1, r_2)$ and $y = (f, g)$ define

$$\mathcal{Q}(\mathbf{x}, q)[r, y] = \big(q\partial_\theta(Br_1\eta_\theta) + \mathscr{E}_{\mathbf{x}}(r_2)\eta, \mathcal{Q}_h(r_2, v)\big).$$

In $\mathcal{Q}_h$ keep the actual state paths in (132), replacing only its explicit $\mathfrak{e}, \mathfrak{e}'$ factors by r_2, r_2'. The same convention was built into $\mathscr{E}$. The increments are linear in y, so this formula is bilinear in (r, y) and gives (170).

Its scalar component has zero integral: one term is an angular derivative and the other is an interior divergence with flat global endpoint flux. The parity of its circle component is the required one. Each product has two weak endpoint factors; the coefficient calculus and flow estimates from Section 7 therefore make $\mathcal{Q}$ smooth tame into $\mathfrak{Y}$. All state neighborhoods in this argument are controlled by one fixed finite order. No output order changes their radius.

The remaining hypotheses have already been verified: the spaces and the normalization slice are tame direct summands; the domain is open; the residual is smooth tame; V is smooth tame; and the reference state is a zero. At a zero the identity itself implies surjectivity of the state derivative. Hamilton's theorem gives a single smooth branch

$$\mathbf{x}(q) = (u_q, \nu_q, \kappa_q, h_{L,q}), \qquad \mathcal{R}(\mathbf{x}(q), q) = 0, \qquad u_q = \Phi_* + O(q) \quad \text{in every } \mathcal{X}_\beta \text{ norm.} \tag{172}$$

This solves the nonlinear system in the weak endpoint scale. The following section extracts stronger information from the equations themselves, rather than starting another iteration in another space.

10 The endpoint gain and the closed surface

The nonlinear construction used the weak weight H^β because the reference inverse is available for exponents below one. The metric controls nevertheless acquire twice that exponent. This is forced by two angular average equations. The first recovers the mean height from a quadratic expression; the second recovers the metric controls from that mean and another quadratic expression. There is no new nonlinear iteration in a stronger function space. Every height in $\mathcal{X}_\beta$ already has a smooth flat endpoint extension. The gain below strengthens the annular result by quantifying the corrections to the mean height and the metric; geometric closure uses the flat extensions established in Section 7.

10.1 The two identities that improve decay

Let $\mathfrak{F}_w(U)$ be the connection period from (27). Its derivative at $U = 0$ is $w\langle U_r\rangle$. Write its bilinear Taylor remainder as

$$\mathcal{B}_{w,U}(v_1, v_2) = \int_0^1 (1-t) D_U^2 \mathfrak{F}_w(tU)[v_1, v_2]\, dt. \tag{173}$$

The normalized period of a solution vanishes, because its radial derivative is the mean scalar residual and its endpoint value is zero. Therefore, with $u^{\mathrm{av}} = \langle u\rangle$,

$$(u^{\mathrm{av}})' = -qw^{-1}\mathcal{B}_{w,qu}(u, u). \tag{174}$$

For the mean circle map use additive coordinates $\mathbf{v} = (Z, \kappa, a)$, where $Z = qu$ and $h_L = \mathrm{id} + a$, and put

$$\mathcal{C}(\mathbf{v}) = \langle \mathfrak{e}(\mathbf{v})\rangle, \qquad \mathcal{B}^\circ_{\mathbf{v}}(v_1, v_2) = \int_0^1 (1-t) D^2\mathcal{C}(t\mathbf{v})[v_1, v_2]\, dt.$$

Its first derivative is the identity on κ, zero on odd chart directions, and R_0° on the radial mean height. Thus the second identity at a solution is

$$0 = \kappa + qR_0^\circ u^{\mathrm{av}} + \mathcal{B}^\circ_{\mathbf{v}}(\mathbf{v}, \mathbf{v}). \tag{175}$$

The two displayed formulas already explain the exponent 2β. We must verify these estimates in the stronger topology, including parameter derivatives, to retain this quantitative conclusion for the entire branch.

10.2 Two weak factors give one strong remainder

Choose a convex neighborhood of the raw state $(U, \nu, \kappa, a) = (0, 0, 0, 0)$ in the weak product spaces, small in one finite norm. On it the denominators $D, Q/w$ are positive and the circle lifts are uniformly increasing. Both Taylor segments above stay in this neighborhood; the circle map is independent of ν. All constants below refer to this one neighborhood.

Lemma 10.1 (Quadratic remainders in the doubled weight)**.** *The maps $w^{-1}\mathcal{B}_{w,U}$ and $\mathcal{B}^{\circ}_{\mathbf{v}}$ are smooth tame with outputs in, respectively, the radial space $\mathcal{X}^0_{2\beta}$ and $\mathcal{K}_{2\beta}$. For every fixed number k of state variations there is a loss d_k, independent of the output order s, for which*

$$\begin{aligned}&\|D^k\mathcal{B}[\eta_1,\dots,\eta_k](\eta_{k+1},\eta_{k+2})\|_{2\beta,s}\\&\quad\le C_{s,k}\left[\sum_{i=1}^{k+2}\|\eta_i\|_{s+d_k}\prod_{j\neq i}\|\eta_j\|_{b+d_k}+(1+\|\mathbf{y}\|_{s+d_k})\prod_j\|\eta_j\|_{b+d_k}\right].\end{aligned}\tag{176}$$

The inputs use their weak product norms; compact scalar directions use their absolute values. The same bound holds for all higher derivatives of $\mathcal{C}$ with at least two arguments.

Proof. On component m every control or chart increment, and every positive height jet, has a factor H_m^{β}. Constants at an endpoint cause no difficulty: their positive jets are supported in the compact positive region, and vanish on all negative control supports. Writing $u = H^{\beta}\widetilde{u}$ gives, for example,

$$H^{-\beta}\mathsf{D}u = \mathsf{D}\widetilde{u} + \beta(\mathsf{D}\log H)\widetilde{u}.$$

Thus positive differentiation does not remove a weak factor. Products and restriction to normalized pieces cost fixed derivatives and reference factors $\mathcal{E}_m^{C_{s,k}}$. Retain two weak factors for the output weight, place the highest derivative on one input or one state coefficient, and absorb the *complete* coefficient product into one extra M order. Finite overlap and $\|ab\|_{\ell^2} \le \|a\|_{\ell^2}\|b\|_{\ell^\infty}$ sum the estimates. It remains to check that the nonlinear operations retain these two factors.

For the period this is visible from its integrand. With $k = w + U_r$, $p = U_\theta/r$ and $E = 1 + w^2$, it is

$$\frac{1 + kp_\theta/(1-p^2)}{\sqrt{E(1-p^2) - k^2}} - 1.$$

It is a smooth function of w and the positive jets $(U_r, U_\theta/r, U_{\theta\theta}/r)$ on the chosen neighborhood. Each second derivative contains one such jet from each argument. Further state differentiation cannot remove either argument. The reciprocal $w^{-1} = w_*^{-1}e^{-\nu\chi_* - \sum\kappa_m\chi_m}$ is included in the coefficient product before its absorption; it does not cost an endpoint exponent.

For the circle map consider a local normalized characteristic flow $\dot\Theta = G(t, \Theta; \mathbf{y})$, using (138) at a root. Its first and second state variations satisfy

$$\dot\Theta_i = G_\theta\Theta_i + G_i,$$

$$\dot\Theta_{12} = G_\theta\Theta_{12} + G_{\theta\theta}\Theta_1\Theta_2 + (G_1)_\theta\Theta_2 + (G_2)_\theta\Theta_1 + G_{12}.$$

The homogeneous integrating factor is uniformly bounded, since the reference speed has zero angular derivative and the perturbation has a small integrated derivative. G_i has one weak factor and G_{12} has two. Variation of constants gives one factor in Θ_i and two in Θ_{12}. Composing flows does not change the component weight.

Inversion preserves this count. If $\iota = h^{-1}$, then

$$\iota_i = -\frac{h_i\circ\iota}{h'\circ\iota}, \qquad \iota_{12} = -\frac{h_{12}\circ\iota + (h_1'\circ\iota)\iota_2 + (h_2'\circ\iota)\iota_1 + (h''\circ\iota)\iota_1\iota_2}{h'\circ\iota}.$$

The denominator is bounded below, and every term in the second numerator contains both variations. Apply this to f^{-1} and h_R^{-1} in (131); parity projection and averaging preserve the weights.

For higher variations label the directions by a set I. In the chain rule for $G(t,\Theta;\mathbf{y})$ a subset $J\subset I$ acts directly on the state and a partition of $I\setminus J$ acts on the flow. Each term is

$$(D_{\mathbf{y}}^{|J|}\partial_\theta^\ell G[\eta_j : j\in J])\circ\Theta\prod_B \Theta_B,$$

where ℓ is the number of blocks. Keep $G_\theta\Theta_I$ on the linear side; all other terms use proper lower variations and contain each label exactly once. Induction retains both designated argument factors. The inverse identity has the same induction. Additional weak factors are bounded in low norms. Local tame product and composition estimates put only one factor at high order. The coefficient formulas, root regularization and finite restrictions have fixed order; k variations add a cost depending on k, not on s. This proves (176).

Finally these are estimates for the actual differentiated formulas. Taylor's integral remainder with one additional variation proves continuity and differentiability in the 2β norm. Iterating proves smoothness in the stronger Fréchet topology, not merely pointwise stronger decay. □

10.3 Integrating the mean without losing the exponent

For $\gamma>0$ a primitive from its own endpoint preserves exponent γ. To see this, orient ϱ toward the endpoint. Then $(\log H)'\to -1$ and $(\log E_0)'\to 0$. For a fixed A, the ratio of $H^\gamma E_0^A$ at $\sigma\ge\varrho$ to its value at ϱ is eventually bounded by $e^{-\gamma(\sigma-\varrho)/2}$. The conjugated integration kernel has bounded row and column integrals; $dr=\ell\,d\varrho$ supplies only a fixed coefficient factor. The L^2 Schur estimate applies. Further derivatives use the integrand itself, while for $c\ge 0$ the weight ratio $M(\varrho)^c/M(\sigma)^c\le 1$. The remaining compact interval has ordinary bounded integration. This proves a tame primitive estimate with fixed loss. For a primitive normalized at the right endpoint, subtract its left constant times χ_L to use the left endpoint estimate.

Let $\mathscr{P}$ denote this right-normalized primitive. Formula (174) identifies the weak mean with the explicit stronger-space map

$$\mathcal{U}(\mathbf{x},q)=-q\mathscr{P}[w^{-1}\mathcal{B}_{w,qu}(u,u)]=u^{\mathrm{av}}\quad\text{on the solution branch.}\tag{177}$$

It is smooth into $\mathcal{X}_{2\beta}$ and has size $O(q)$ in every norm. In particular its endpoint scalar is also $O(q)$. The fixed row (166) maps its strong radial part into $\mathcal{K}_{2\beta}$ with fixed loss, and annihilates the endpoint step. The circle identity therefore gives

$$\kappa_q=-qR_0^\circ\mathcal{U}(\mathbf{x}(q),q)-\mathcal{B}^\circ_{\mathbf{v}(q)}(\mathbf{v}(q),\mathbf{v}(q)).\tag{178}$$

The weak branch has $\mathbf{v}(q)=O(q)$. The two terms on the right are consequently $O(q^2)$ in every $\mathcal{K}_{2\beta}$ norm. This identity also proves smoothness of the existing controls in that stronger topology and $\dot\kappa_0=0$.

The compact scalar is quadratic as well. Differentiate the scalar equation at $q=0$:

$$\mathscr{L}_{w_*}\dot u_0+\dot\nu_0 k_*+K_*\dot\kappa_0+\partial_q\mathcal{F}_q(w_*,\Phi_*)|_{q=0}=0,\qquad \ell_*\dot u_0=0.$$

The final term is quadratic in second-mode jets. Products of $\cos 2\theta$ and $\sin 2\theta$ have frequencies zero and four, so its second-mode projection vanishes. Since $\dot\kappa_0=0$, the second-mode equation is the homogeneous bordered reference equation. Its left inverse gives $\dot\nu_0=0$, hence $\nu_q=O(q^2)$.

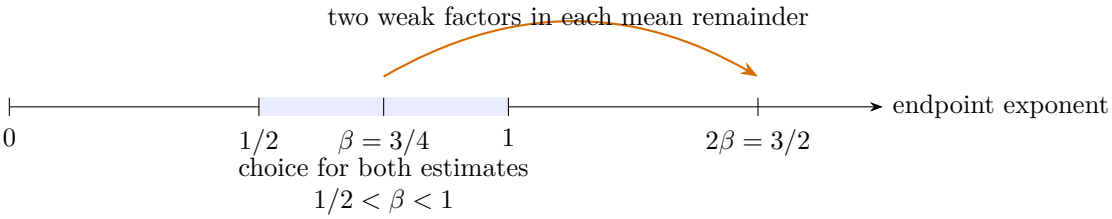


Figure 13: The choice $\beta = 3/4$ serves two estimates. Neighboring exponents below one control the reference inverse. The mean equations multiply two weak factors and produce an exponent above one, giving the strengthened metric bound in Theorem 10.2. Flat extension already follows from any positive exponent.

10.4 The annular conclusion

Theorem 10.2 (A nonradial height with flat ends)**.** *For the fixed reference of Proposition 4.2 there is $q_0 > 0$ and a smooth branch $(u_q, \nu_q, \kappa_q, h_{L,q})$, $|q| < q_0$, through $(\Phi_*, 0, 0, \mathrm{id})$, with the following properties. Set $w_q = w_* e^{\nu_q \chi_* + \sum_m \kappa_{q,m}\chi_m}$ and $z_q = h_{w_q} + q u_q$, choosing a fixed additive normalization for $h'_{w_q} = w_q$. Then*

$$\mathcal{N}_{h_{w_q}}(z_q) = 0, \qquad |dz_q|_{(1+w_q^2)dr^2 + r^2 d\theta^2} < 1, \qquad \ell_* u_q = 1.$$

The profile is positive in the annular interior and extends smoothly by zero across both endpoints. The height is even and π-periodic in angle, extends smoothly by a constant at each endpoint, and is nonradial for $q \neq 0$. The two constants may differ. All roots of w'_q and their radial germs agree with the reference. The two characteristic returns of the true height derivative obey

$$h_{R,q,m} f_{q,m,\pm} h_{L,q,m}^{-1} = R_{\pm T_m} \tag{179}$$

for increasing odd π-equivariant charts and the fixed reference angles. On the same interval of q, for every s,

$$\|u_q - \Phi_*\|_{\mathcal{X}_\beta, s} \le C_s |q|, \qquad |\nu_q| + \|\kappa_q\|_{\mathcal{K}_{2\beta}, s} \le C_s q^2. \tag{180}$$

The branch is smooth with these topologies. In particular, for every $a \ge 0$ and $\epsilon > 0$,

$$|\partial_r^a (w_q - w_*)| \le C_{a,\epsilon} q^2 H^{1-\epsilon}$$

at the endpoints, and $w_q - w_* = O(q^2)$ in every smooth norm.

Proof. The weak branch was constructed in Section 9. Equations (177) and (178) give the strengthened estimates and smooth dependence just proved. Choose q_0 so that $[-q_0, q_0]$ lies inside its domain. Every seminorm of this smooth branch is bounded on that one compact interval; the constants depend on the order, but q_0 does not.

On the tails the compact control vanishes. Differentiating the formula for w_q gives $|\partial_r^a (w_q - w_*)| \le C_a q^2 H^{2\beta} E_0^{C_a}$. Since $2\beta > 1$, (122) implies the displayed endpoint estimate. In the compact middle use smoothness and the quadratic bound for ν_q. All controls vanish near each turning circle, preserving its full metric germ; the common low neighborhood preserves the signs elsewhere.

At a zero both residual corrections vanish. The exact return identities for $\widehat{L}$ therefore hold for the true derivative. The common inequality $D > 0$ is equivalent to $|dz_q|_g < 1$. The scalar residual vanishes, and the curvature formula identifies this with flatness of $g - dz_q^2$. Since $\ell_* u_q = 1$, the second angular component is nonzero, so z_q is nonradial for $q \neq 0$.

The height space already gives a flat extension after subtracting its endpoint constants, including all parameter derivatives. Extend w_q by zero and z_q by those constants and enlarge the annulus slightly. The enlarged annulus has zero-profile open collars around its original positive-profile core. The remainder metric stays positive and flat on these collars, giving the data needed for geometric closure. This proves the theorem. □

10.5 Fixing the metric and moving the surface

We finish by distinguishing the two parameters. The small parameter q was used to construct one nonradial realization of one rotational metric. Choose a sufficiently small nonzero value and keep it fixed. The annular theorem then supplies exactly the data of Theorem 2.4: a positive flat metric $g - dz_q^2$, the half-turn symmetry, and two constant-height collars.

That theorem develops the graph, fills its inner disk, and attaches the fixed marked complementary cap. Its independent phase parameter τ gives the jointly smooth periodic family

$$F_\tau(r,\theta) = R_{-2\pi\tau} F(r, \theta + 2\pi\tau)$$

on the moving annulus and the fixed map elsewhere. The collars cancel the two rotations pointwise. The marked cap makes two resulting closed images noncongruent. The same geometric theorem proves embeddedness, constant area, constant enclosed volume and constant total oriented mean curvature. This proves Theorem A.

A Why a finite angular ansatz does not close

Section 3 ruled out a monotone flat end for the linearized height. A different tempting simplification is to keep a finite number of angular modes in the nonlinear height. The obstruction below explains why the inverse must accommodate all even frequencies, even though its initial seed has only frequency two. It is independent of the existence argument.

Proposition A.1. *Let $E > 0$ be C^1 on a connected interval $I \subset (0,\infty)$. If a real C^2 solution of the Darboux equation for $E\,dr^2 + r^2\,d\theta^2$ satisfies*

$$z^2 = \sum_{|j|\le J} b_j(r) e^{ij\theta}, \qquad b_{-j} = \overline{b_j}, \qquad b_j \in C^2(I),$$

and is radial on a nonempty open radial interval, then it is radial throughout I. This includes every finite Fourier height, and every smooth square root of a finite trigonometric polynomial.

The issue is continuation through a zero of a Fourier coefficient. A nonzero coefficient has a complex phase, so a real-valued uniqueness argument would not suffice. We give the continuation argument rather than assume that the resulting singular ordinary differential equation has ordinary uniqueness.

Proof. Set $v = z^2$ and $K = E'/(2rE^2)$. The Hessian identity $\nabla^2 v = 2z\nabla^2 z + 2dz \otimes dz$, together with the two-dimensional rank-one determinant identity, gives

$$2v \det_g \nabla^2 v - \operatorname{cof}_g(\nabla^2 v)[dv, dv] = 8Kv^2 - 2Kv|dv|_g^2. \tag{181}$$

To check it without dividing by z, expand the two terms on the left: the terms containing $z^3 \operatorname{cof}_g(\nabla^2 z)[dz, dz]$ cancel, leaving $8z^4 \det_g \nabla^2 z$. Substitution of the Darboux equation gives the right side. Thus the identity also holds where $z = 0$.

Let $b = b_J$ for the highest positive frequency J, and put $a = EJ^2/r$, $d = E'J^2/(2r)$. The coefficient of frequency $3J$ in (181) contains only three copies of b. It is

$$\begin{aligned} 0 = (2bb' - ab^2)b'' - (b')^3 + ab(b')^2 - \frac{2a}{r}b^2b' + \frac{2a}{r^2}b^3 \\ + db^2b' - \frac{2d}{r}b^3. \end{aligned} \tag{182}$$

For a direct coefficient check, write the three Hessian entries of v as $A = v_{rr} - E'v_r/(2E)$, $B = v_{r\theta} - v_\theta/r$, $C = v_{\theta\theta} + rv_r/E$. After multiplication by Er^2, the left side is $2v(AC - B^2) - (Cv_r^2 - 2Bv_rv_\theta + Av_\theta^2)$. Replace the jets by $b, b', b'', ibJ, ibJ', -J^2b$ and collect terms; the quadratic term on the right has no frequency $3J$.

On a component where $b \neq 0$, take a continuous complex square root $y^2 = b$. Substitution reduces (182) to

$$(4y' - ay)y'' + a(y' - y/r)^2 + d(yy' - y^2/r) = 0. \tag{183}$$

Fix a compact subinterval and $K_1 > 1 + \sup|a|$. The quantity

$$U = \max(K_1|y|, |y'|) = \max\left(K_1\sqrt{|b|}, \frac{|b'|}{2\sqrt{|b|}}\right)$$

satisfies $|U'| \leq CU$ almost everywhere. Indeed, where the second term is larger, $|4y' - ay| \geq 3|y'|$, and (183) gives $|y''| \leq C|y'|$. Where the first is larger, its derivative has absolute value at most $K_1|y'| \leq K_1U$. The maximum is locally Lipschitz, so the same estimate holds through its switching points. It bounds U both above and away from zero at every finite endpoint of a nonzero component.

At a zero of b, (182) gives $b' = 0$. At a finite endpoint of a nonzero component, $y \to 0$, so $U = |y'|$ nearby. The estimate for y'' yields a limit $y' \to \lambda \neq 0$, and consequently

$$b'' = 2(y')^2 + 2yy'' \longrightarrow 2\lambda^2 \neq 0. \tag{184}$$

Thus such an endpoint is an isolated quadratic zero. Across it U has the continuous value $\sqrt{|b''|/2}$ and the same differential bound.

Consider the open set $\mathcal{O} = \{b \neq 0\} \cup \{b = 0, b'' \neq 0\}$. If a component of $\mathcal{O}$ had an interior boundary point, then $b = b'' = 0$ there. An infinite accumulation of quadratic zeros would contradict the positive lower bound for U and continuity of b''; a last nonzero interval would contradict (184). Hence $\mathcal{O}$ is either empty or all of I. Since b vanishes on the given open radial interval, it must vanish everywhere.

Remove this highest coefficient and repeat down the finite expansion. The resulting z^2 is radial. At each radius a real continuous function on the connected circle whose square is constant is itself constant. Therefore z is radial. □

This result does not say that a single angular mode is useless. It is exactly the right seed for the linearized construction. It says that the nonlinear completion cannot be confined to a finite angular ansatz once it is required to meet a radial collar.

Use of AI

AI systems assisted with mathematical exploration, writing, algebraic verification and critical reading. Separate AI readers were used to review the manuscript; their reports are revision records, not human peer review. The author is responsible for the mathematical claims and their presentation.

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